INFORMATICS INSTITUTE OF TECHNOLOGY

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DEPARTMENT OF BUSINESS MANAGEMENT

DOC327 – WORKING WITH DATA
COURSEWORK – 2022/23
(Foundation September 2022)

Module Leader:	Chathura Wickramasinghe
Team No:	I-2
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Student Id	Student Name
20221031	A.K. Sooriyarachchi -
20221044	W.K.A.Madusha
20221025	D.S.R.D.S Gunasekara
20221041	Azra Safrullah
20221024	H.S.Tennakoon

Acknowledgement.

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Without Ms. Dilki we would not have acquired the necessary knowledge to complete this document; hence, we would like to dedicate a small portion of this text to them. We also want to express our gratitude to all the other IIT lecturers who inspired us and helped us along the way. Finally, we want to extend our gratitude to our parents and friends for helping us in our endeavors.

Question 2

DDL statements required to create a database

CREATE DATABASE Ryanair

Screenshot of the DDL statement:

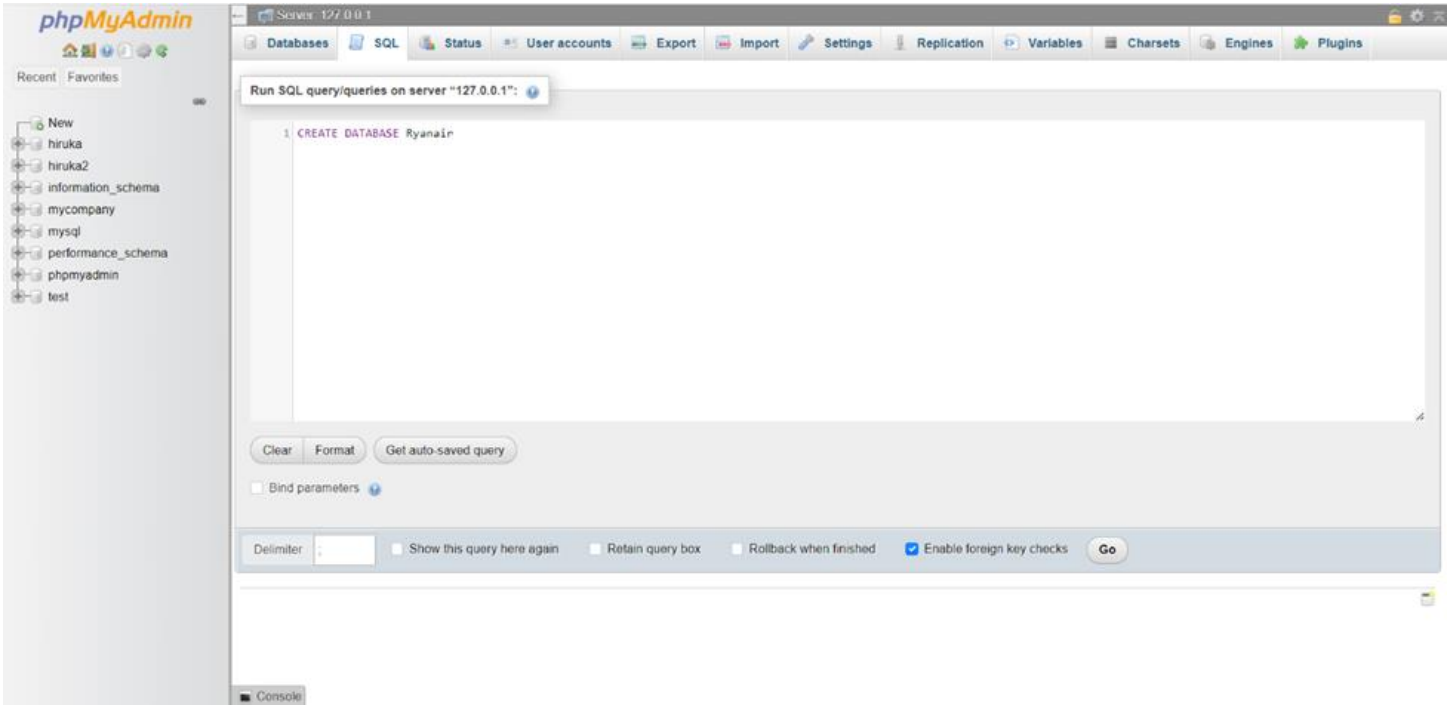


Figure 1 Create Database

Screenshot of the result :

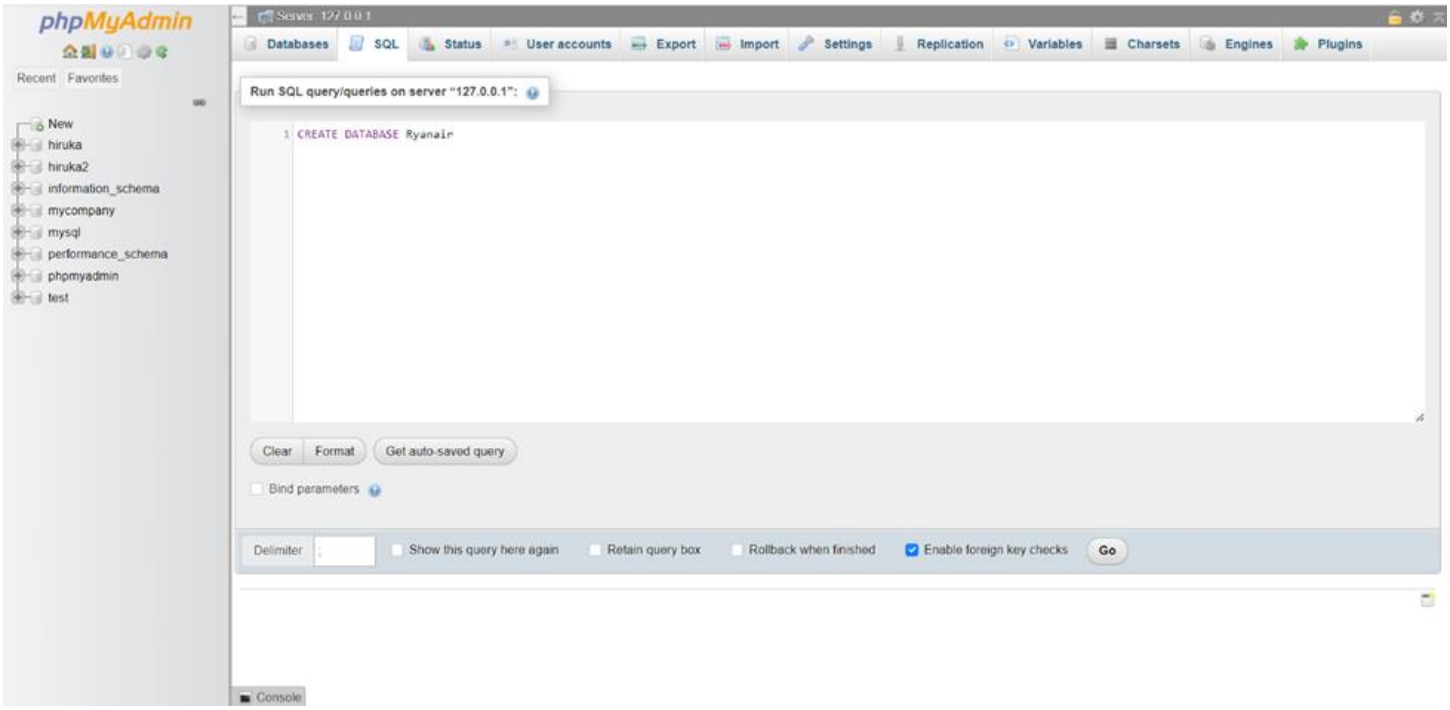


Figure 2 Create Database results

DDL statements required to create the Country Office table

```
CREATE TABLE Country_Office (
   country_office_code INT PRIMARY KEY,
   country_name VARCHAR(50),
```

```
    location VARCHAR(50),
    adress VARCHAR(100),
    number_of_staff INT,
    manager_name VARCHAR(50)
);
```

Screenshot of the DDL statements :

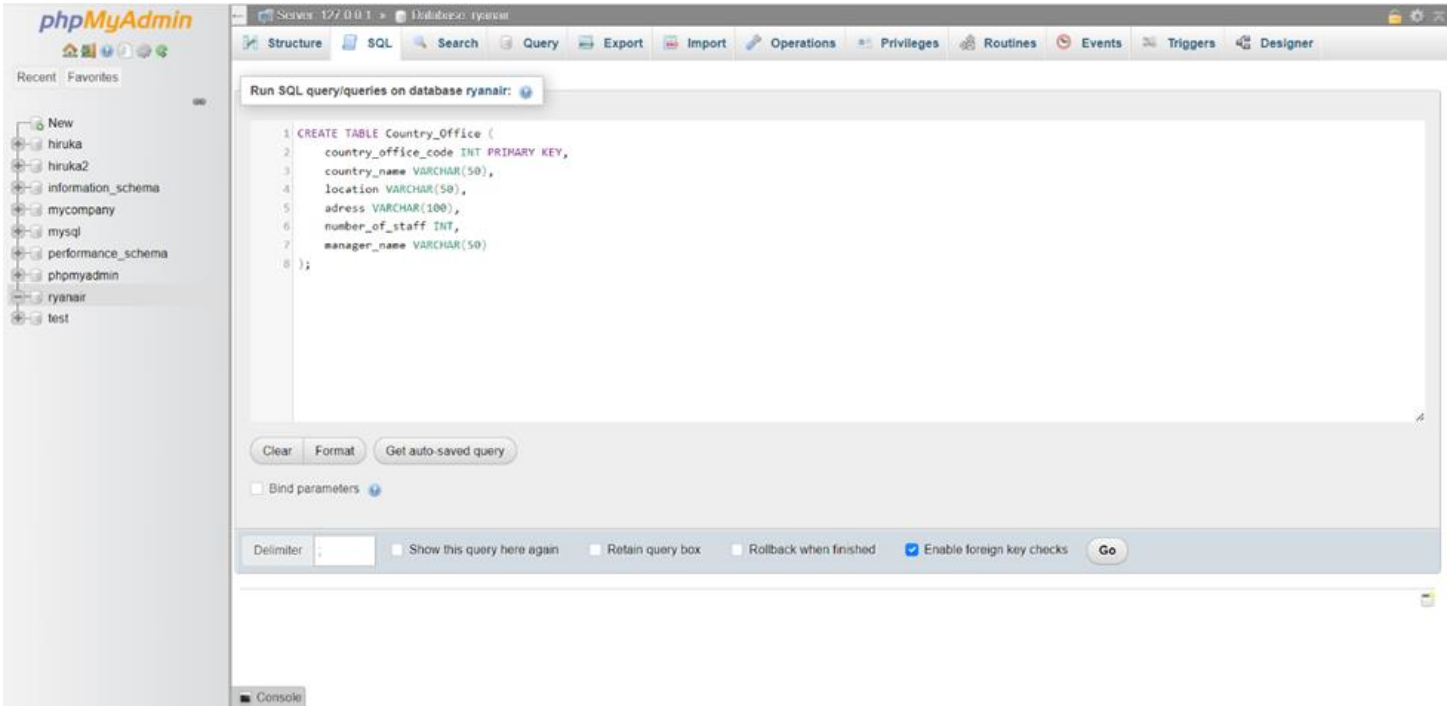


Figure 3 Country Office Table query

Screenshot of the result :

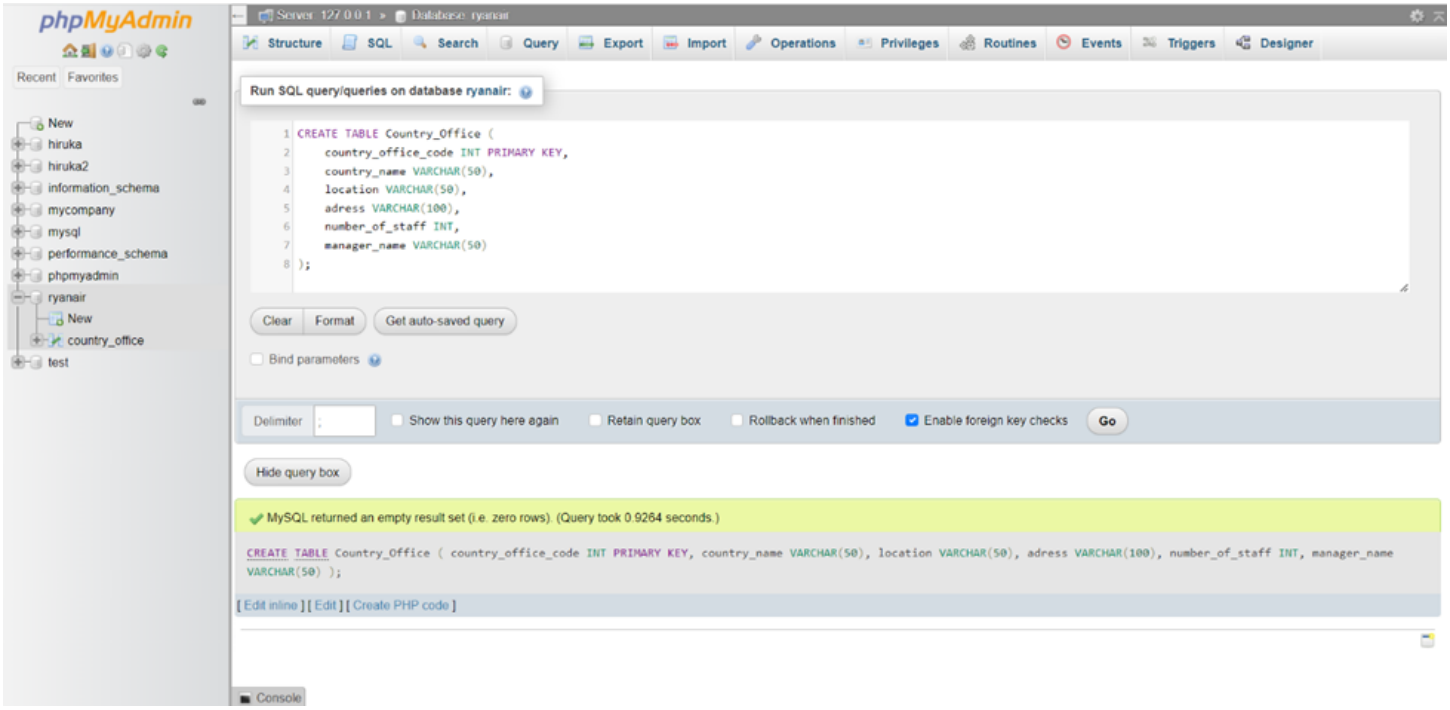


Figure 4 Result of Country Office Table

Screenshot of the table output :

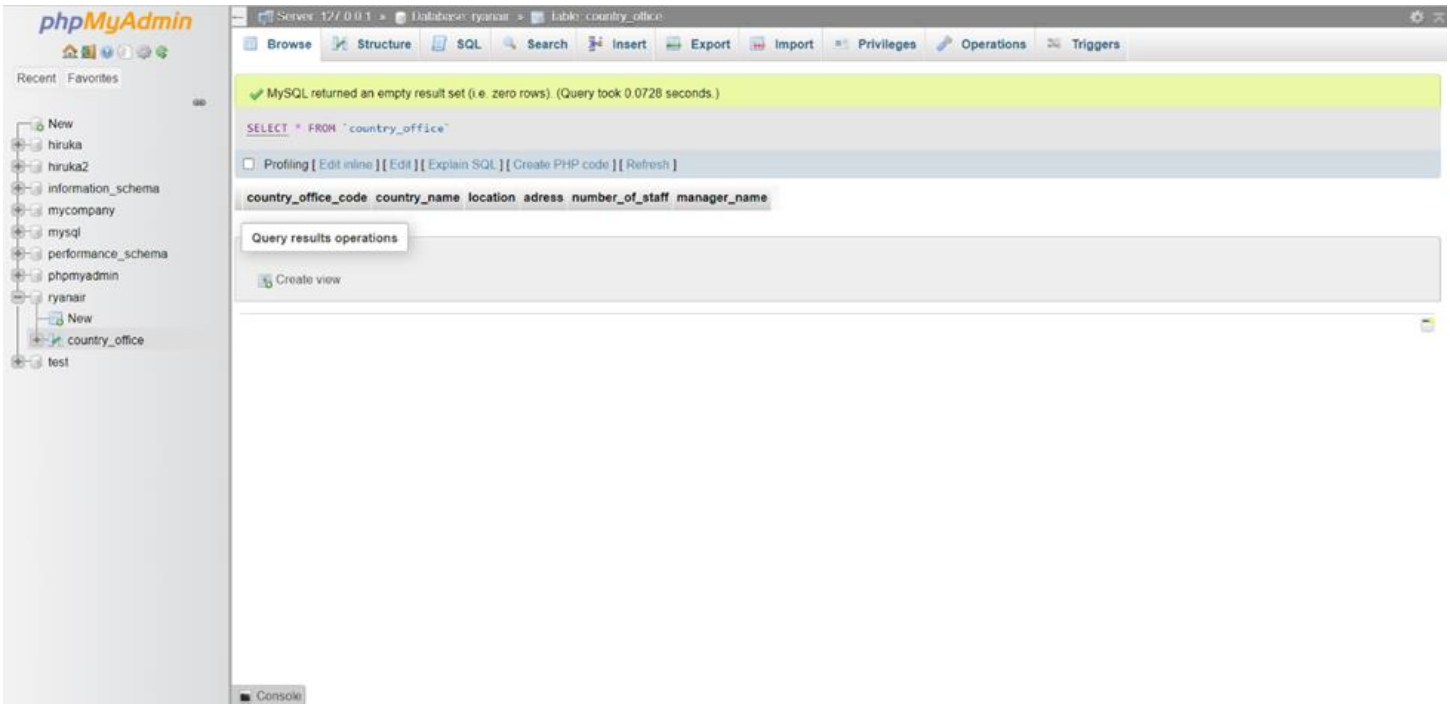


Figure 5 Country Office Output Table

DDL statements required to create the Staff Member table

```
CREATE TABLE Staff_Member (
  staff_member_id INT PRIMARY KEY,
  country_office_code INT,
  first_name VARCHAR(50),
  middle_name VARCHAR(50),
  family_name VARCHAR(50),
  passport_number VARCHAR(20),
  date_of_birth DATE,
  gender VARCHAR(10),
  designation VARCHAR(50),
  salary_grade VARCHAR(10),
  joined_date DATE,
  mobile_number VARCHAR(20),
  email_address VARCHAR(100),
  bank_account_number VARCHAR(20)
);
```

Screenshot of the DDL statements :

Screenshot of the result:
Screenshot of the table output :

DDL statements required to create the Guardian table

```
CREATE TABLE Guardian (
  guardian_id INT PRIMARY KEY,
  name VARCHAR(50),
  passport_number VARCHAR(20),
  address VARCHAR(100),
  contact_number VARCHAR(20),
  bank_account_number VARCHAR(20)
);
```

Screenshot of the DDL statements :

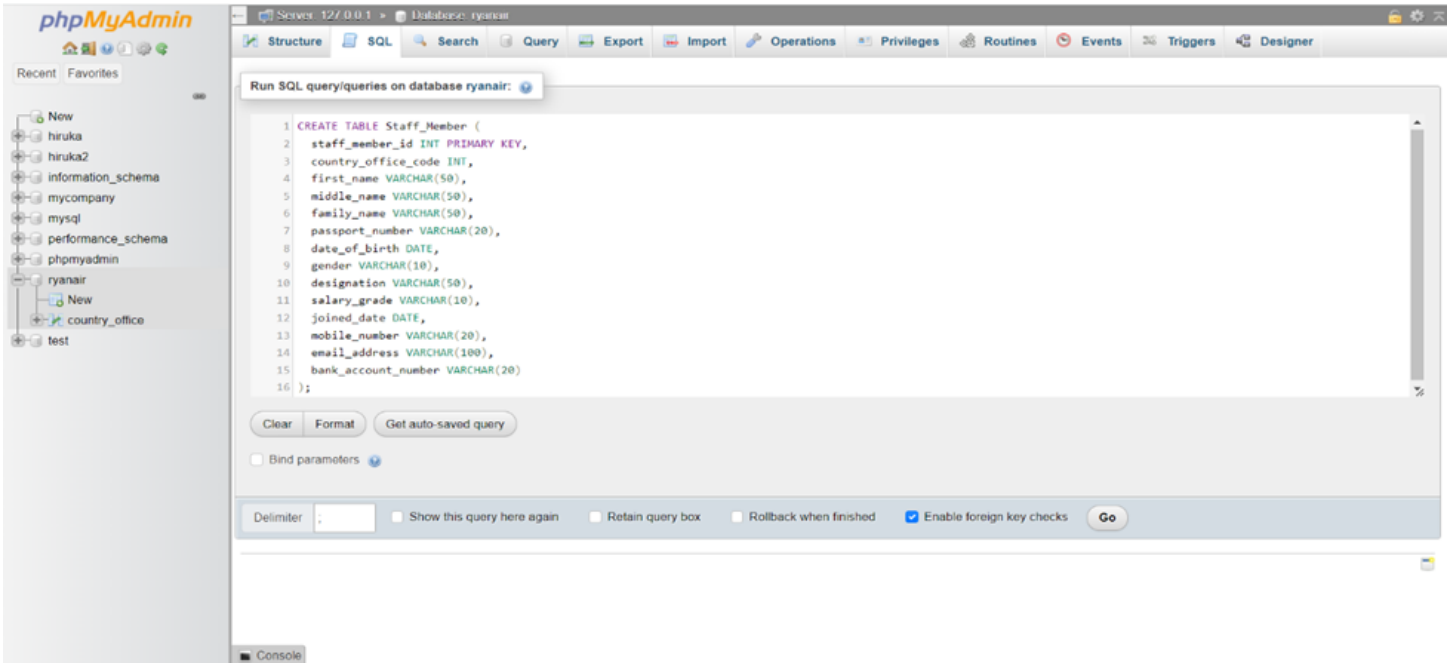


Figure 6 Staff Member Table query

Screenshot of the result :

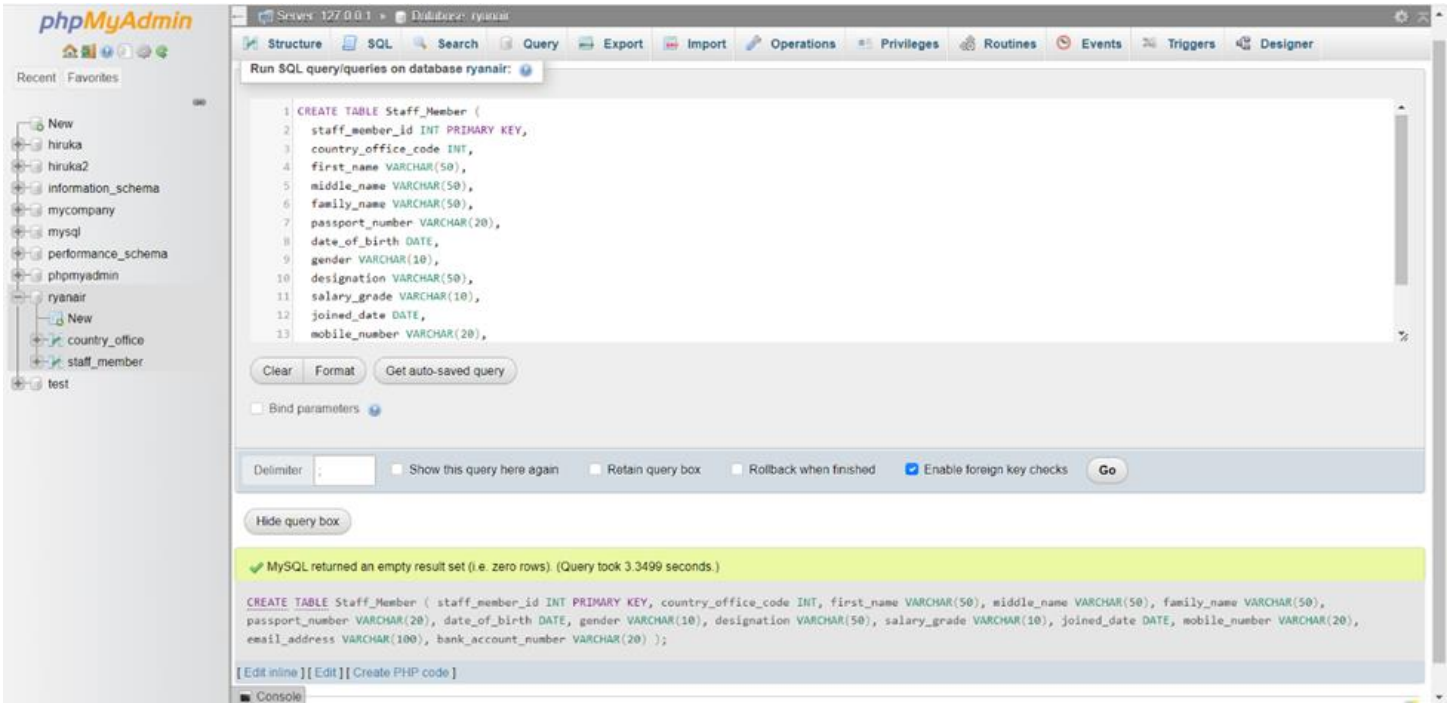


Figure 7 Result of Staff Member Table

Screenshot of the table output :

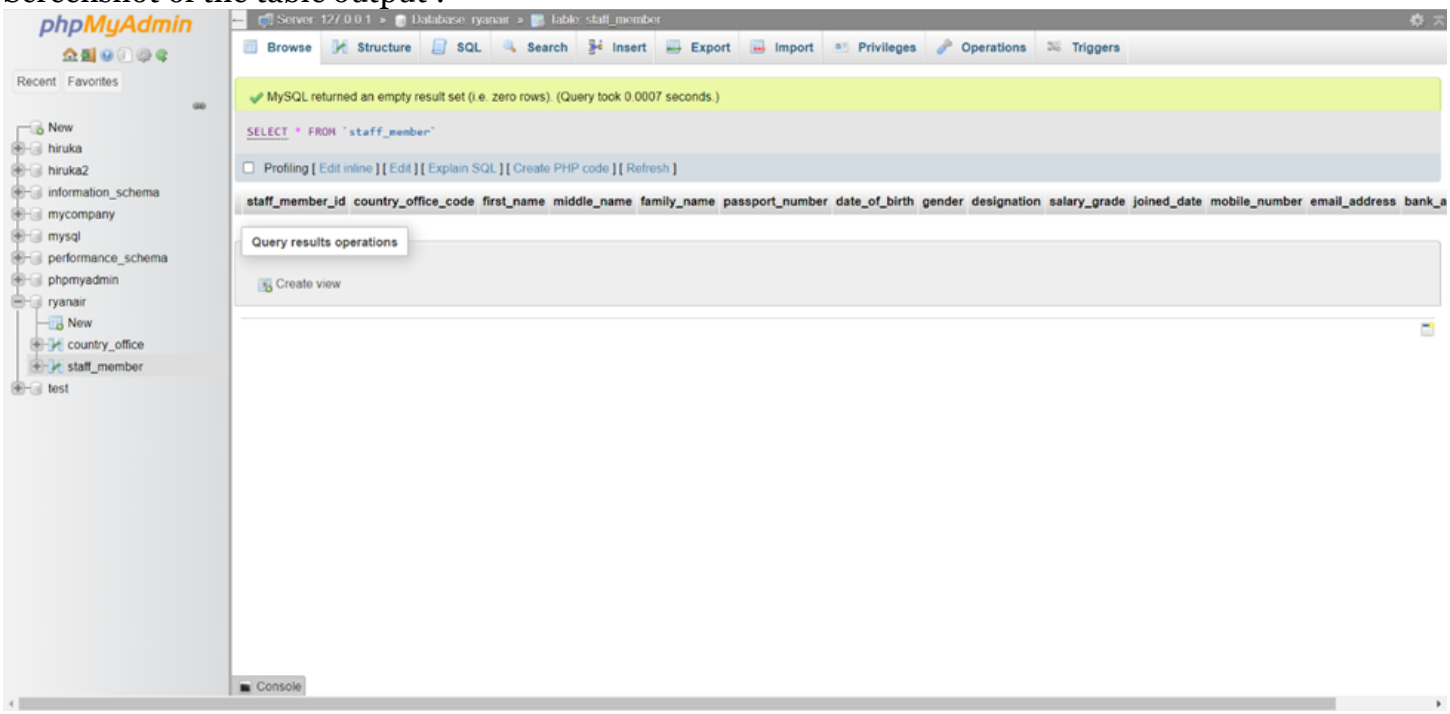


Figure 8 Staff Member Output Table

DDL statements required to create the Airport table

```
CREATE TABLE Airport (
  airport_code VARCHAR(10) PRIMARY KEY,
  airport_name VARCHAR(50),
  airport_city VARCHAR(50)
);
```

Screenshot of the DDL statements :

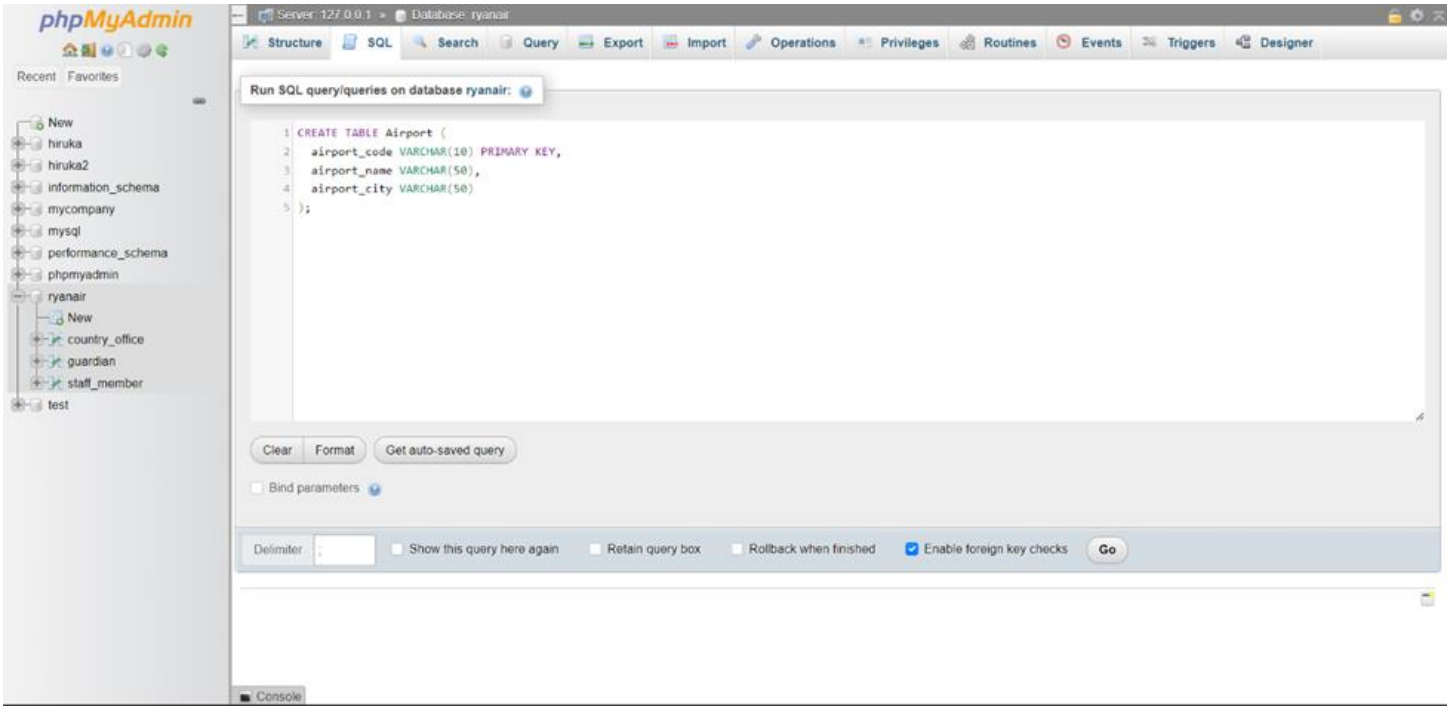


Figure 9 Airport Table query

Screenshot of the result :

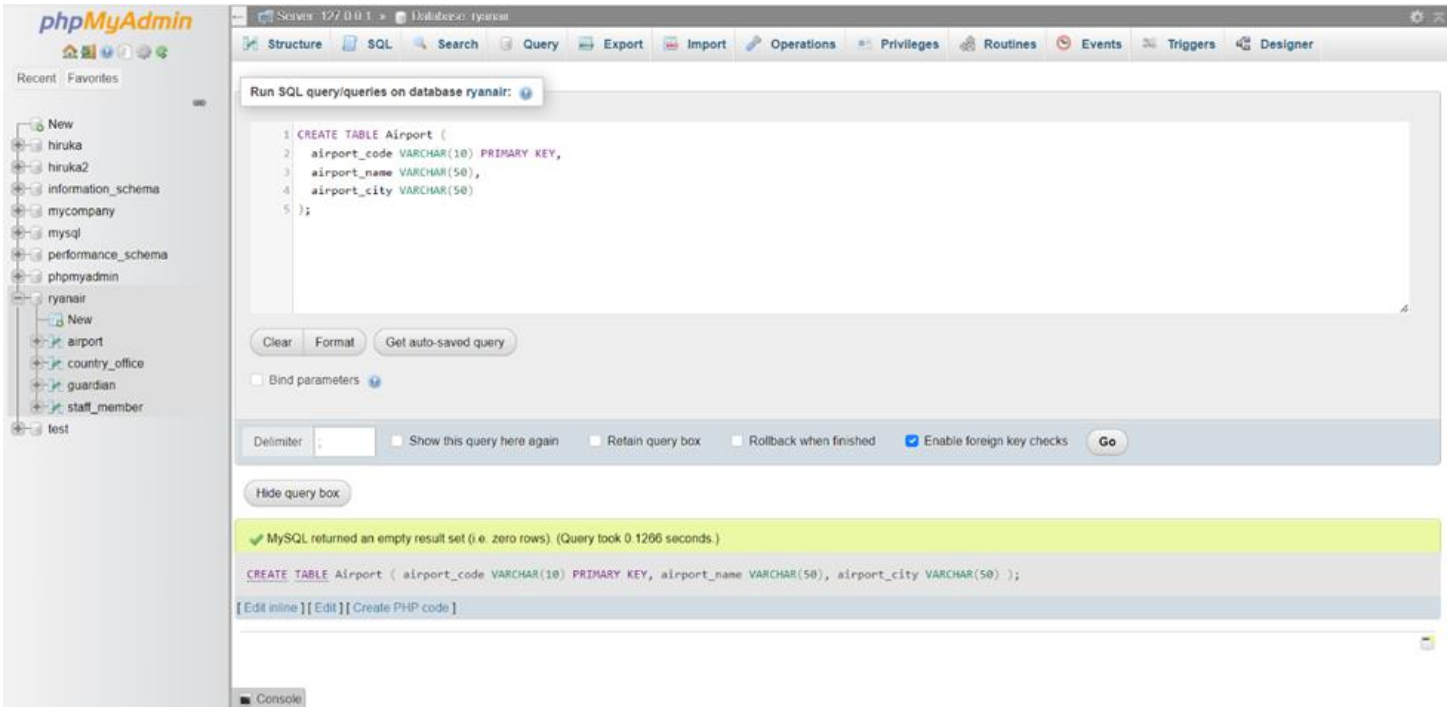


Figure 10 Result of Airport Table

Screenshot of the table output :

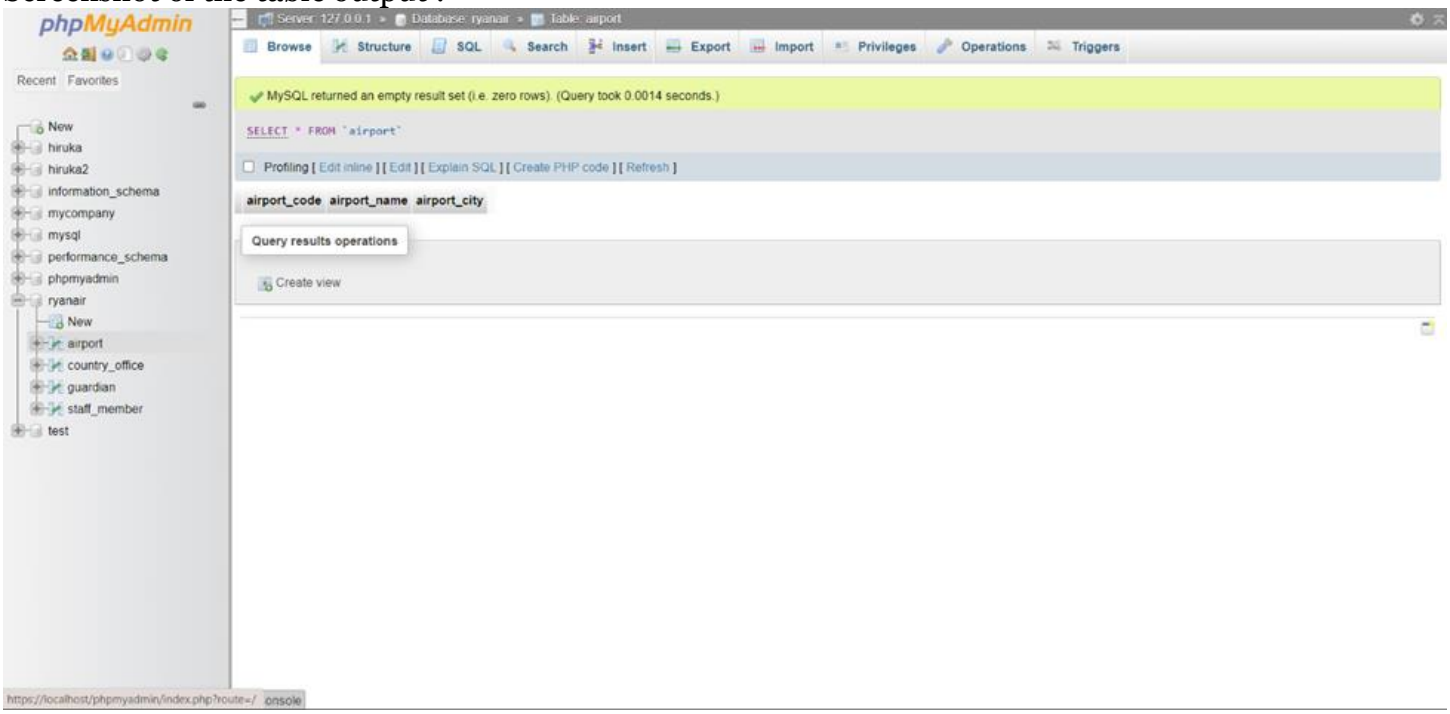


Figure 11 Airport Output Table

DDL statements required to create the Airplane table

```
CREATE TABLE Airplane (
  registration_number VARCHAR(20) PRIMARY KEY,
  passenger_capacity INT,
  model_number VARCHAR(50),
```

```
  manufacturer VARCHAR(50),
  refuel_range INT,
  date_of_acquisition DATE
);
```

Screenshot of the DDL statements :

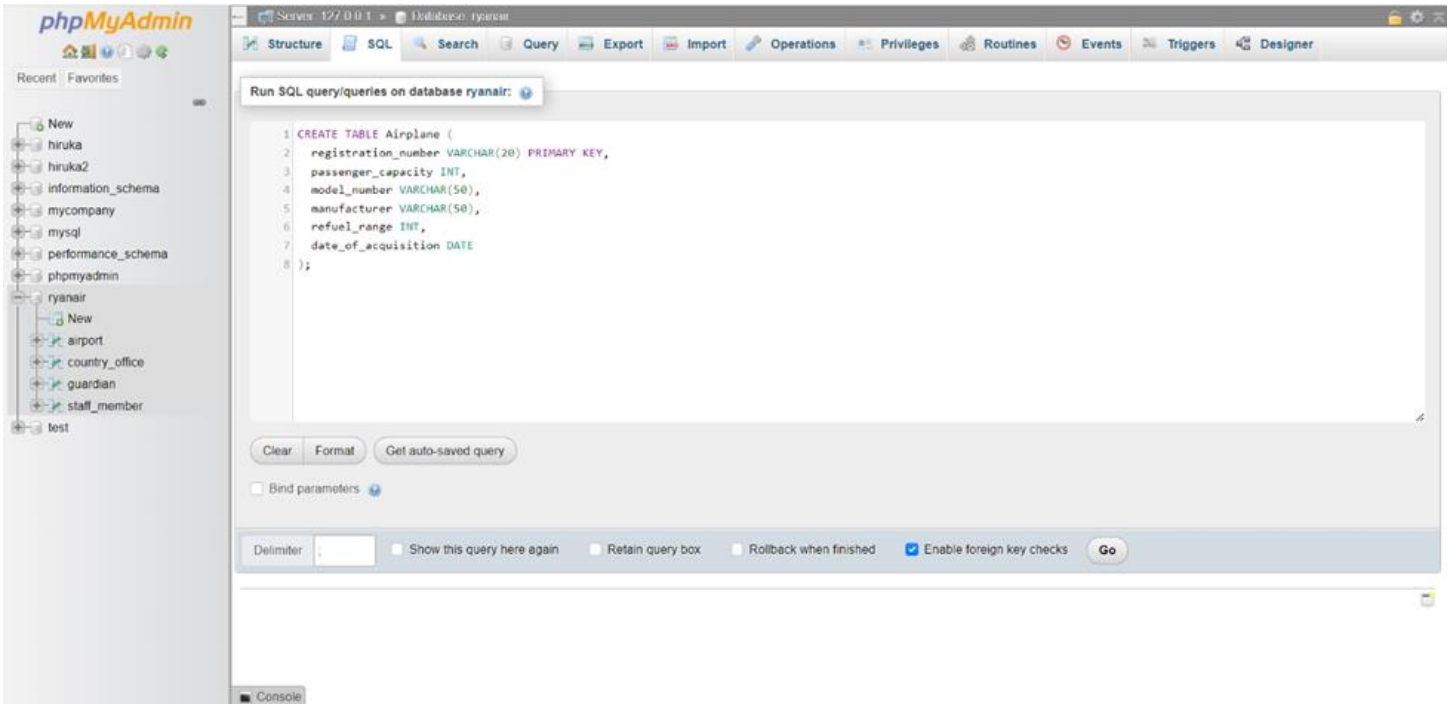


Figure 12 Airplane Table query

Screenshot of the result:

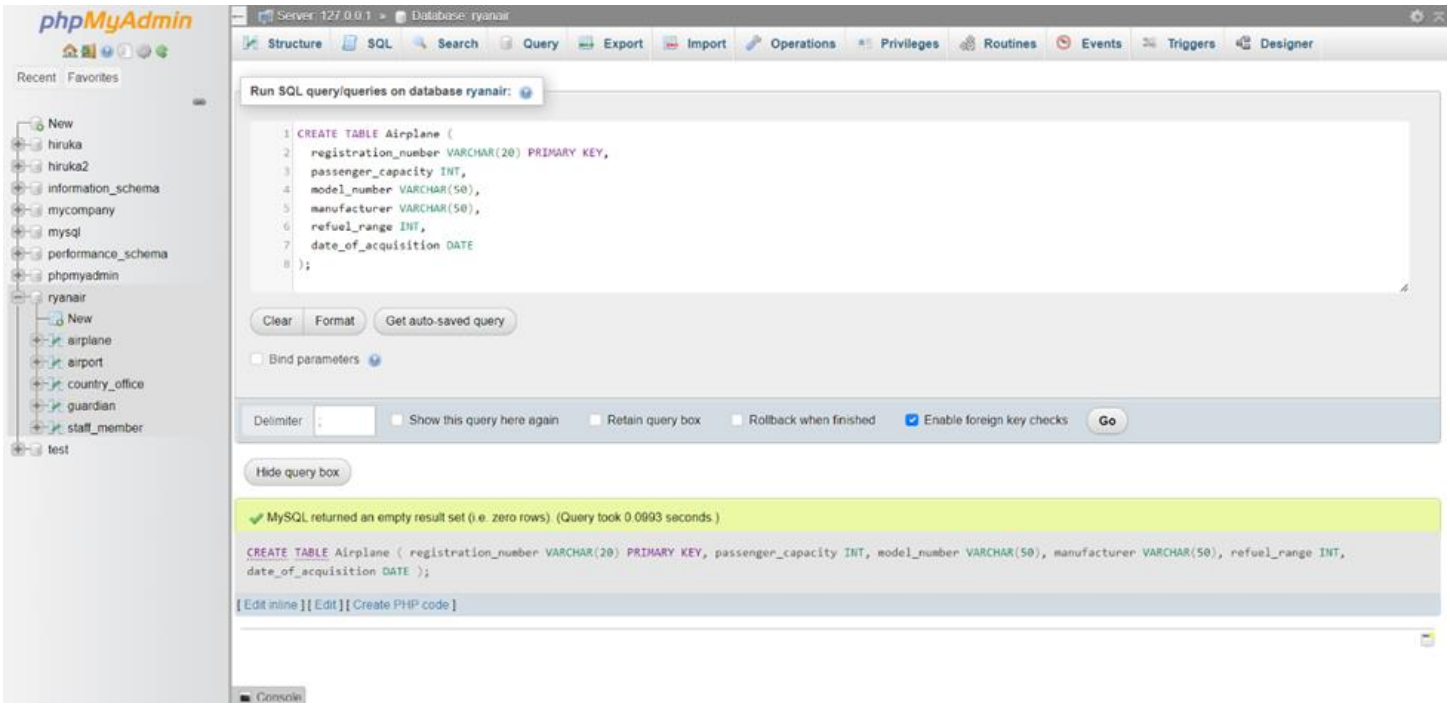


Figure 13 Result of Airplane Table

Screenshot of the table output :

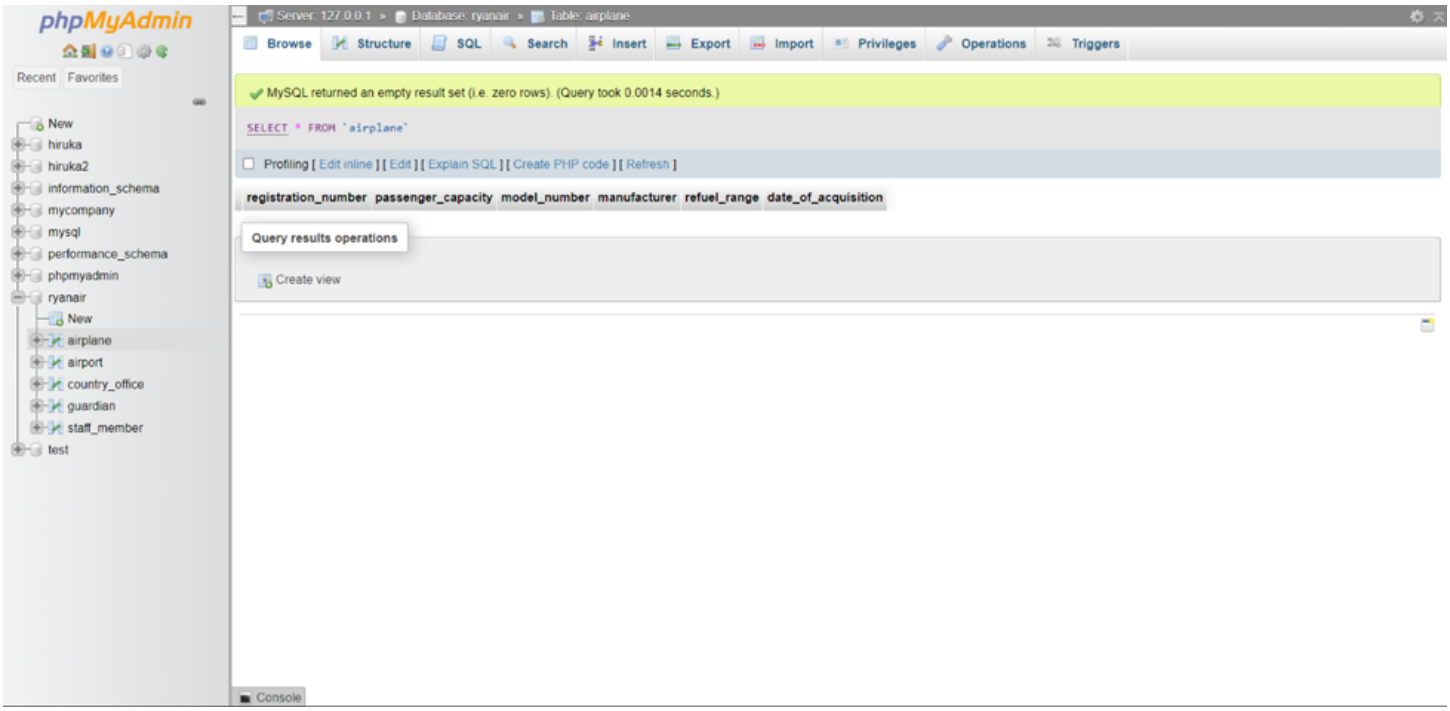


Figure 14 Airplane Output Table

DDL statements required to create the Flight Route table

```
CREATE TABLE Flight_Route (
  route_id INT PRIMARY KEY,
  departure_airport VARCHAR(10),
  arrival_airport VARCHAR(10),
  total_distance INT,
  FOREIGN KEY (departure_airport) REFERENCES Airport(airport_code)
);
```

Screenshot of the DDL statements :

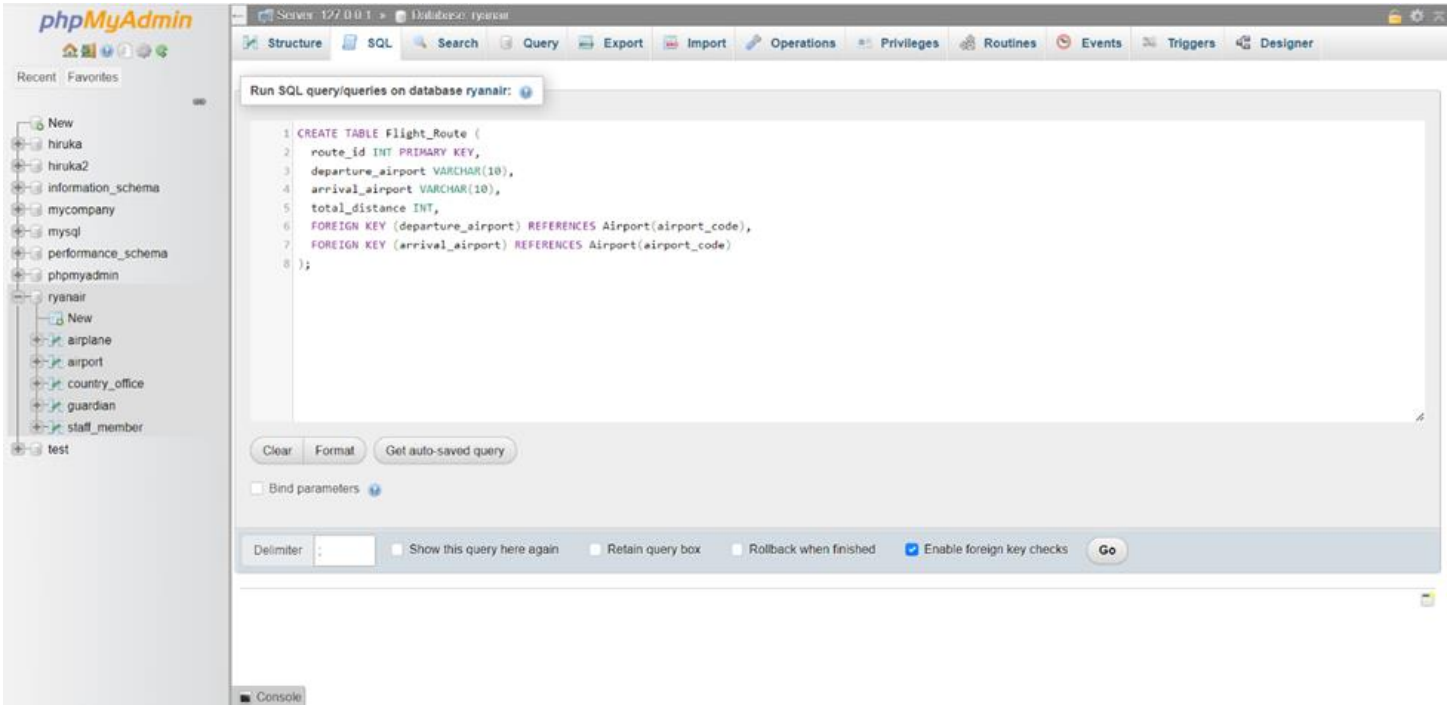


Figure 15 Flight Route Table query

Screenshot of the result:

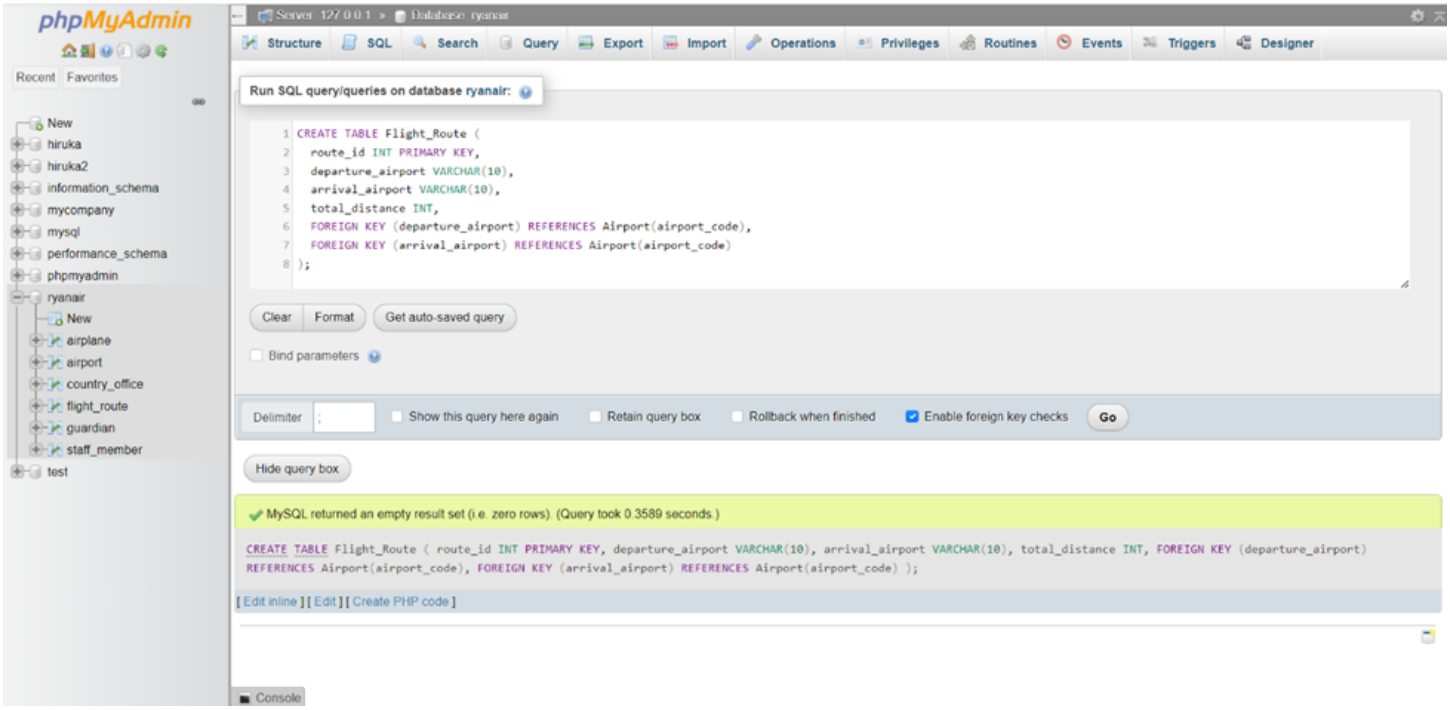


Figure 16 Result of Flight Route Table

Screenshot of the table output :

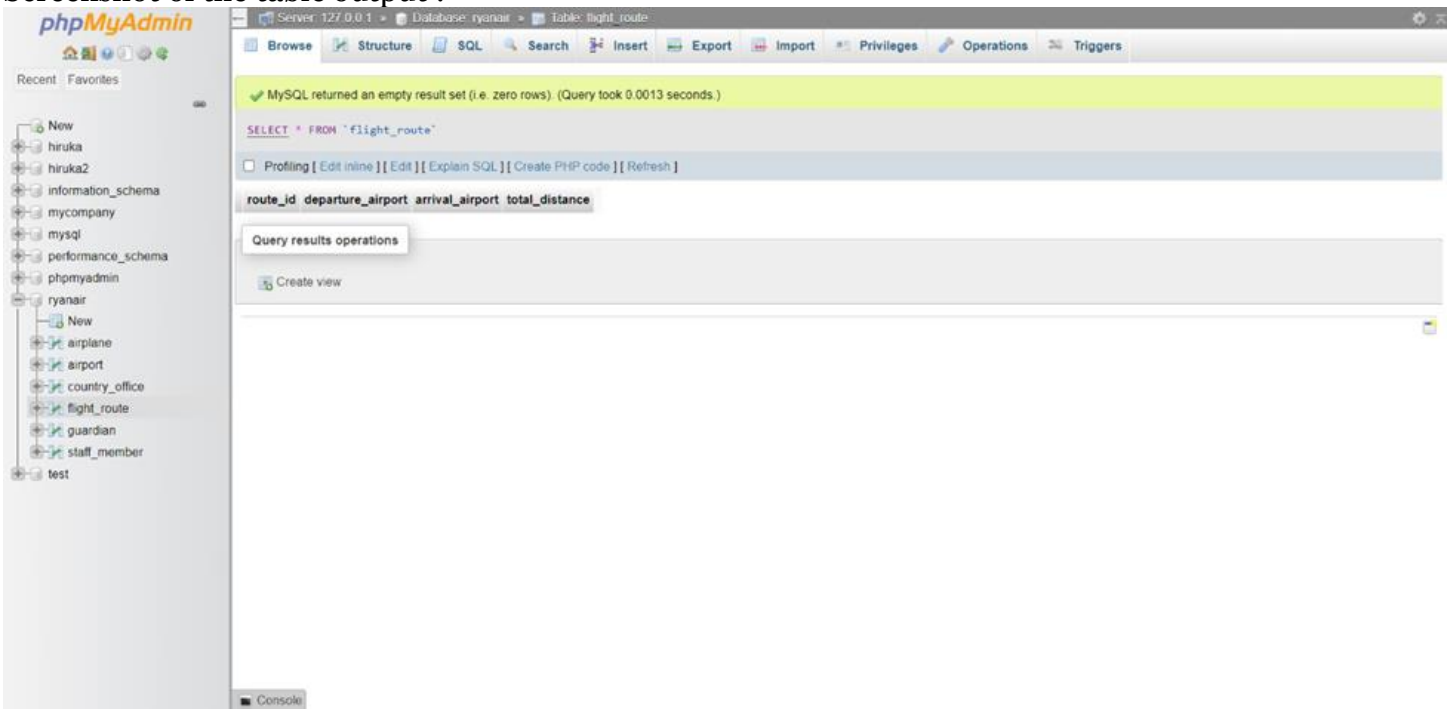


Figure 17 Flight Route Output Table

DDL statements required to create the Passenger table

```
CREATE TABLE Passenger (
  passenger_id INT PRIMARY KEY,
  first_name VARCHAR(50),
  middle_name VARCHAR(50),
  family_name VARCHAR(50),
  address VARCHAR(100),
  passport_number VARCHAR(20),
```

```
  date_of_birth DATE,
  mobile_number VARCHAR(20),
  email_address VARCHAR(100)
);
```

Screenshot of the DDL statements :

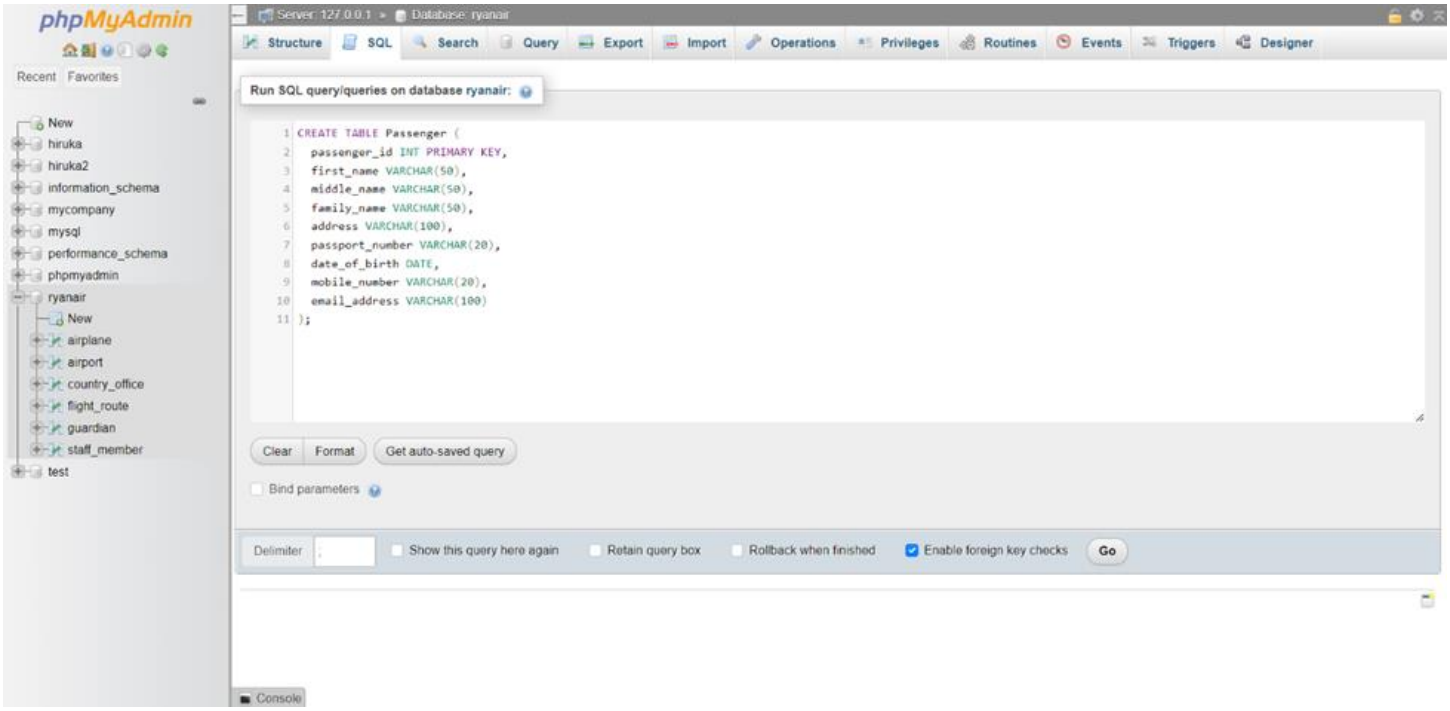


Figure 18 Passenger Table query

Screenshot of the result :

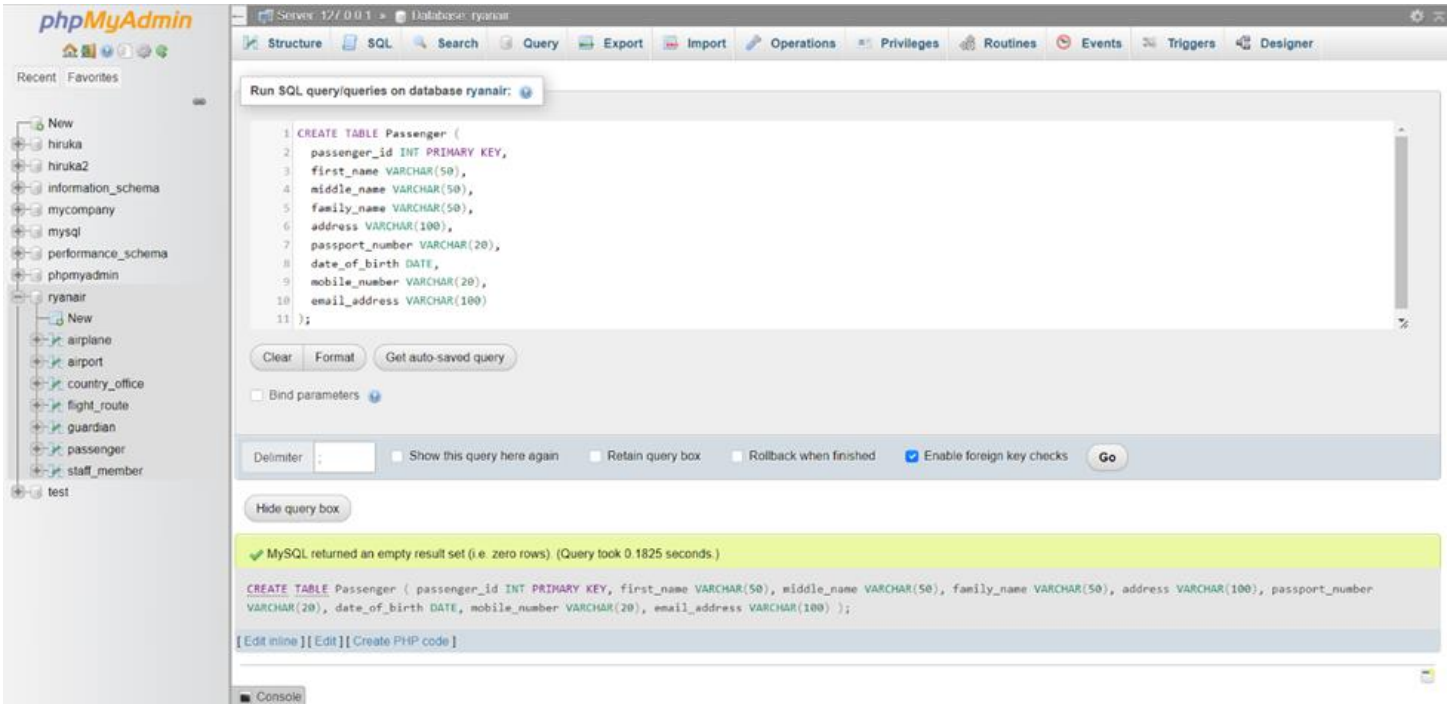


Figure 19 Result of Passenger Table

Screenshot of the table output :

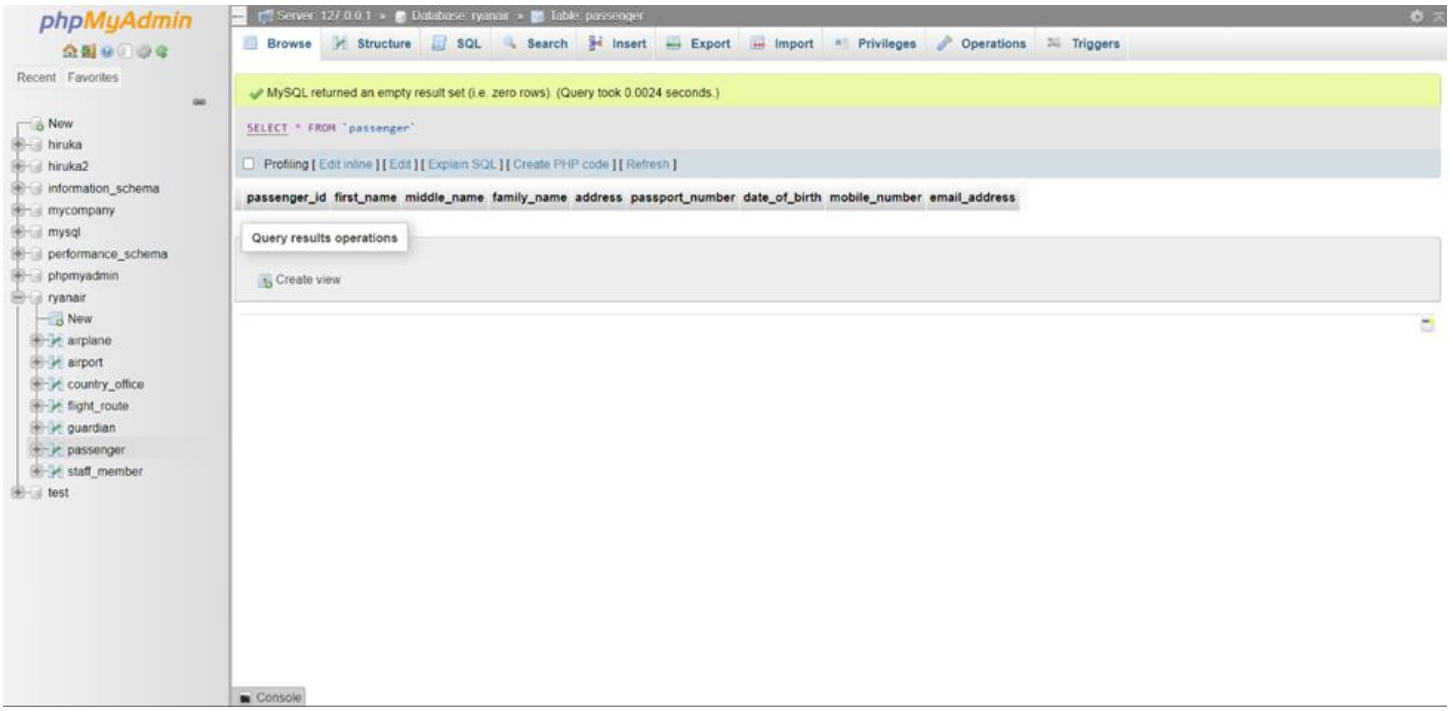


Figure 20 Passenger Output Table

DDL statements required to create the Flight table

```
CREATE TABLE Flight (
  flight_number VARCHAR(20) PRIMARY KEY,
  departure_time DATETIME,
  route_id INT,
  available_seats INT,
  arrival_time DATETIME,
  flight_category VARCHAR(50)
);
```

Screenshot of the DDL statements :

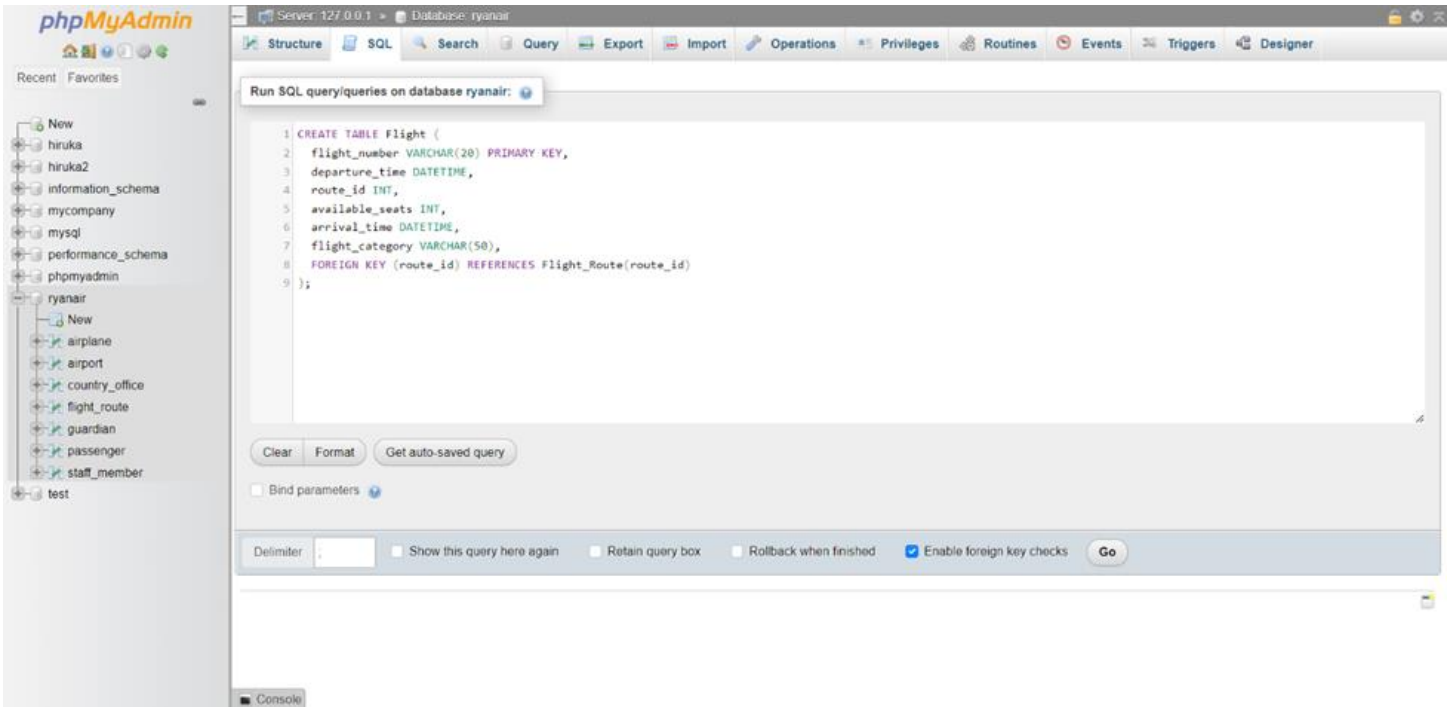


Figure 21 Flight Table query

Screenshot of the result :

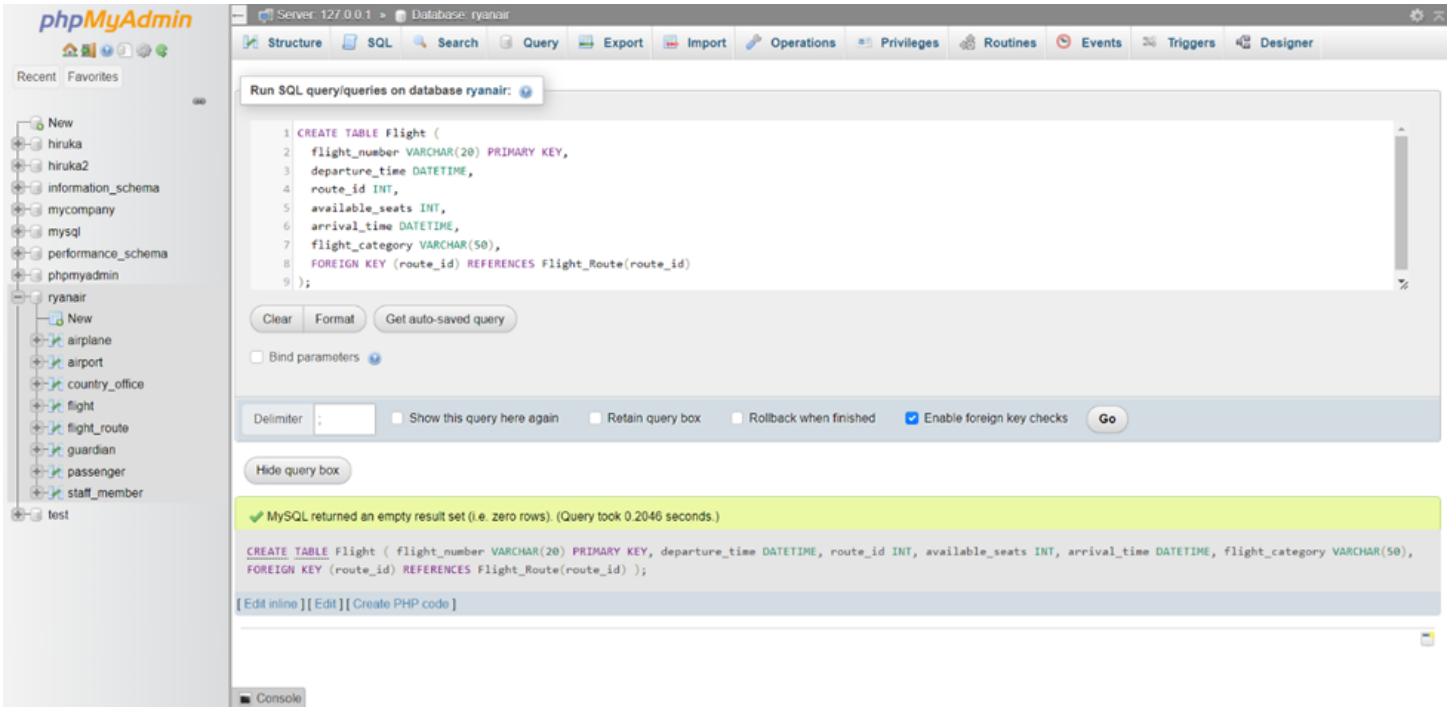


Figure 22 Result of Flight Table

Screenshot of the table output :

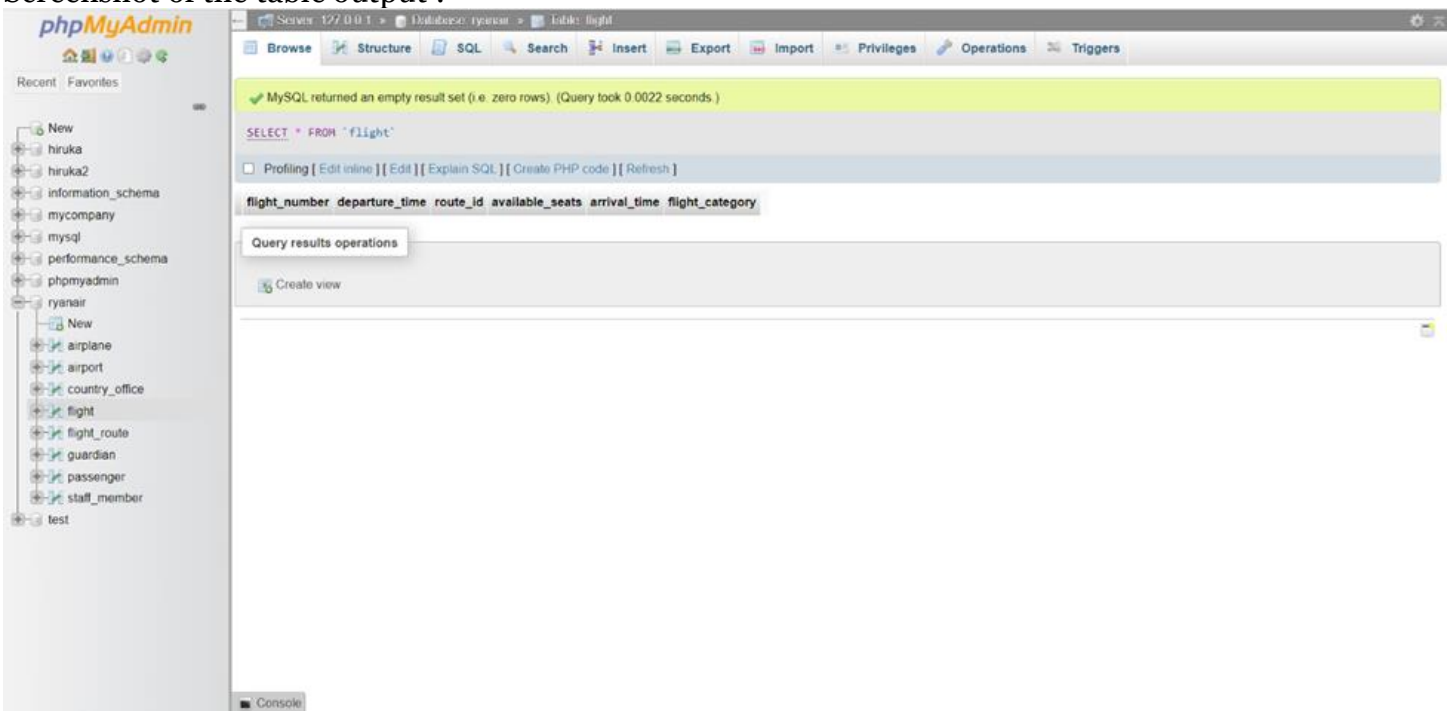


Figure 23 Flight Output Table

DDL statements required to create the Booking table

```
CREATE TABLE Flight_Booking (
  flight_booking_id INT PRIMARY KEY,
  flight_number VARCHAR(20),
  passenger_id INT,
  flight_fare DECIMAL(10, 2),
  booking_date DATETIME,
  FOREIGN KEY (flight_number) REFERENCES Flight(flight_number),
  FOREIGN KEY (passenger_id) REFERENCES Passenger(passenger_id)
```

);

Screenshot of the DDL statements :

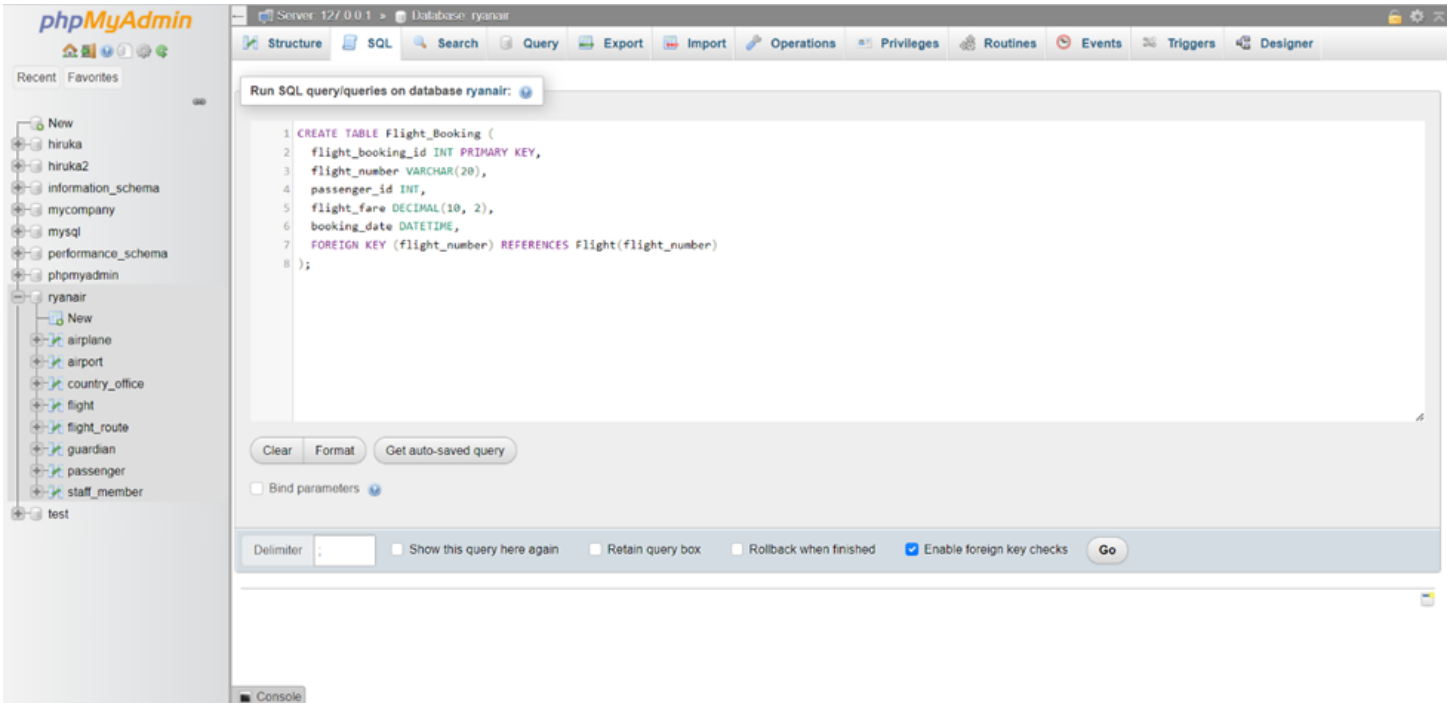


Figure 24 Flight Booking Table query

Screenshot of the result :

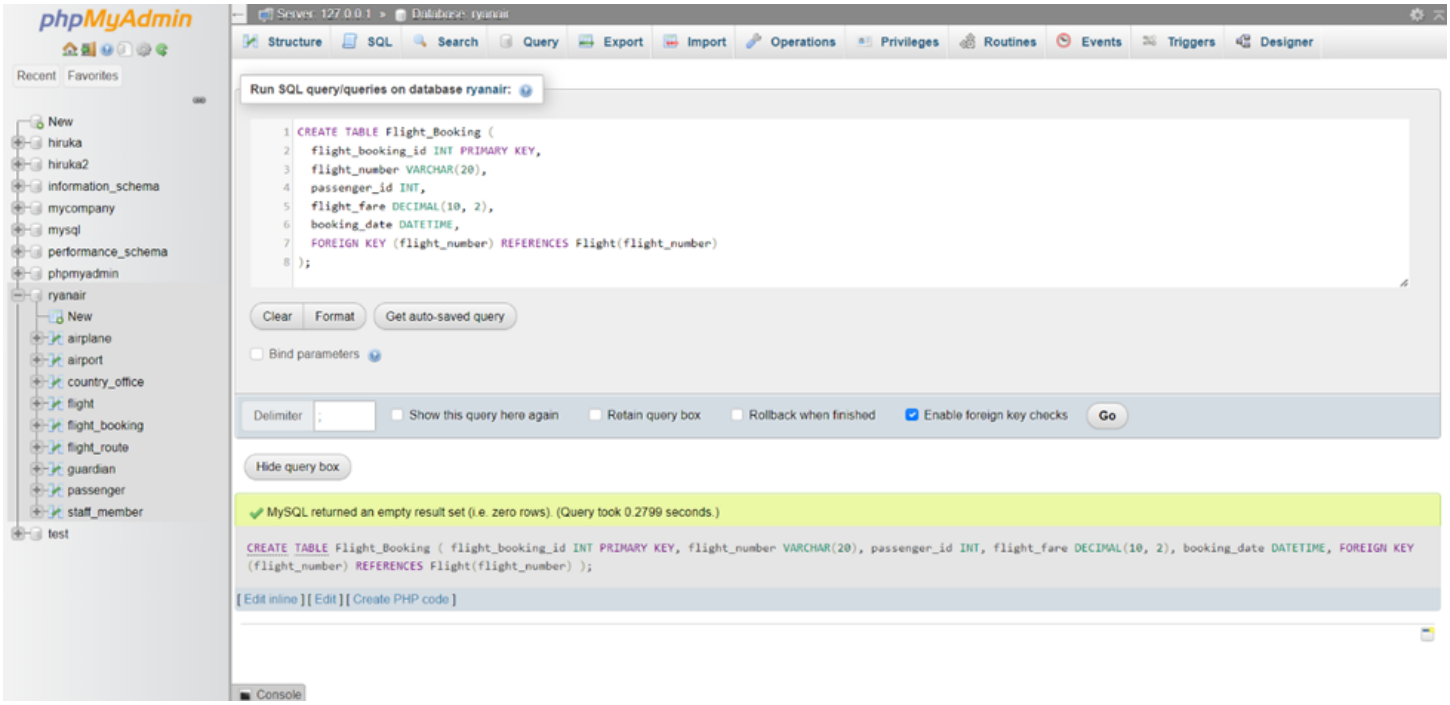


Figure 25 Result of Flight Booking Table

Screenshot of the table output :

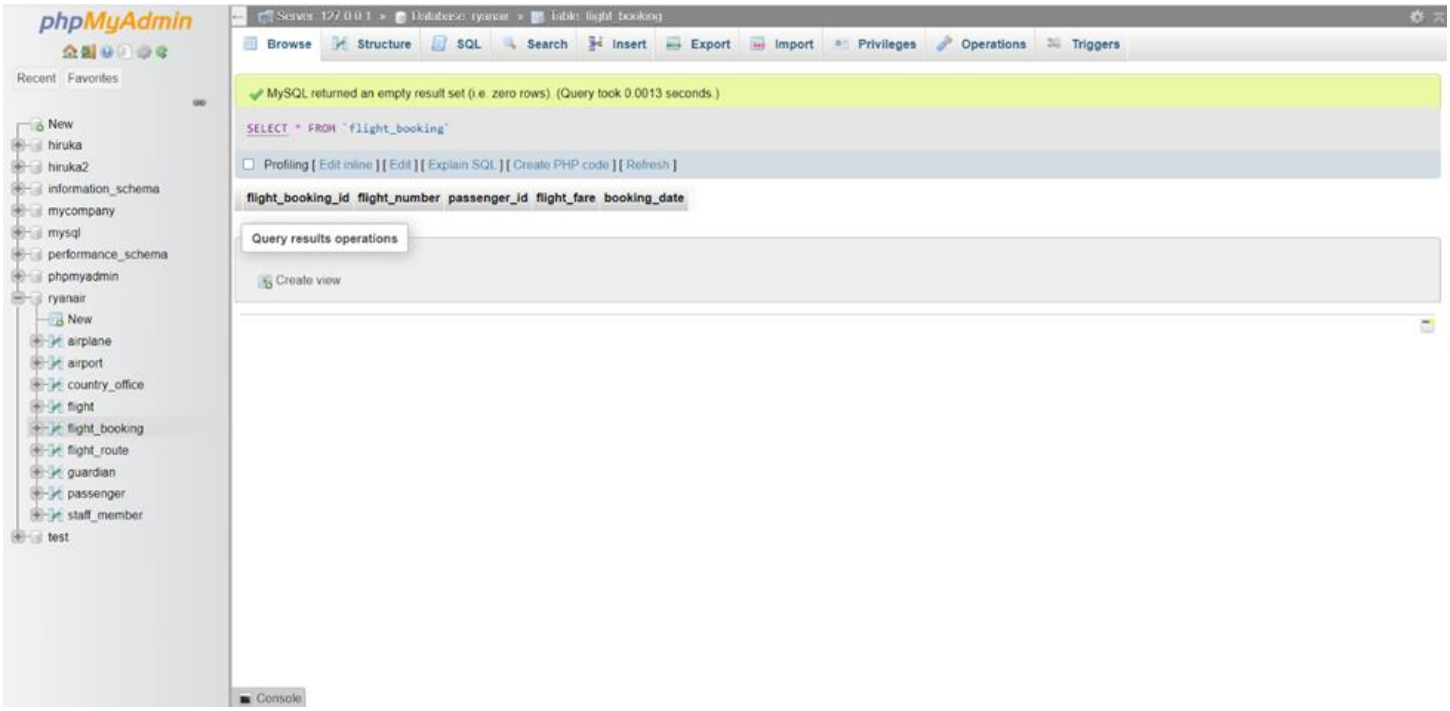


Figure 26 Flight Booking Output Table

DDL statements required to create the Guardian table

```
CREATE TABLE Guardian (
  guardian_id INT PRIMARY KEY,
  name VARCHAR(50),
  passport_number VARCHAR(20),
  address VARCHAR(100),
  contact_number VARCHAR(20),
  bank_account_number VARCHAR(20)
);
```

Screenshot of the DDL statements :

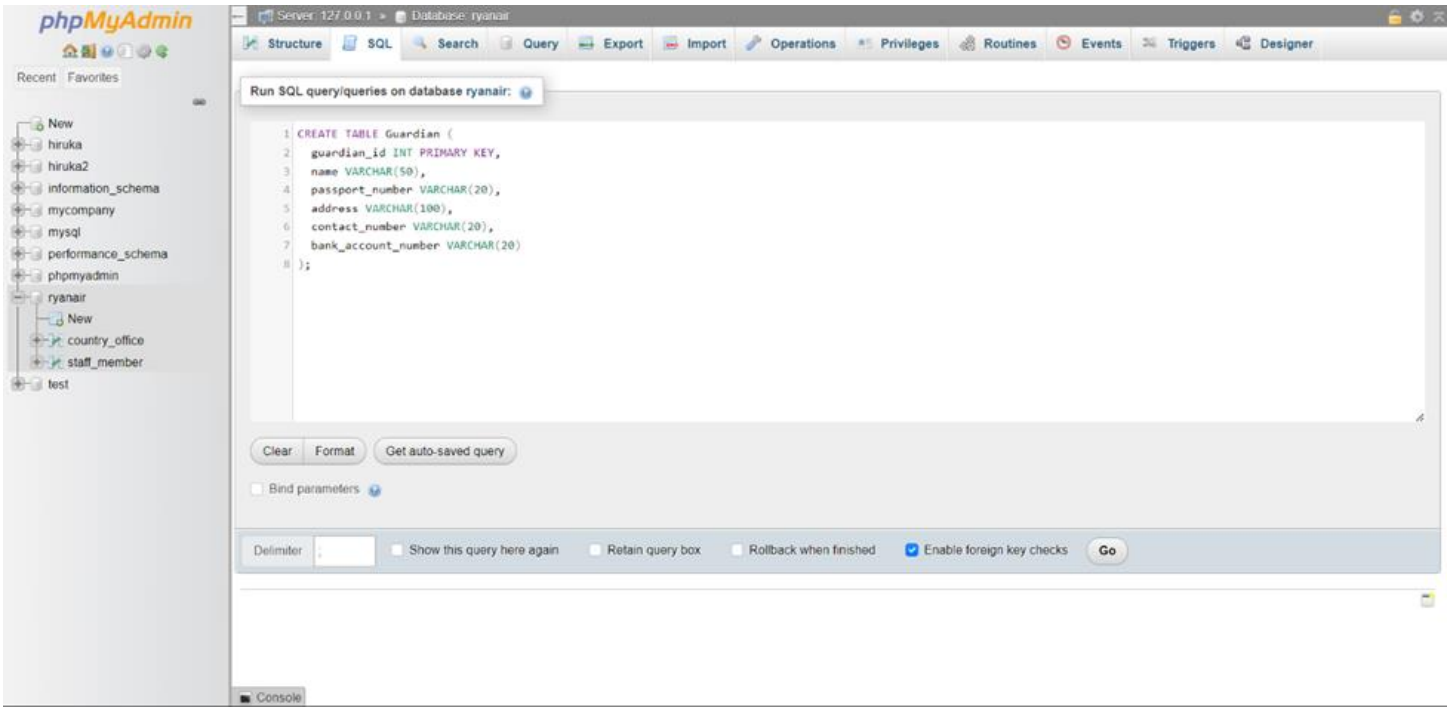


Figure 27 Guardian Table query

Screenshot of the result :

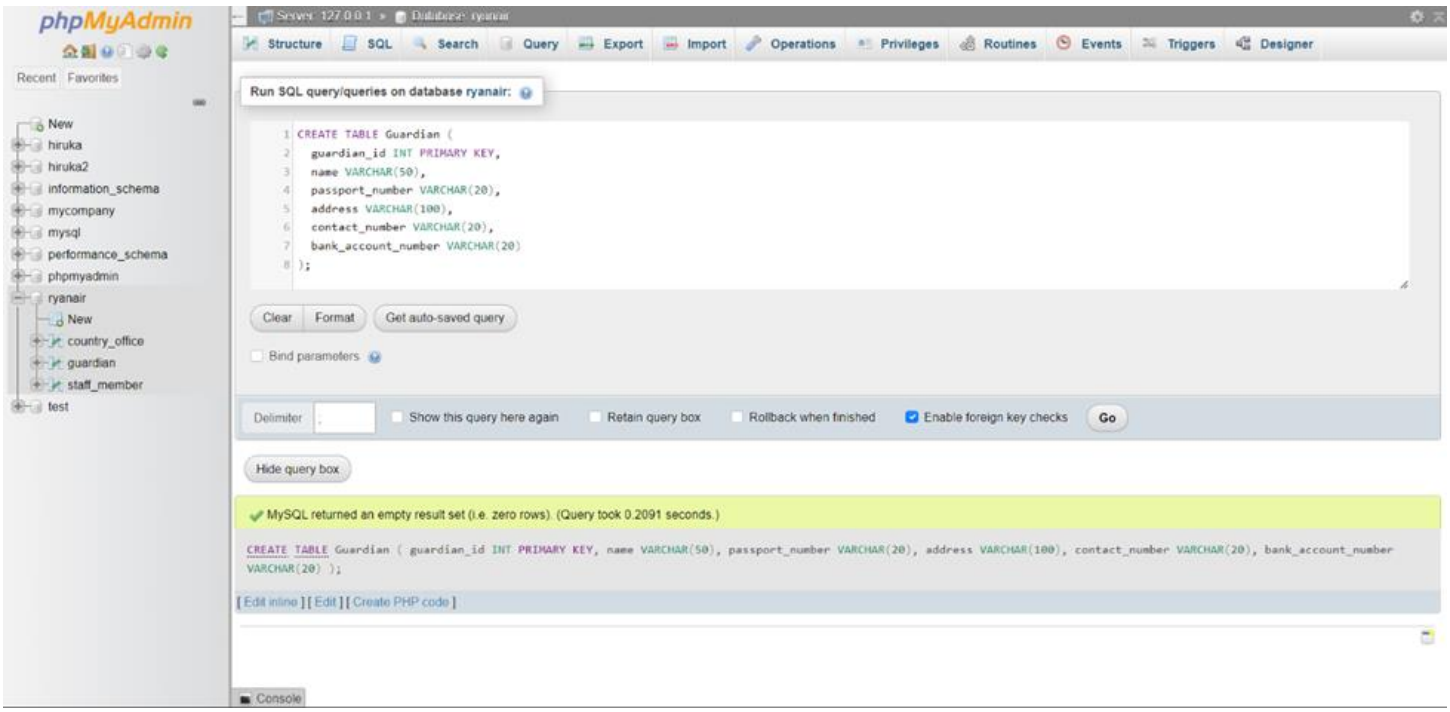


Figure 28 Result of Guardian Table

Screenshot of the table output :

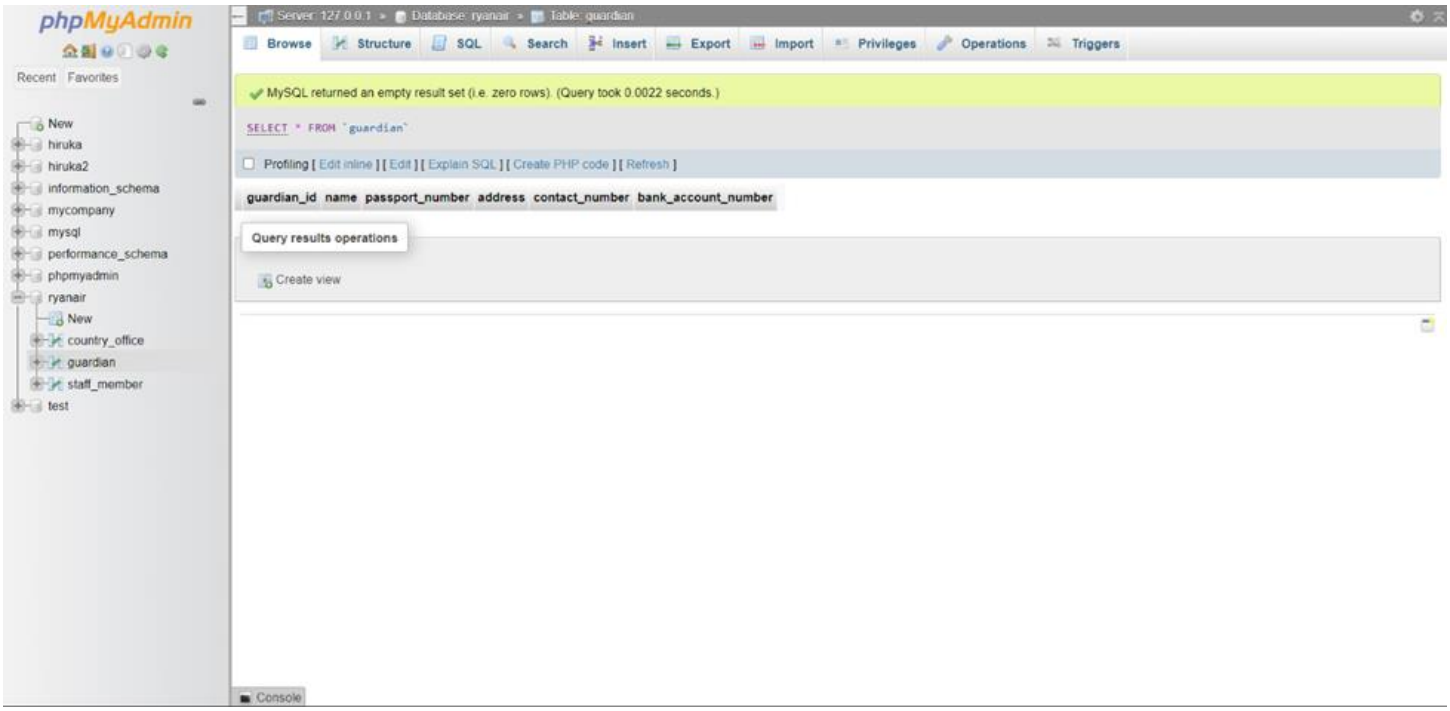


Figure 29 Guardian Output Table

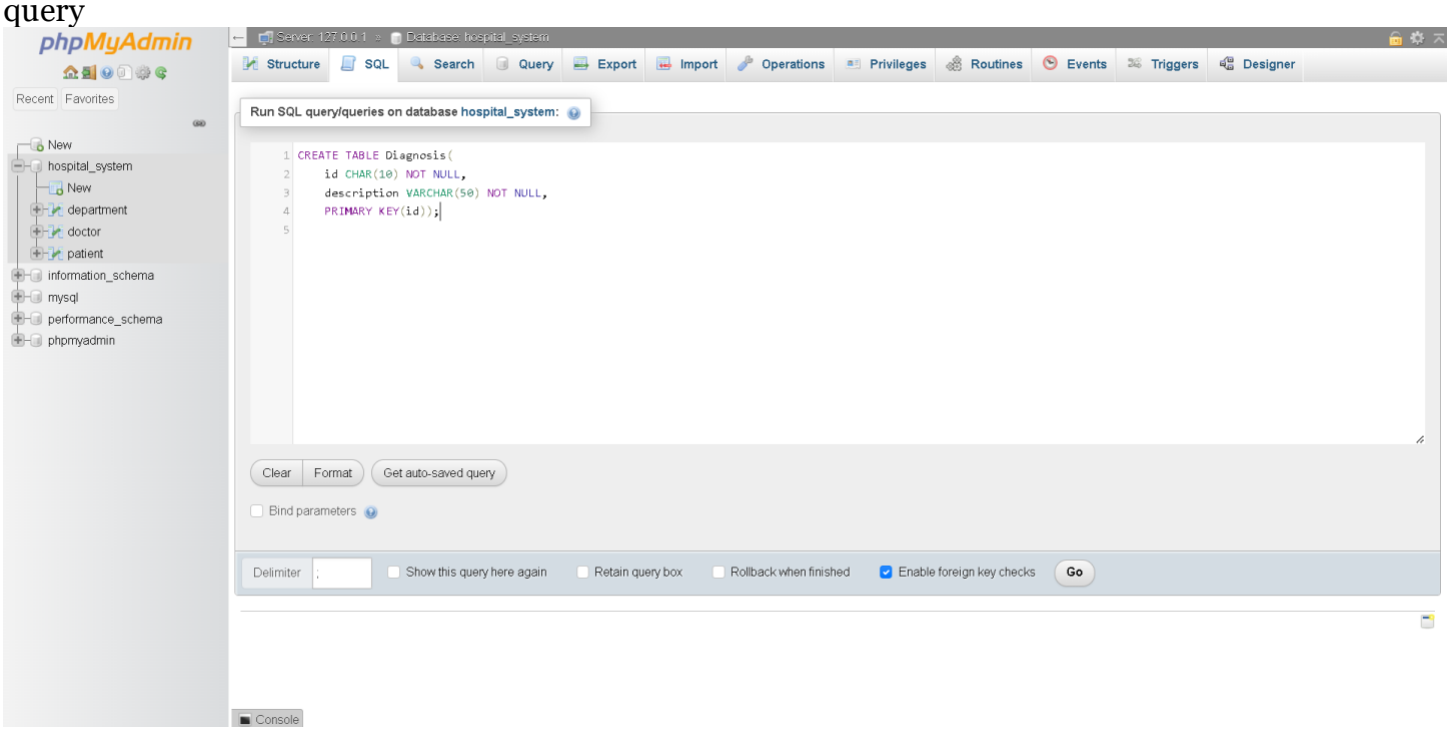
Part B

Question 1.

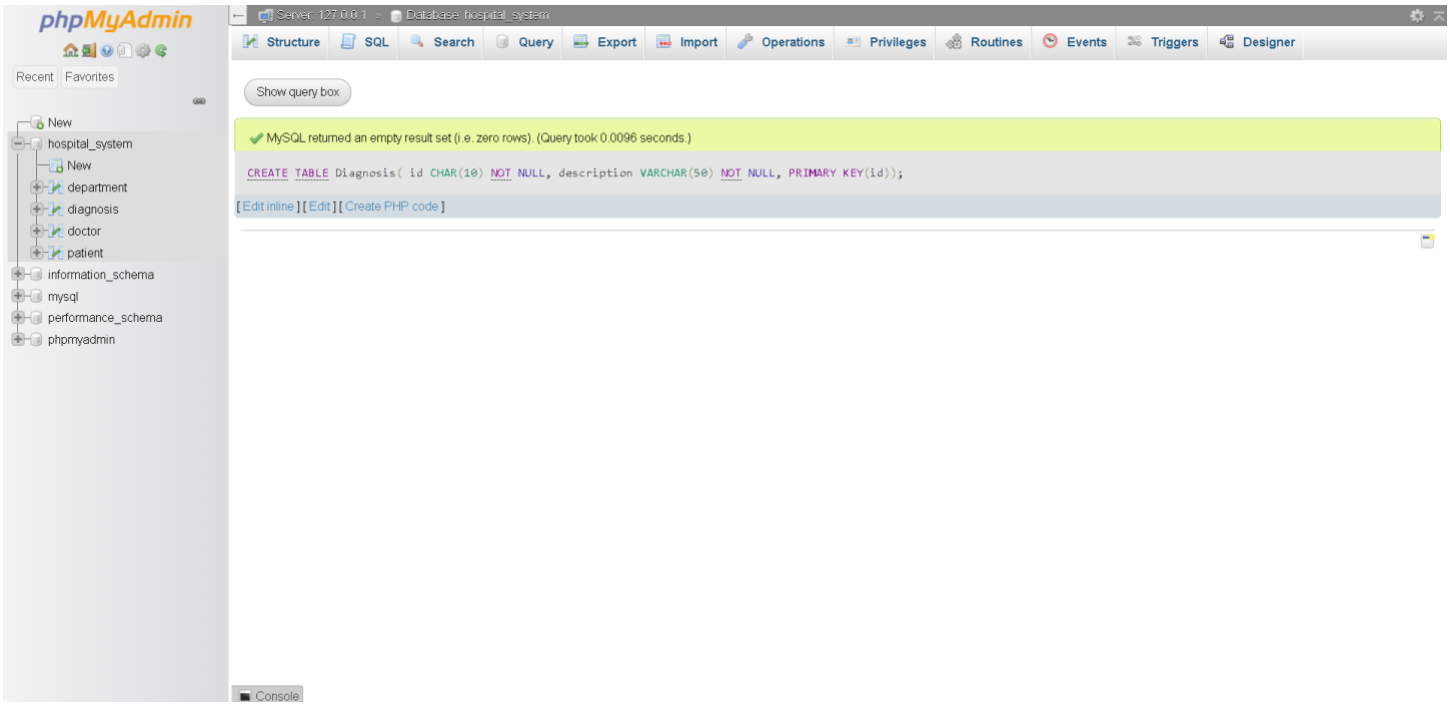
Give the DDL (Data Definition Language) statements required to create the tables. Include primary and foreign key definitions.

1.. Diagnosis table

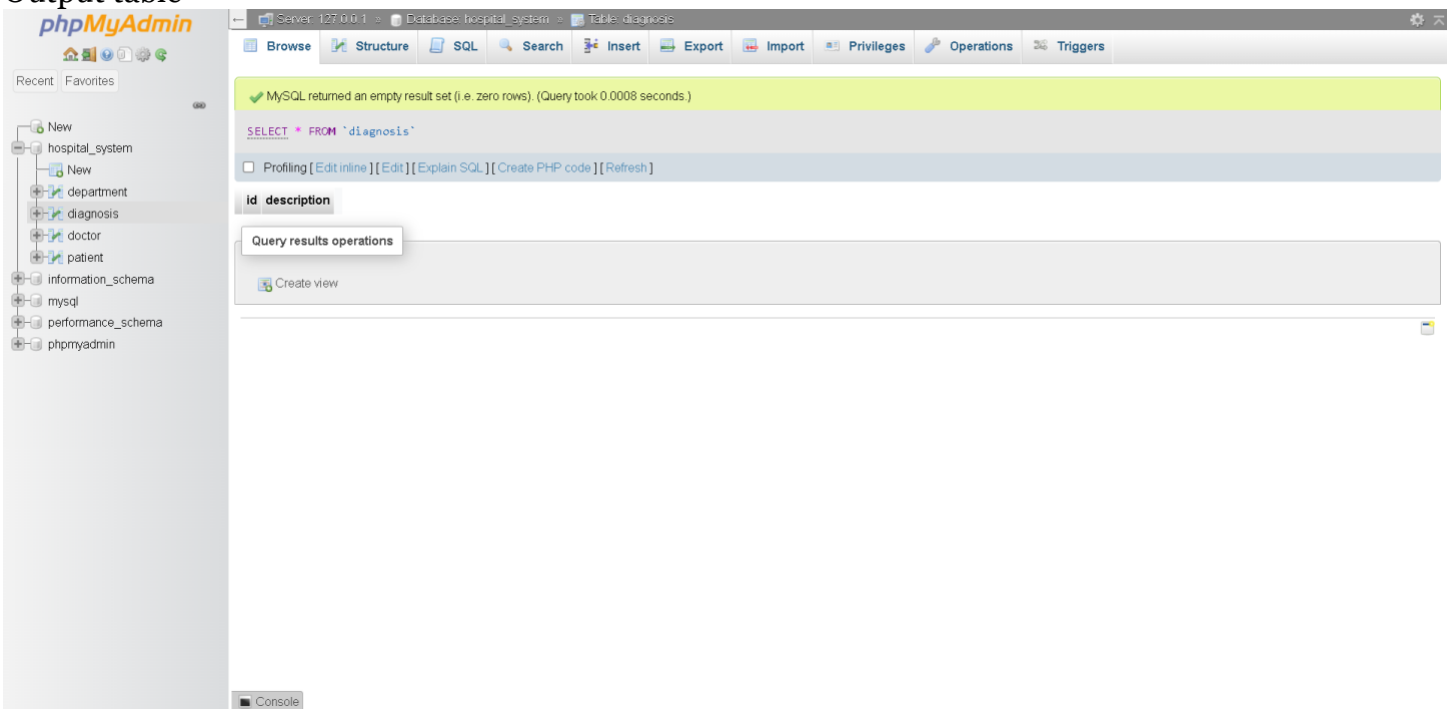
query



Result

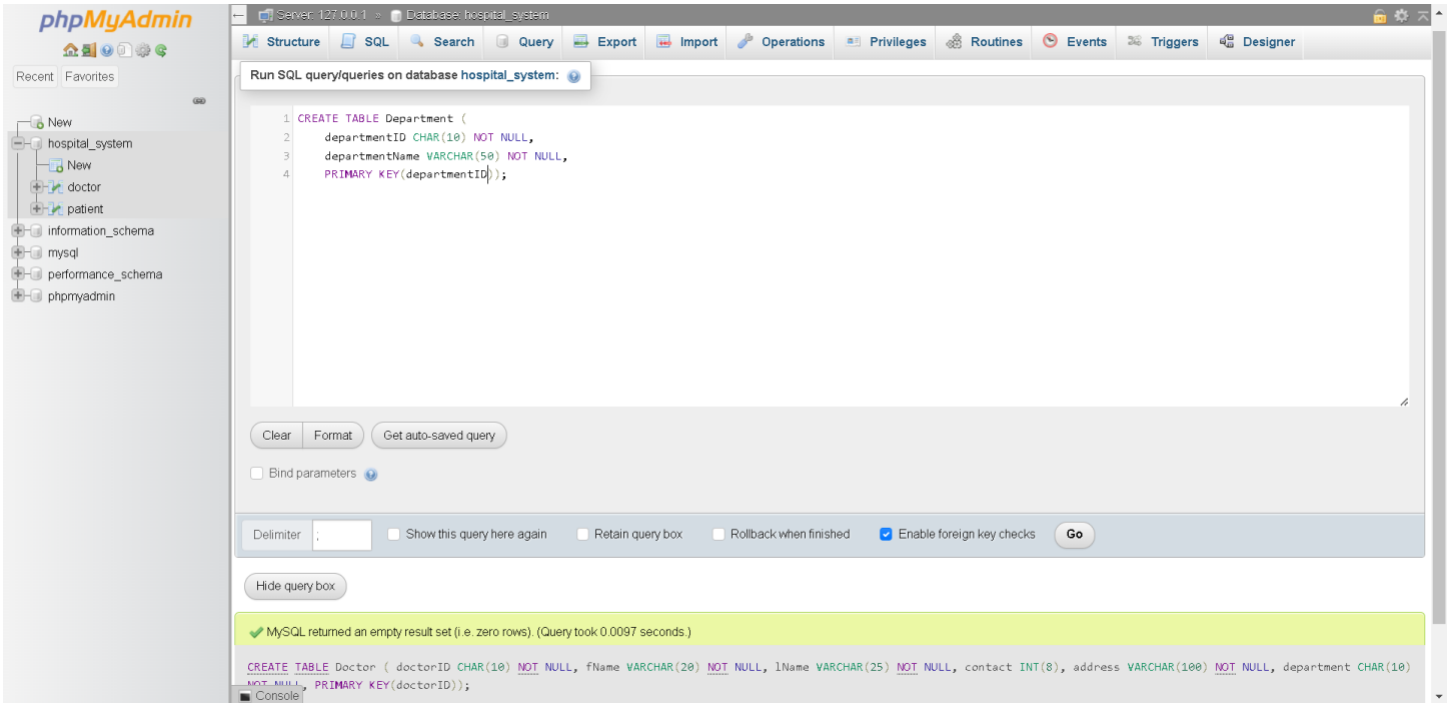


Output table

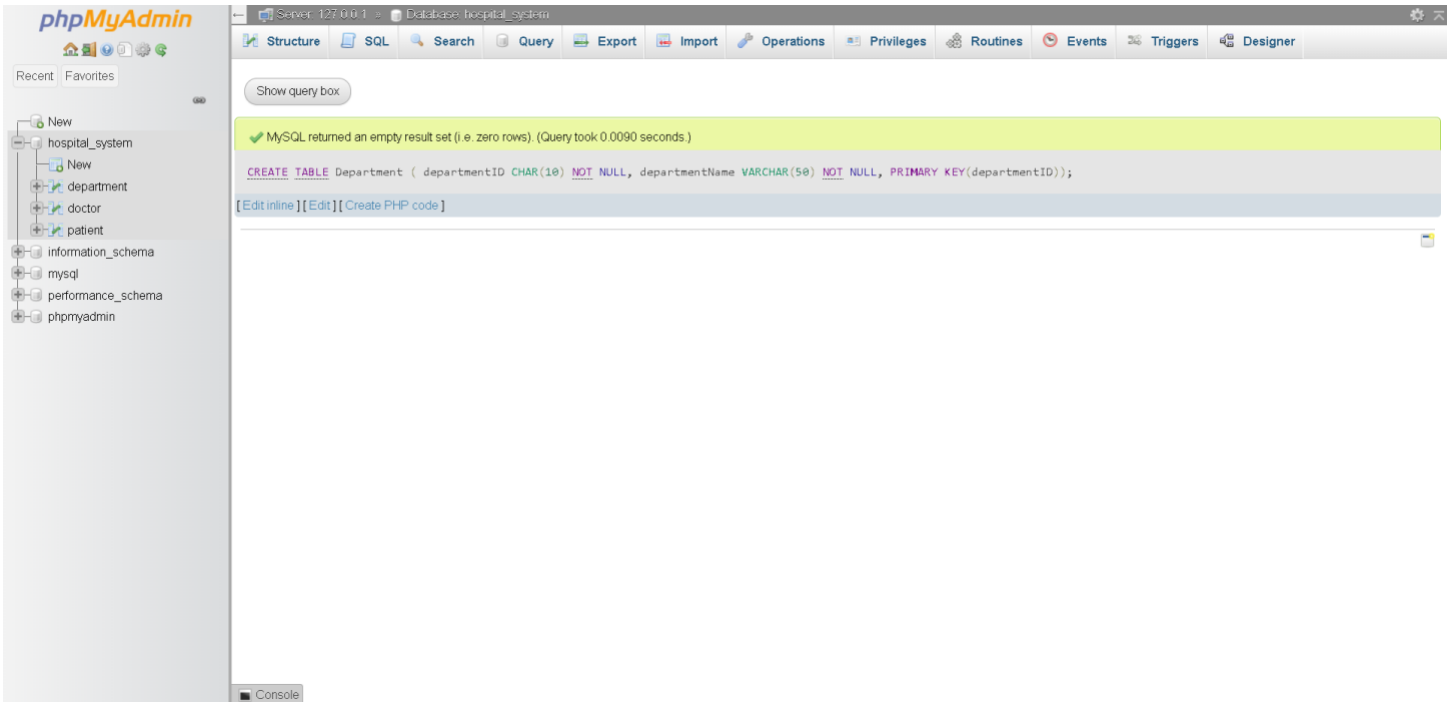


2.Department table

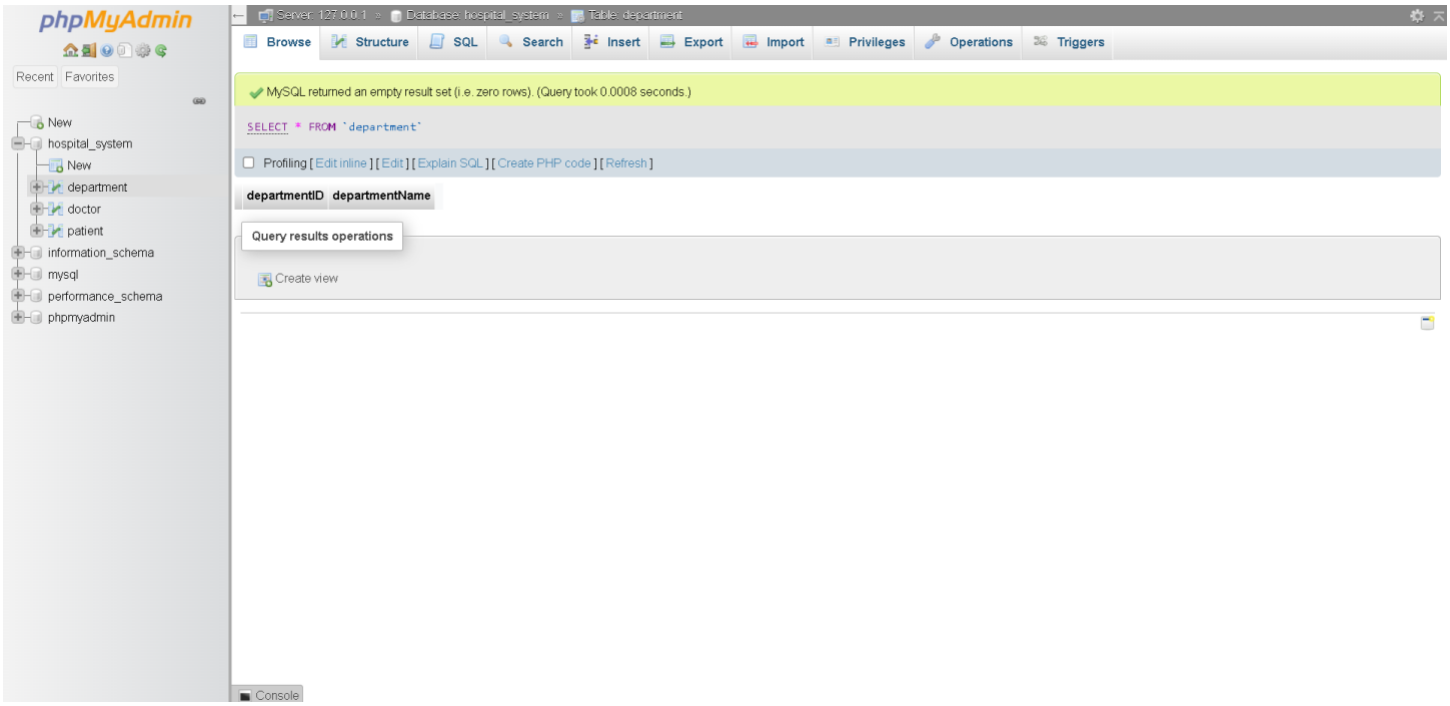
Query



Result

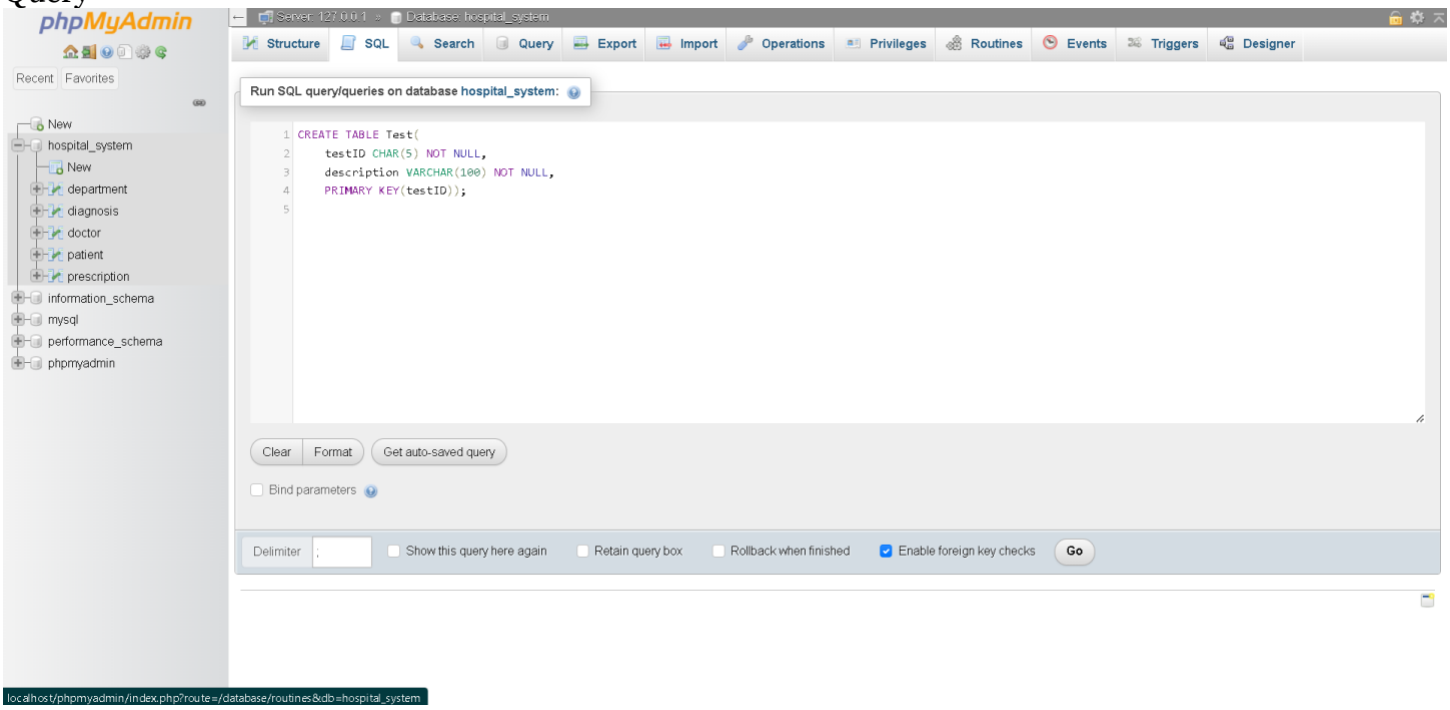


Output table

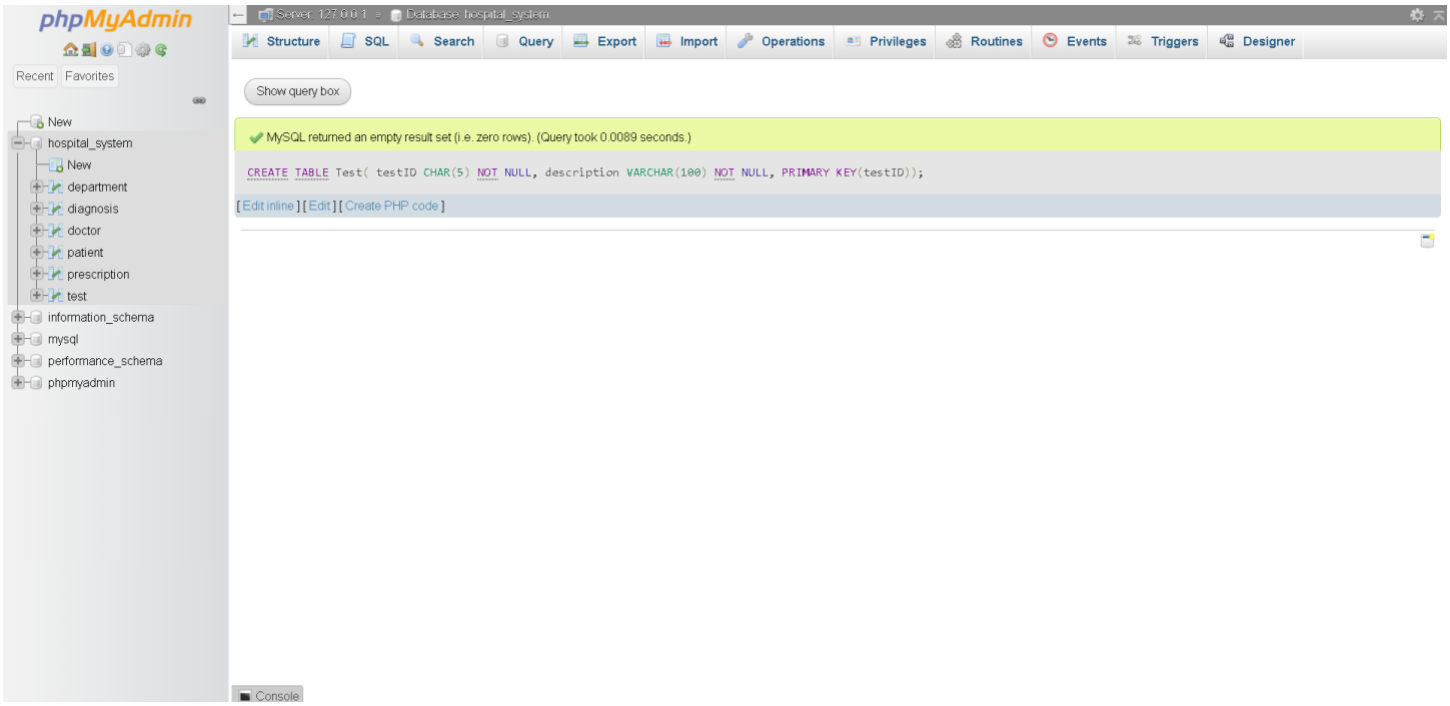


3.Test table

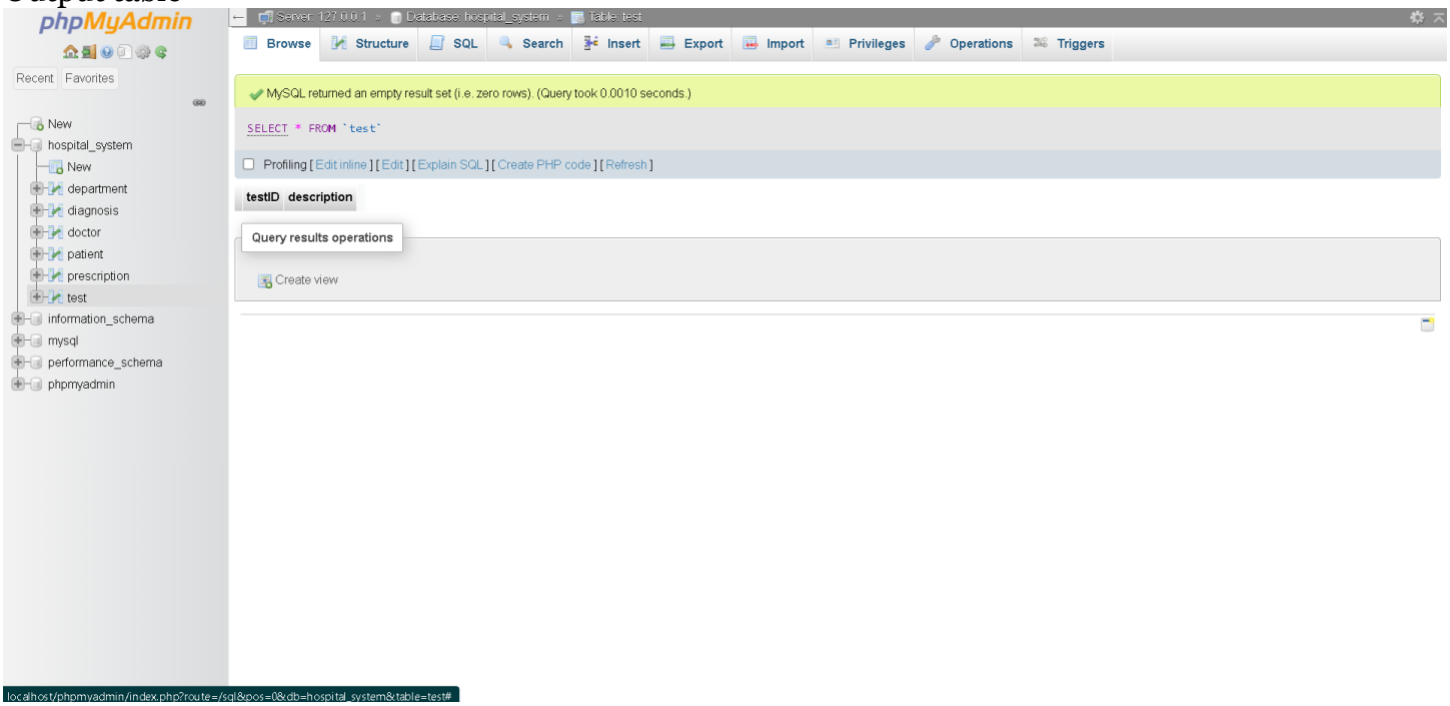
Query



Result

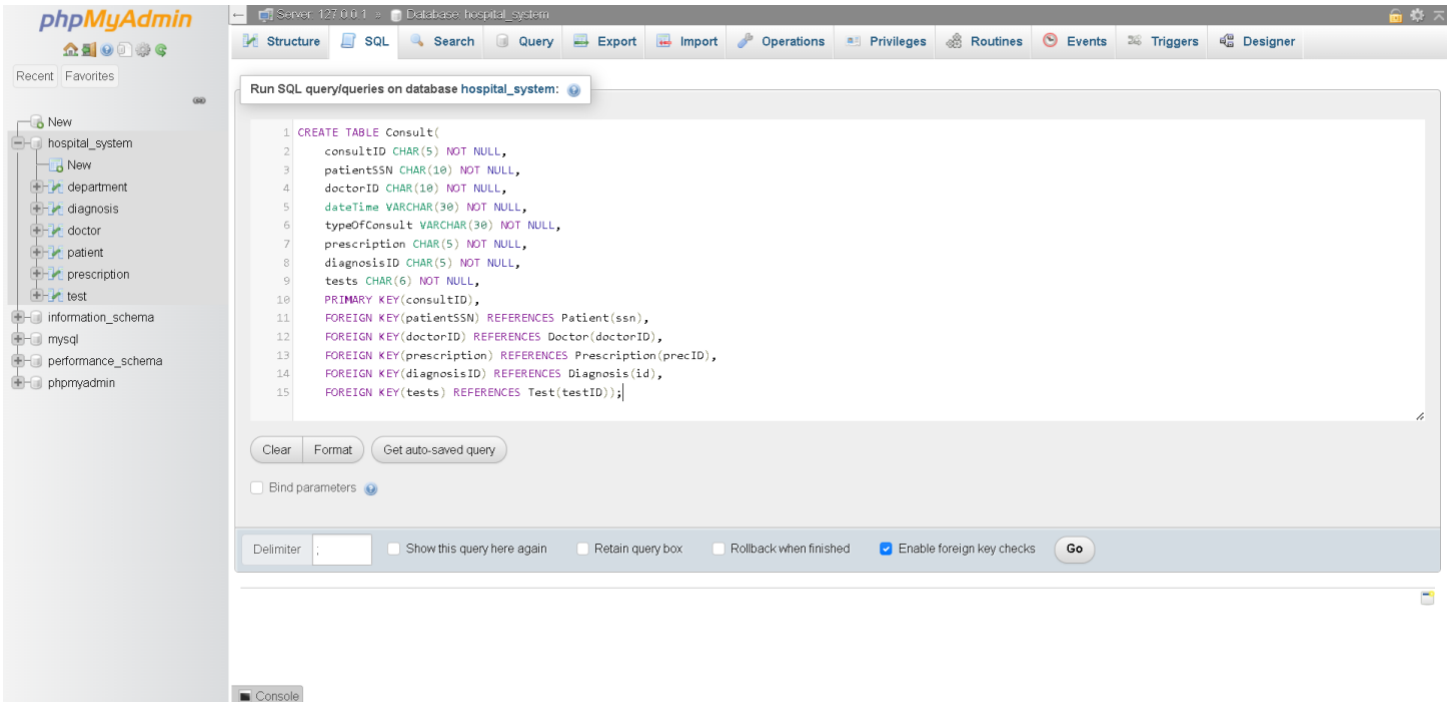


Output table

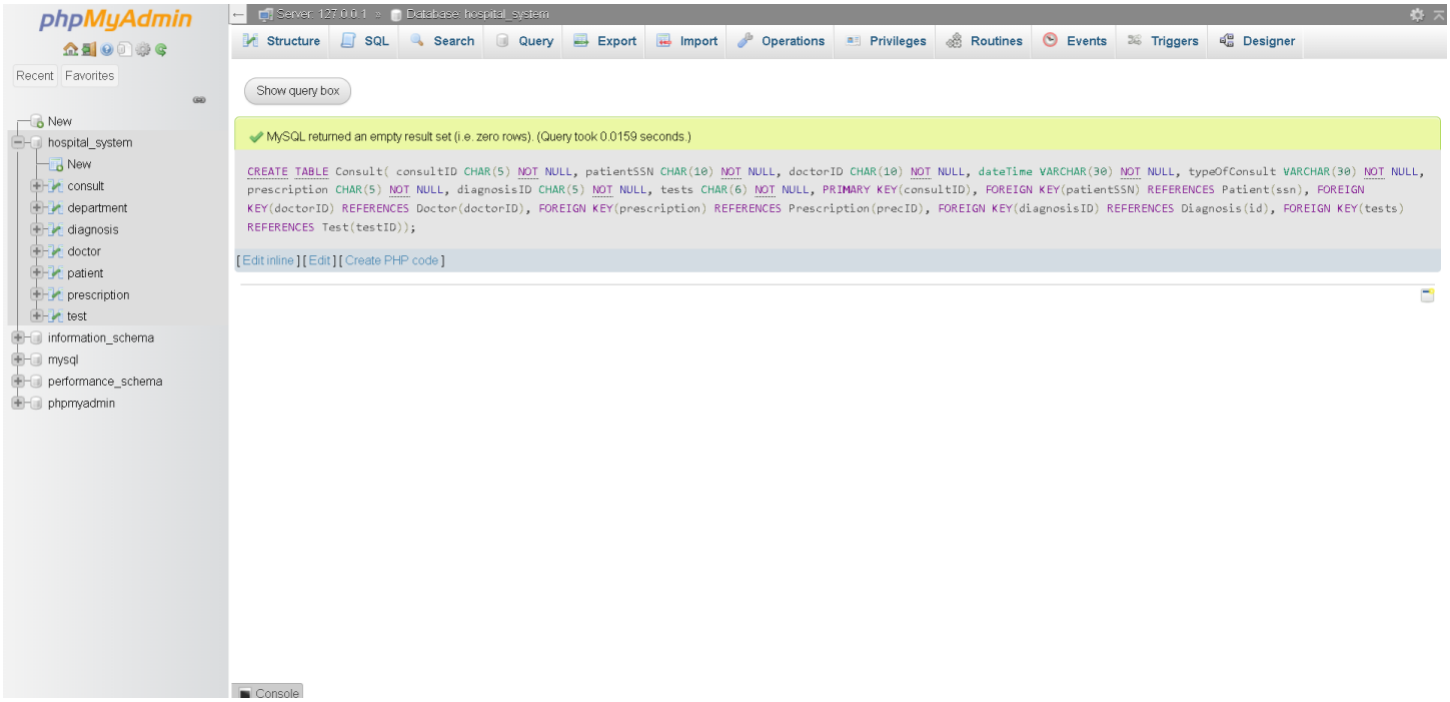


4.Consult Table

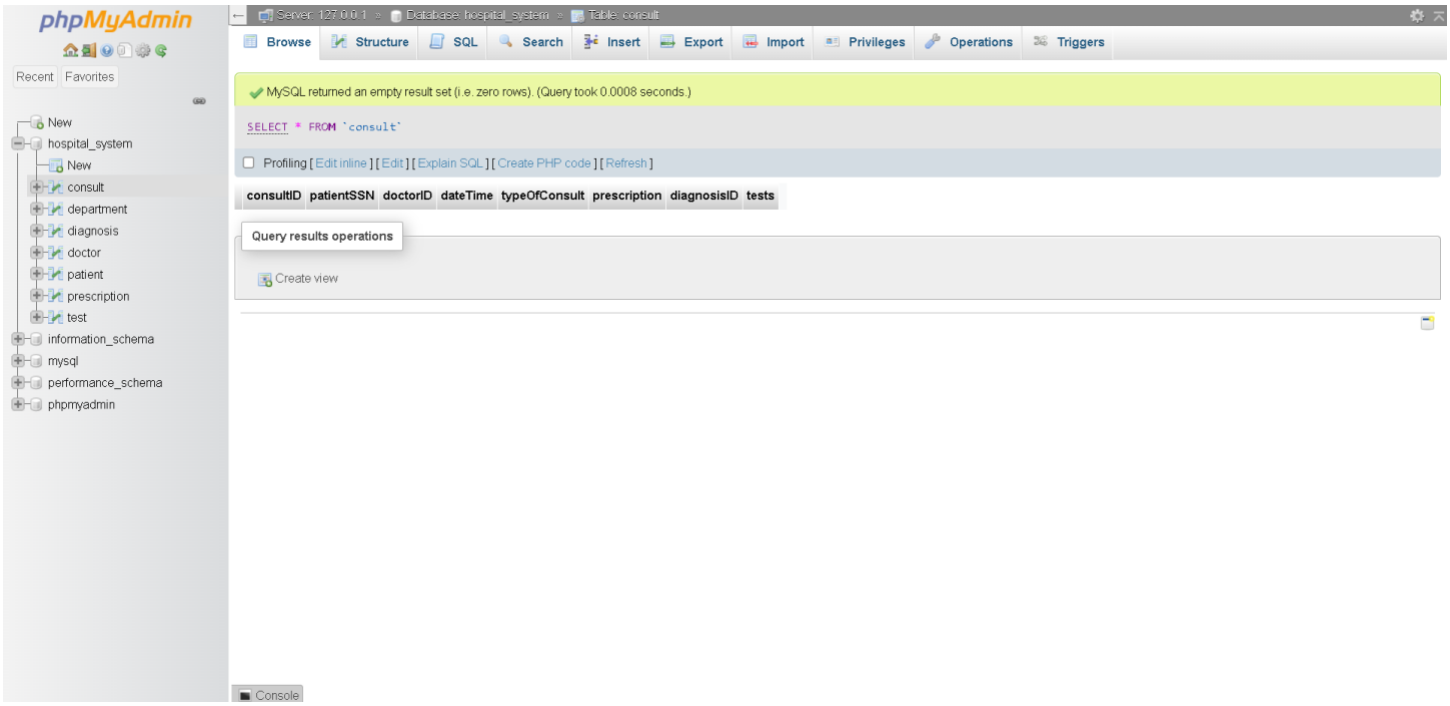
Query



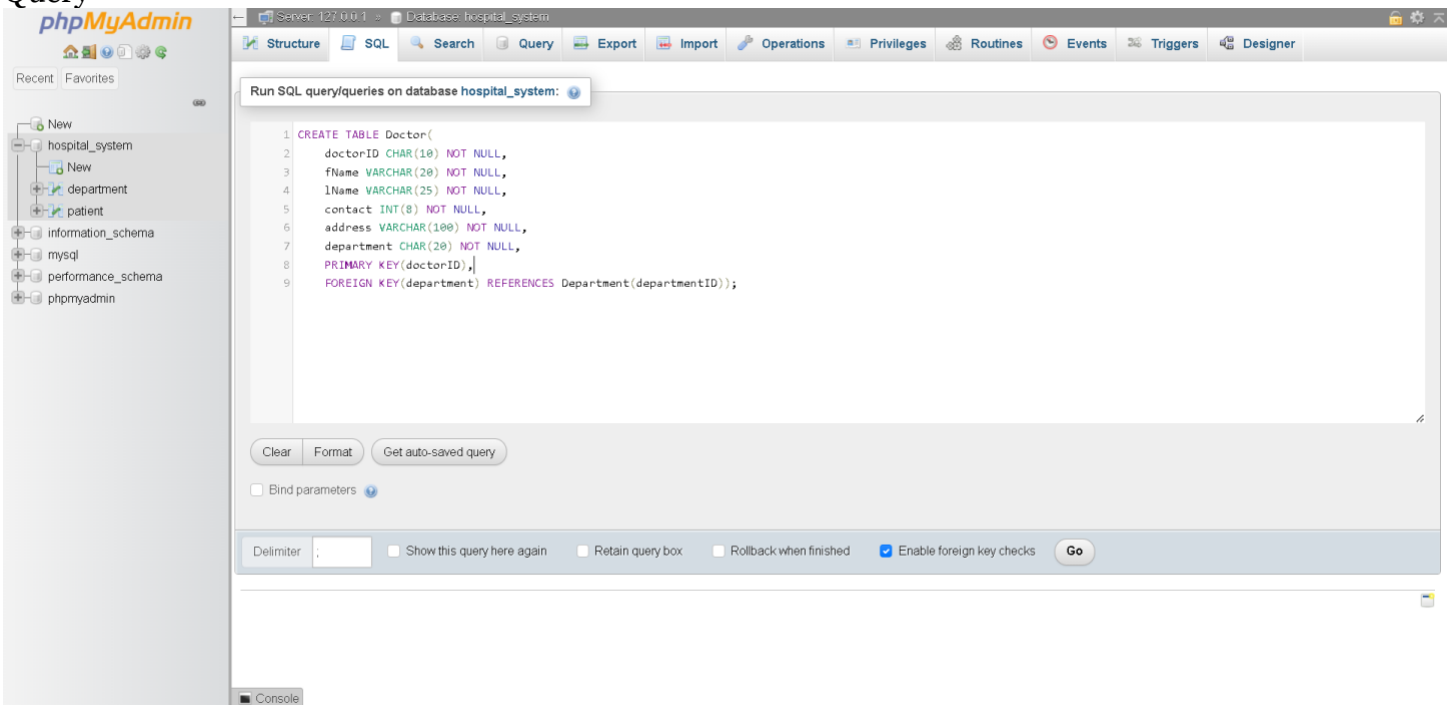
Result



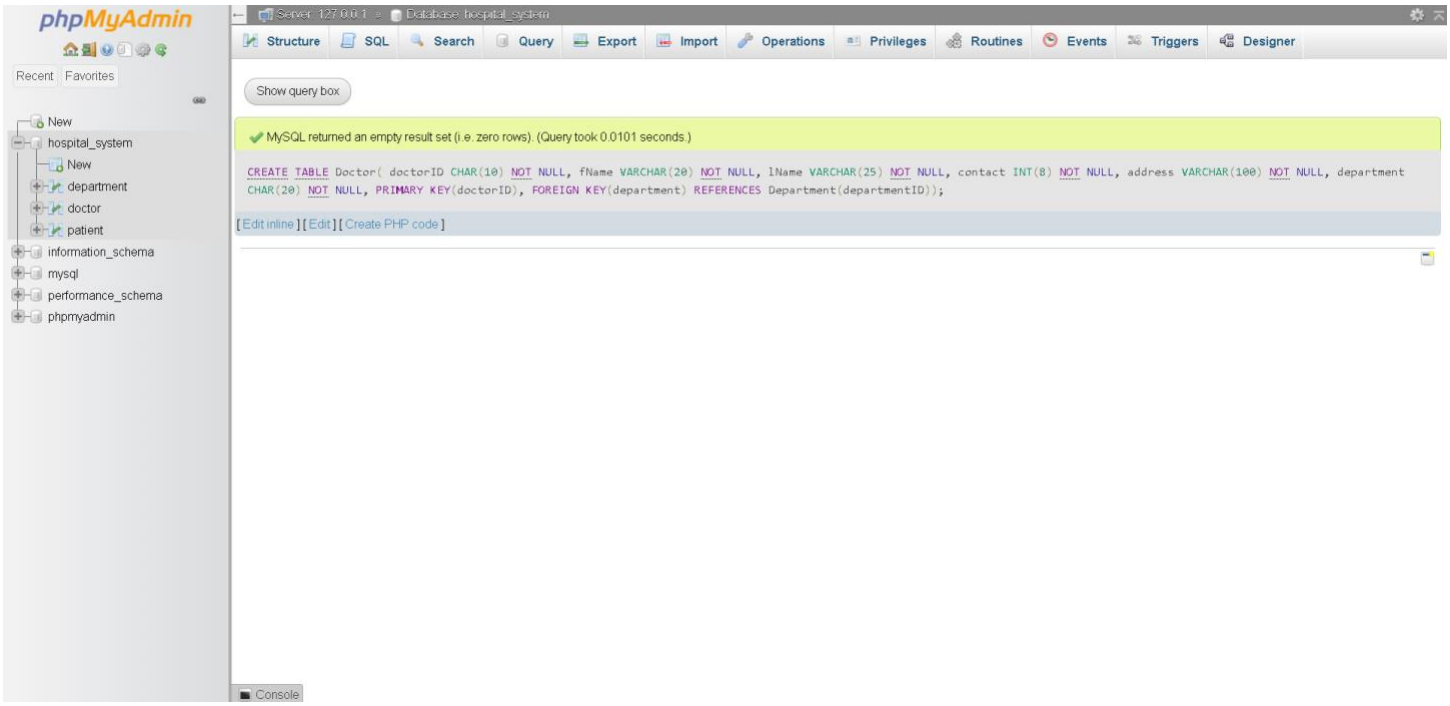
Output table



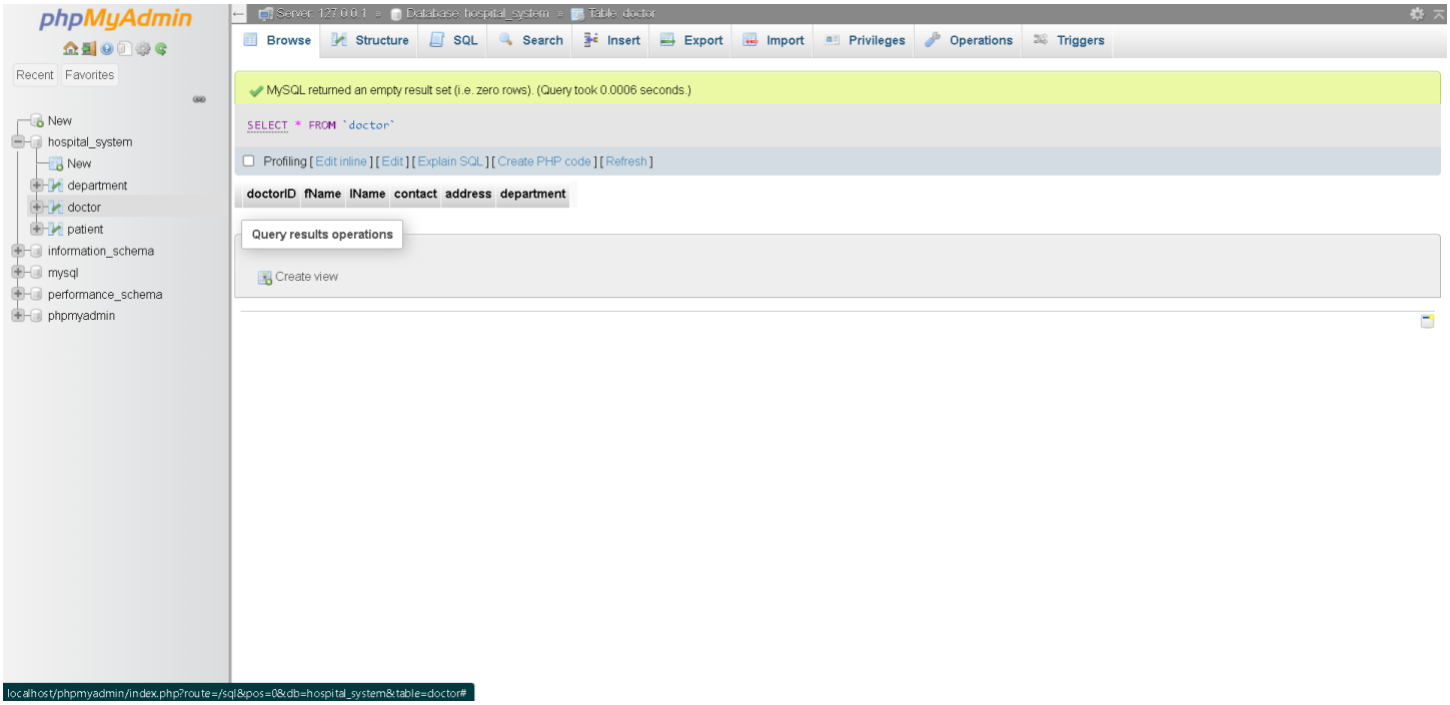
5.Doctor table
Query



Result

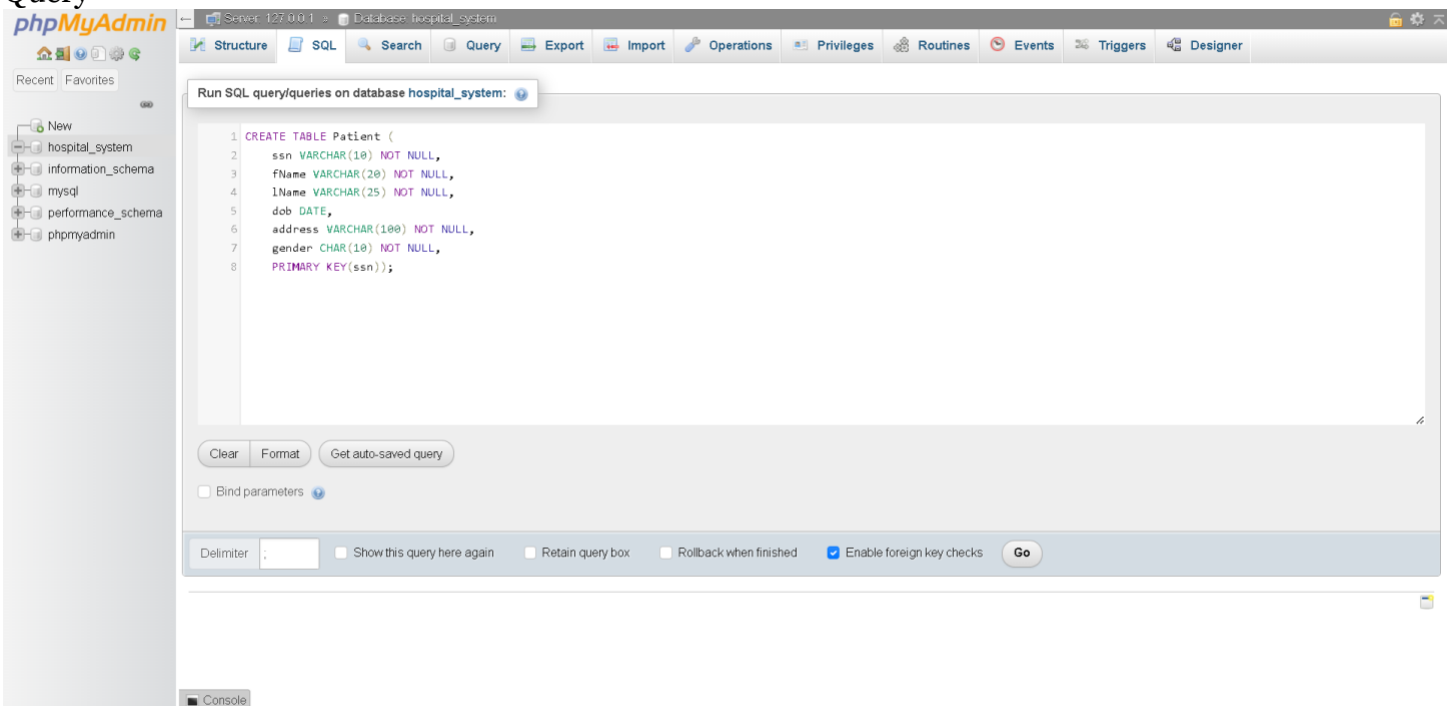


Output table

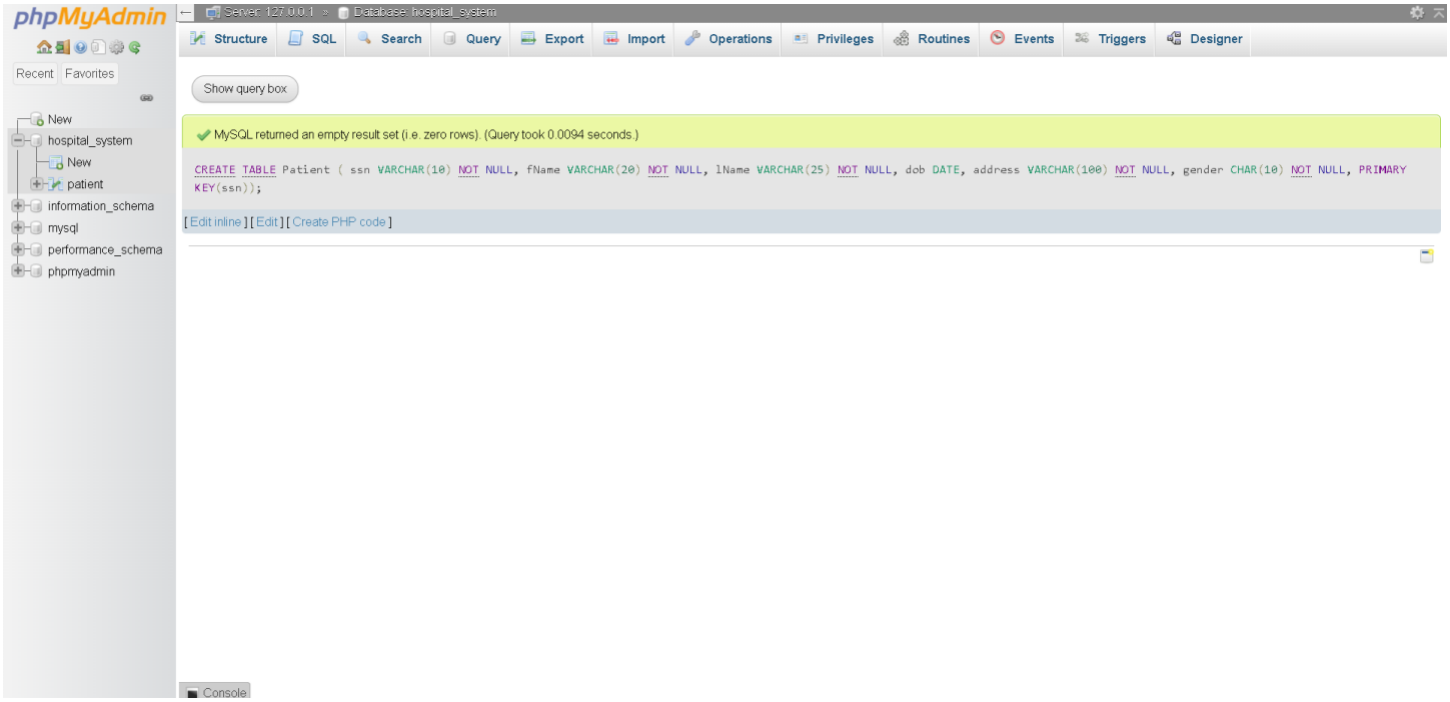


6.Patient table

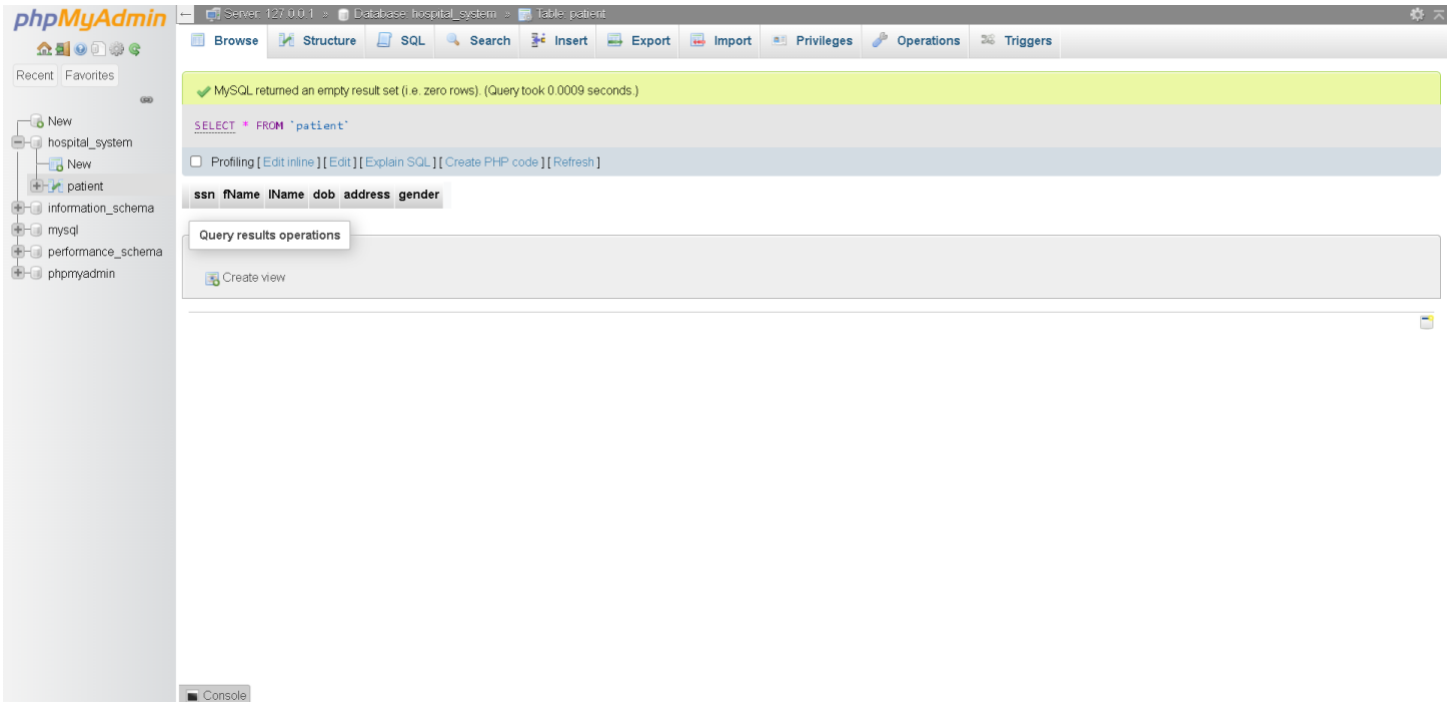
Query



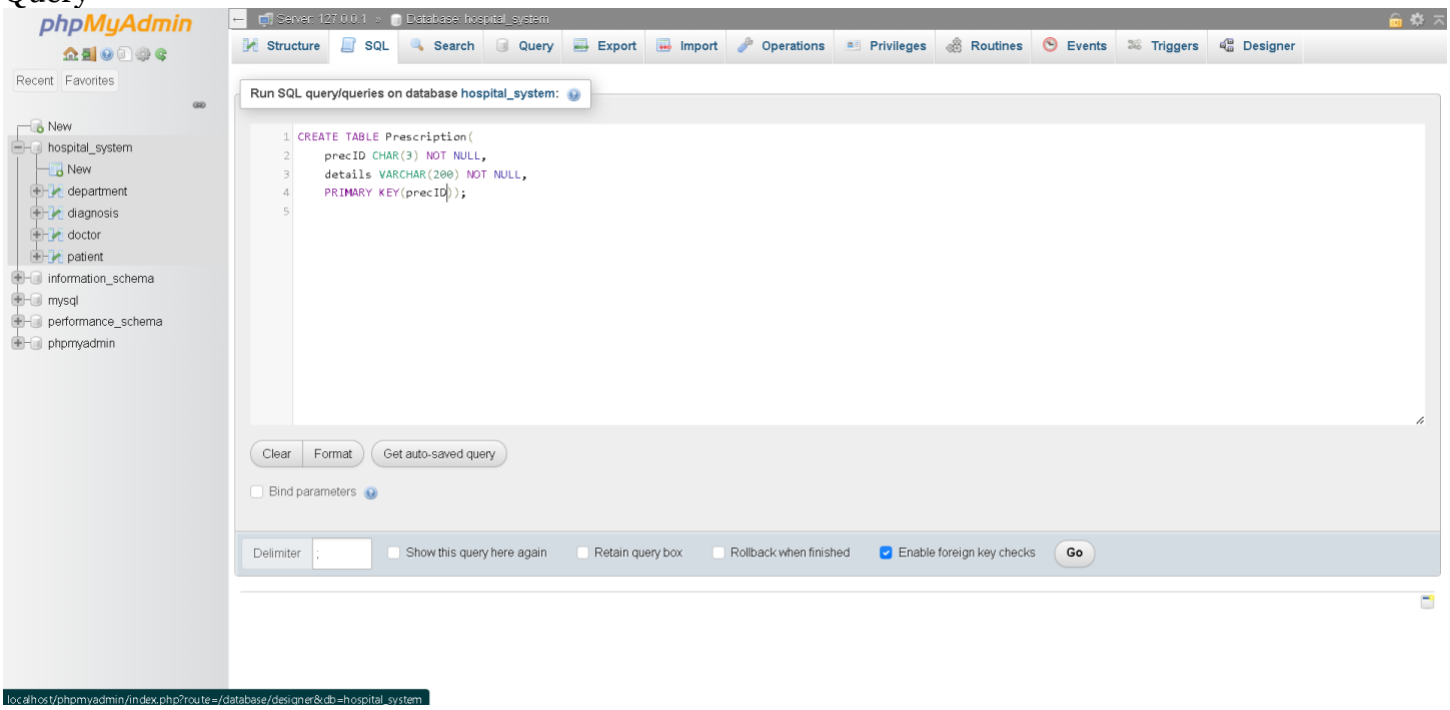
Result



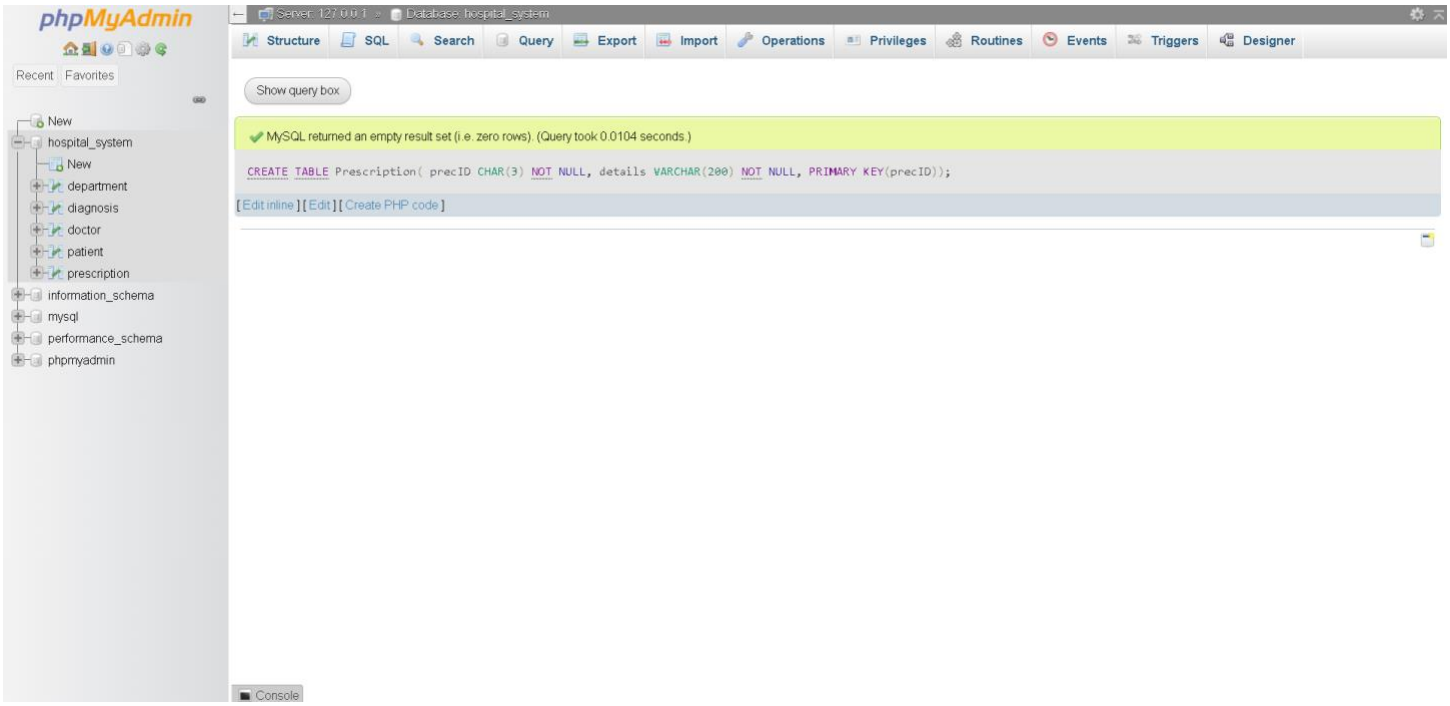
Output table



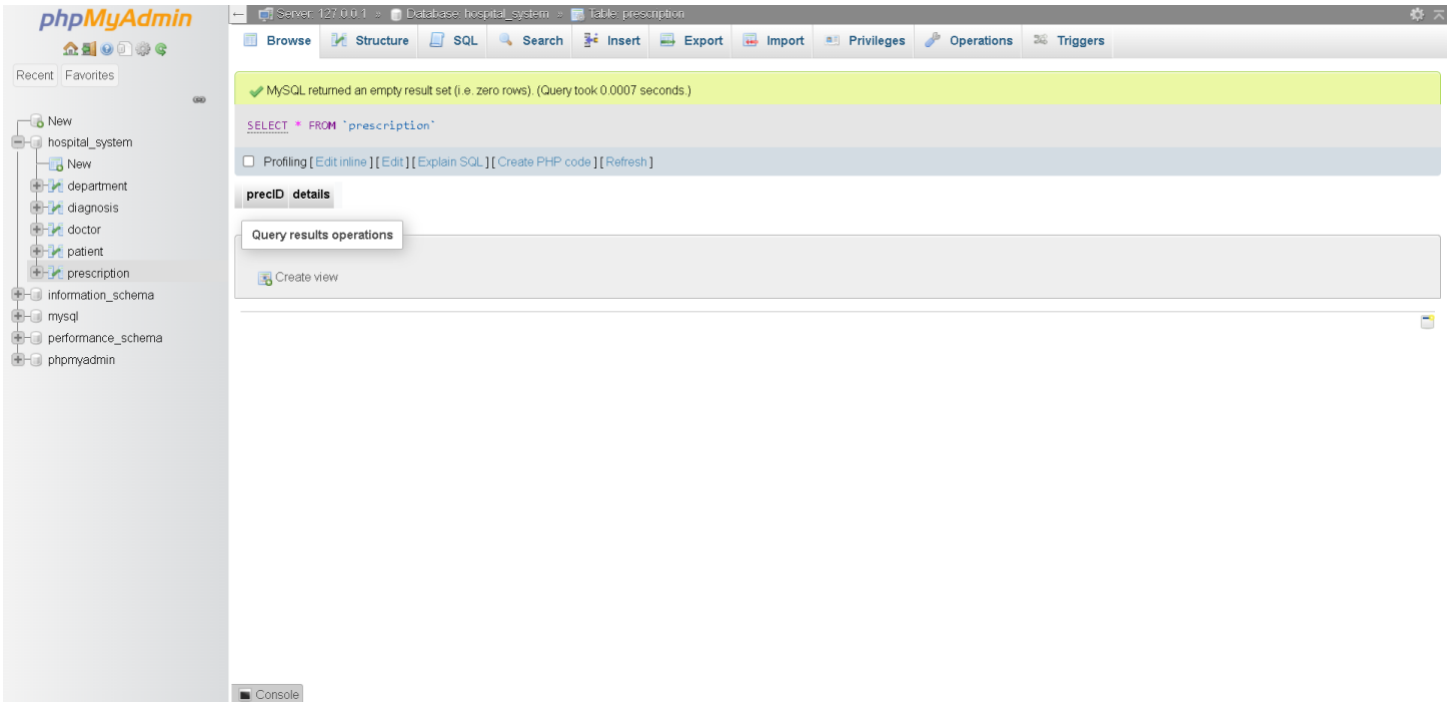
7.Prescription table
Query



Result

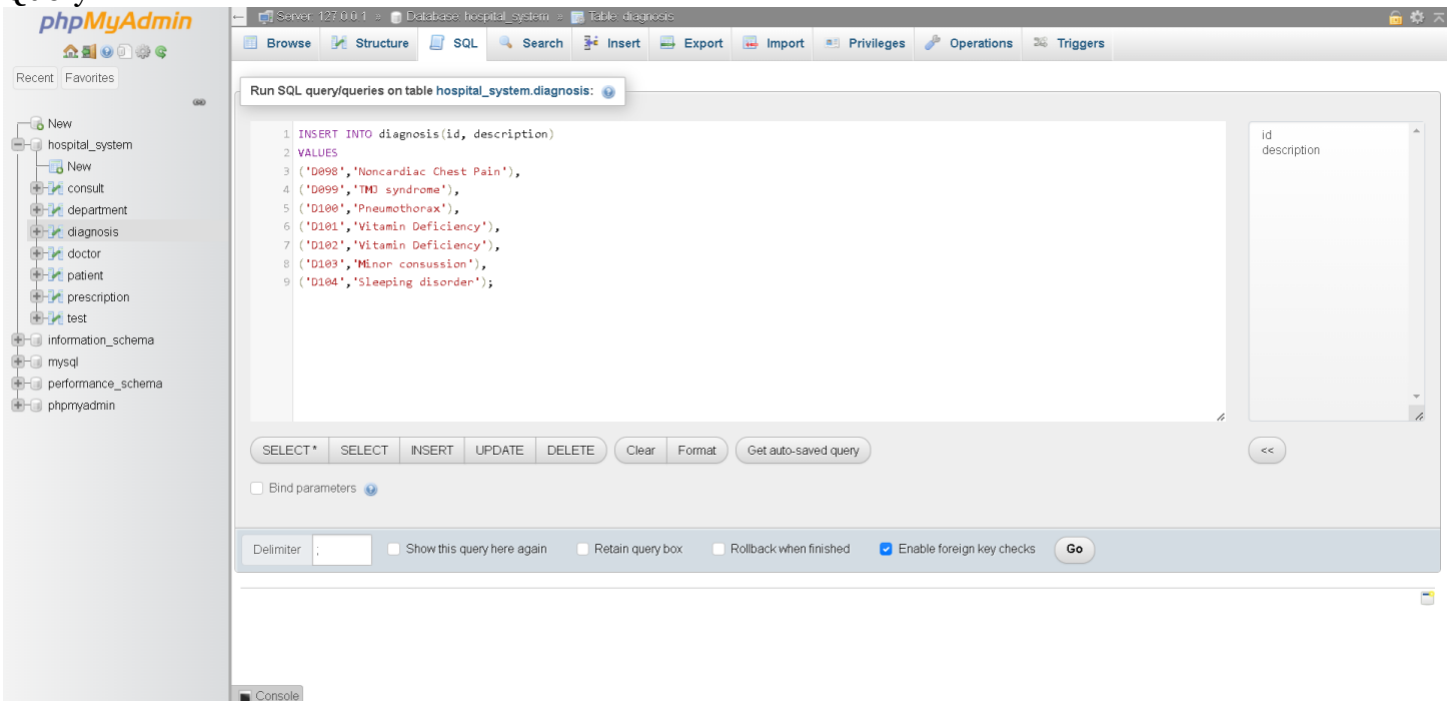


Output table

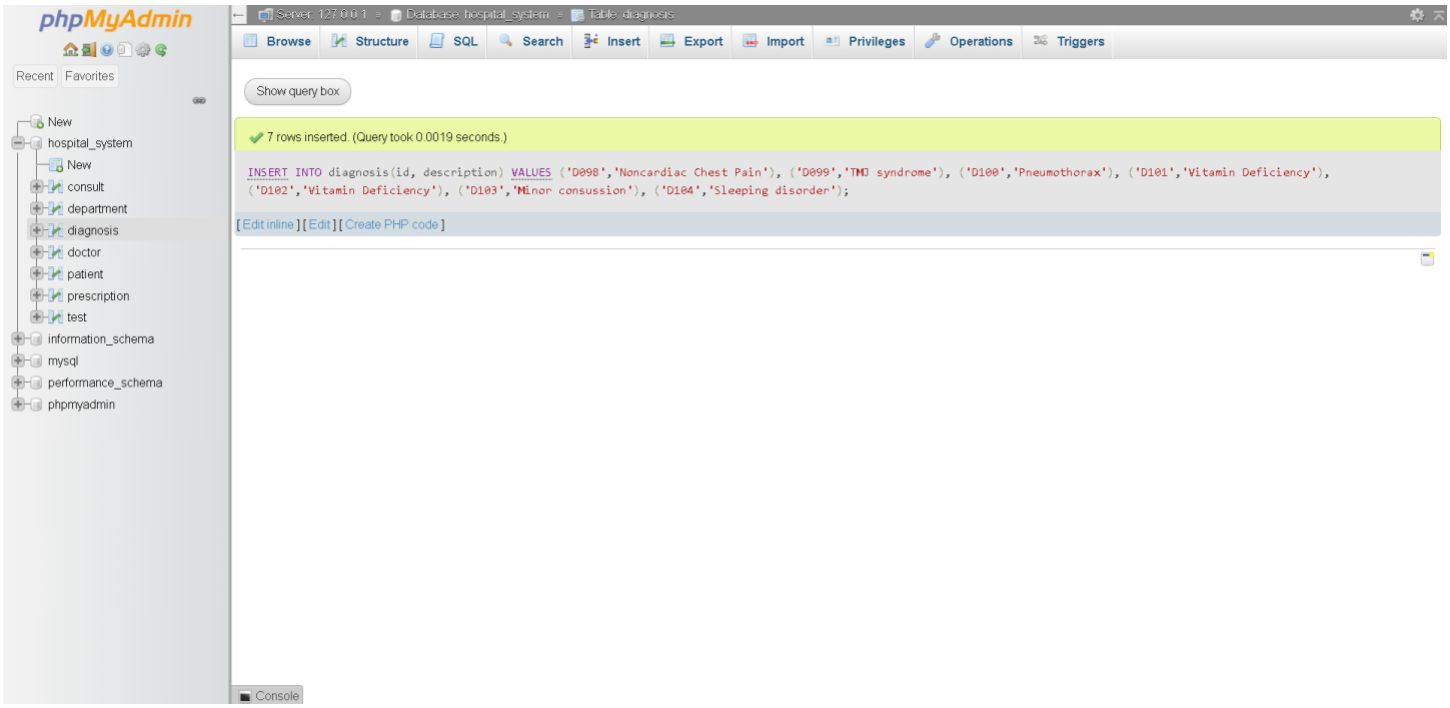


Question 2. Insert data into the tables.

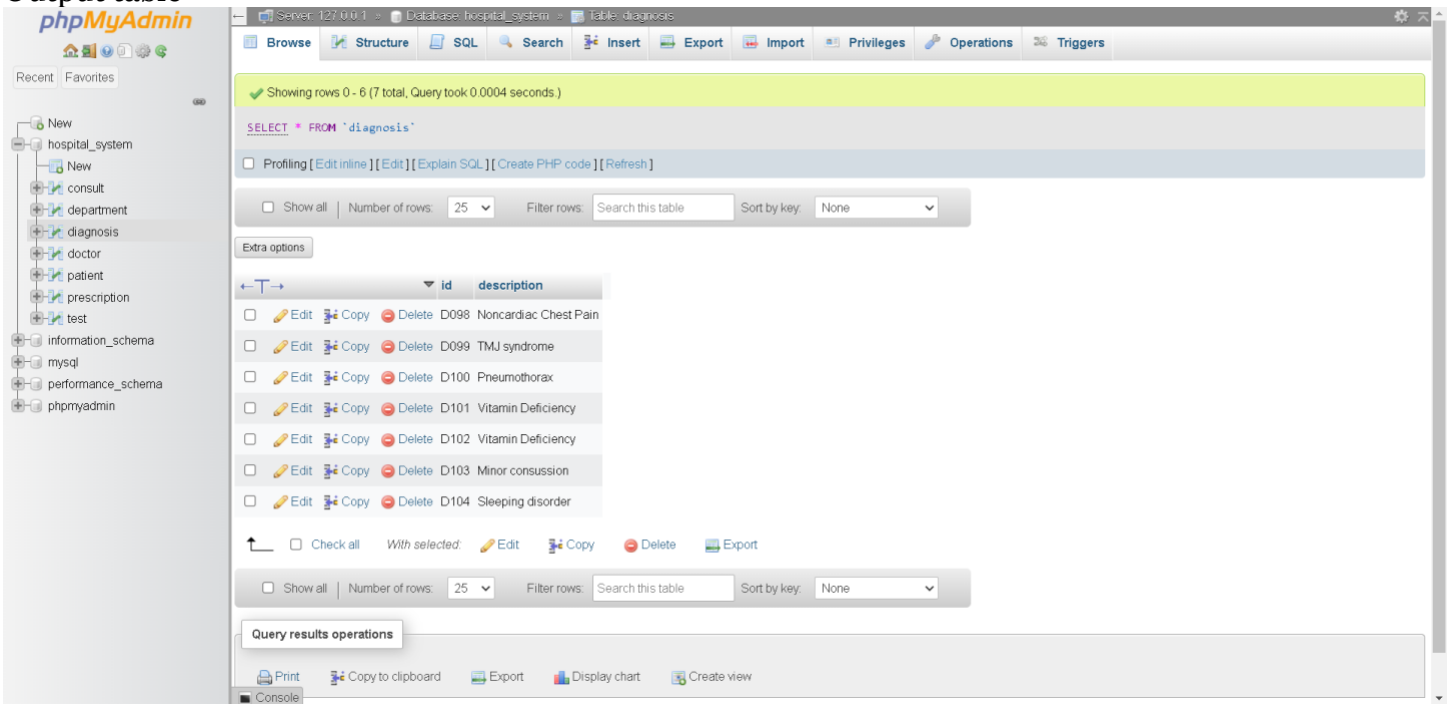
1.Diagnosis Table
Query



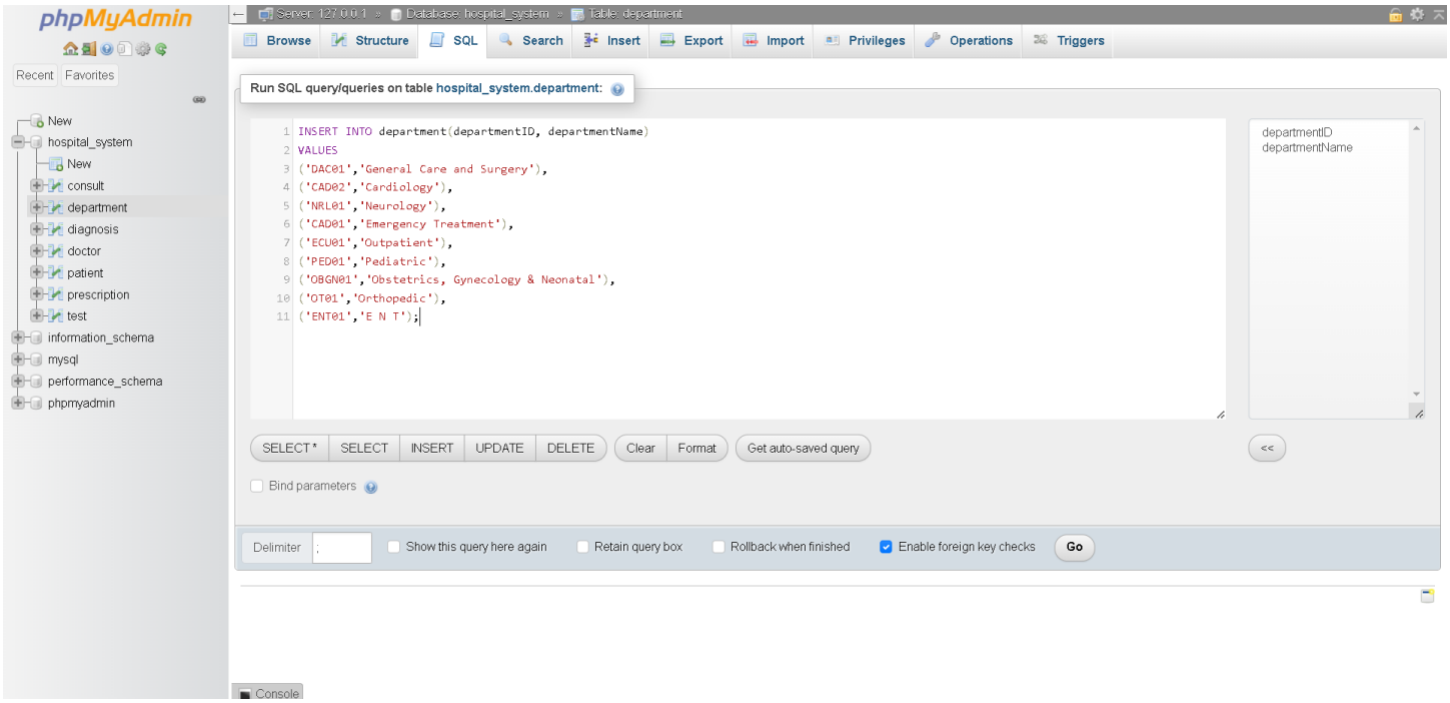
Result



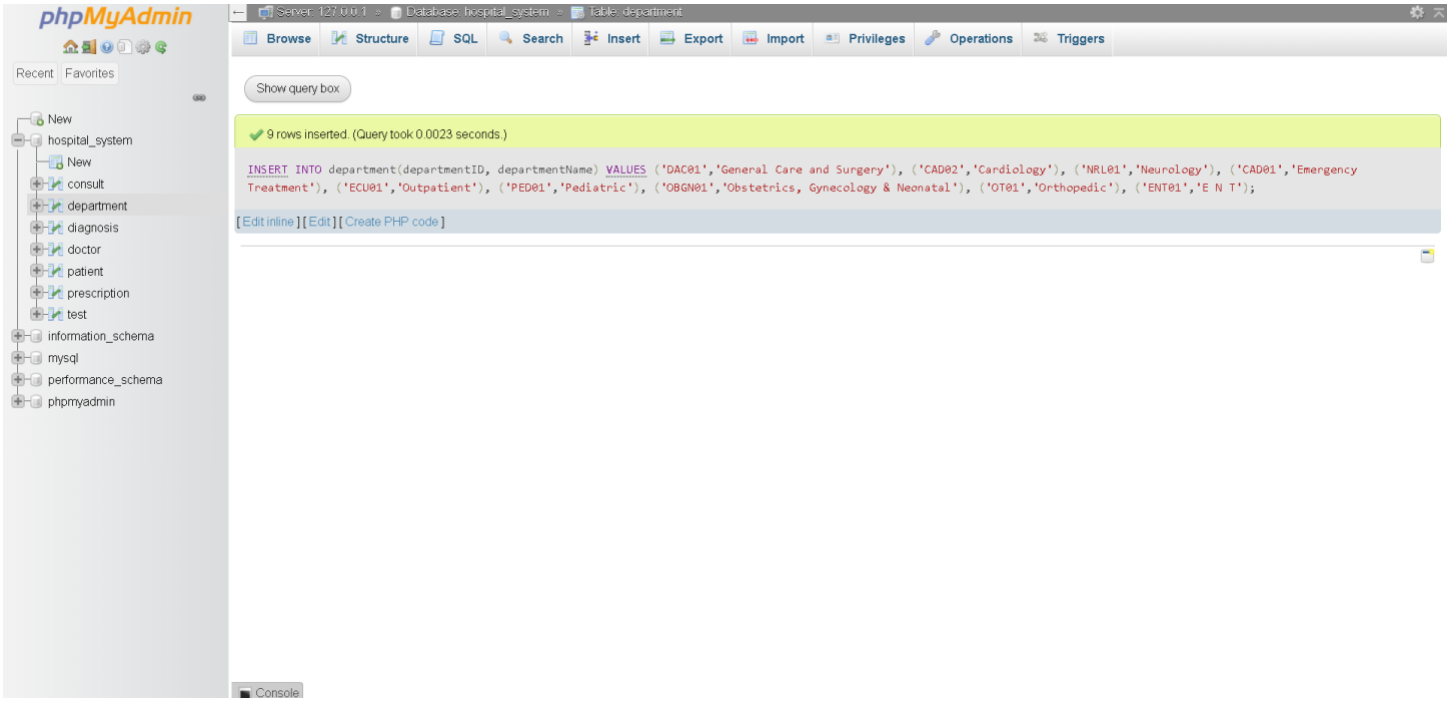
Output table



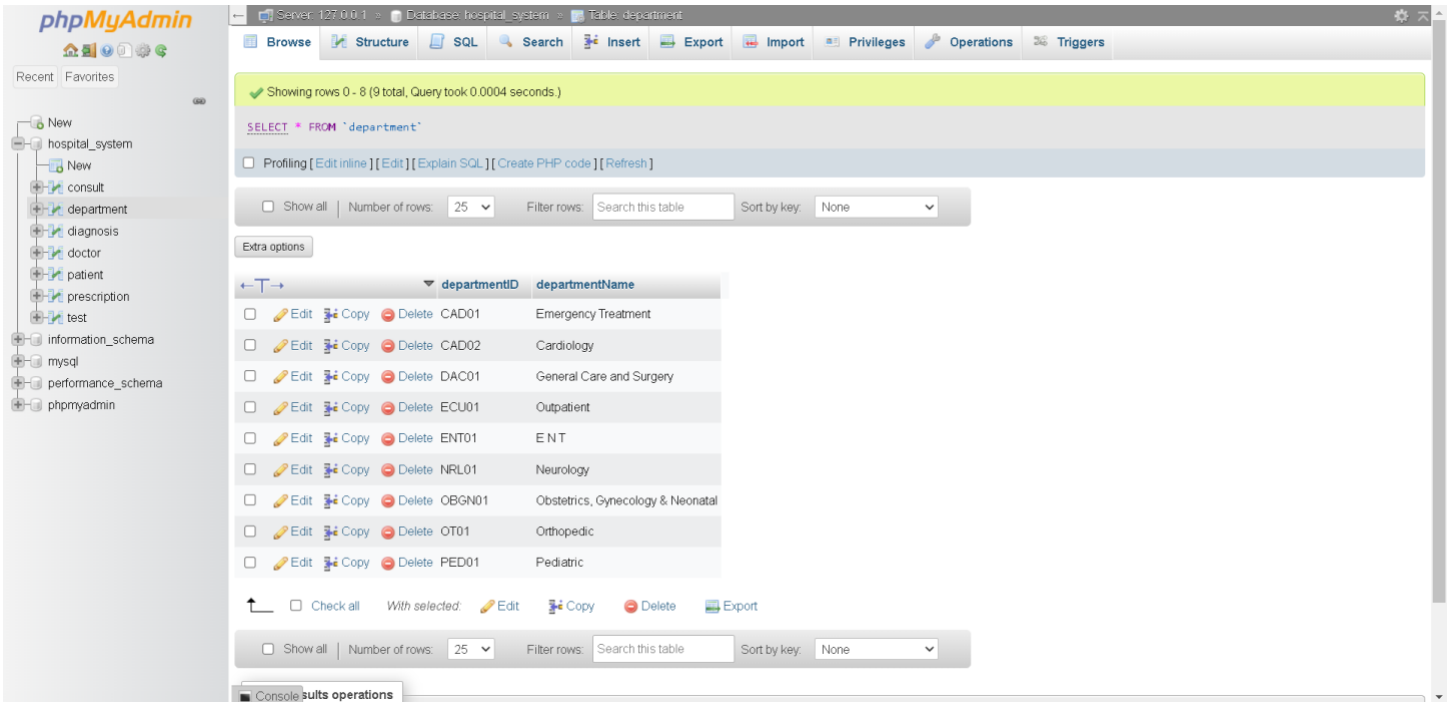
2.Department table
Query



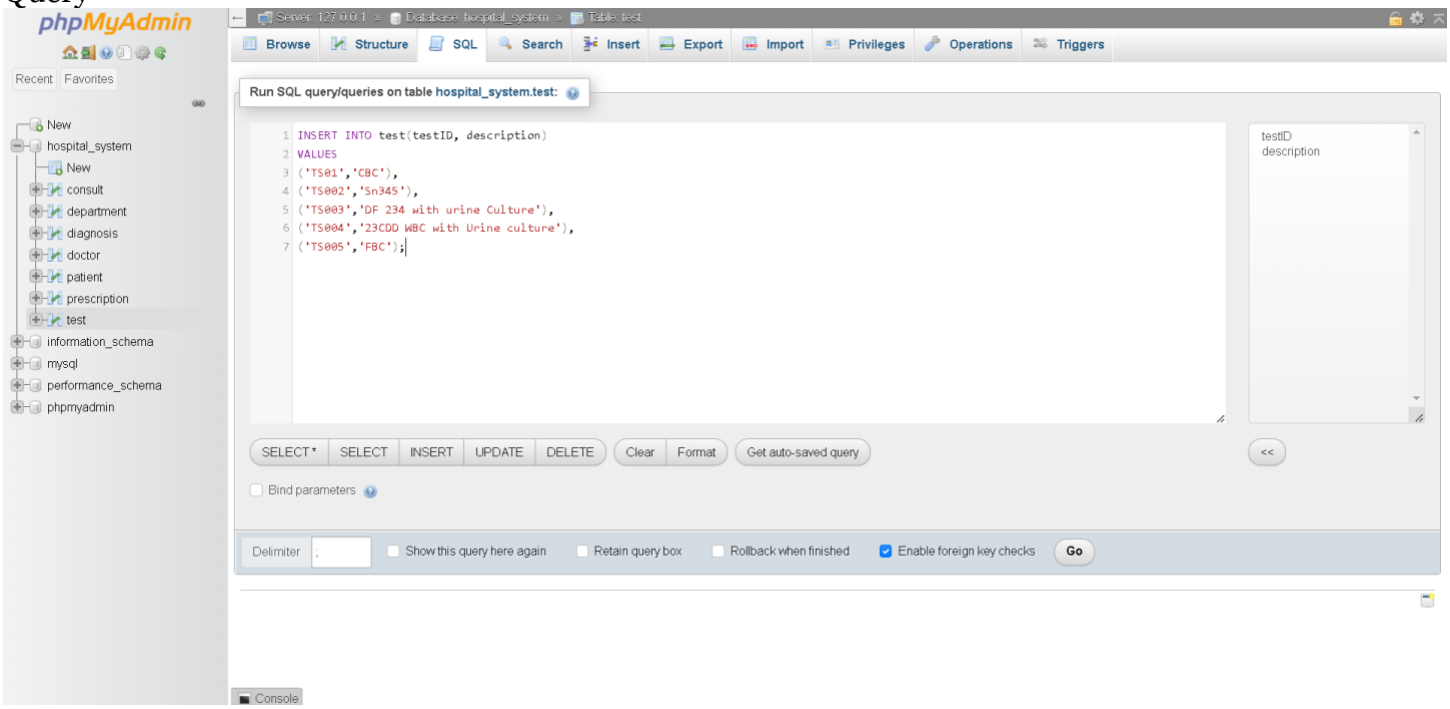
Result



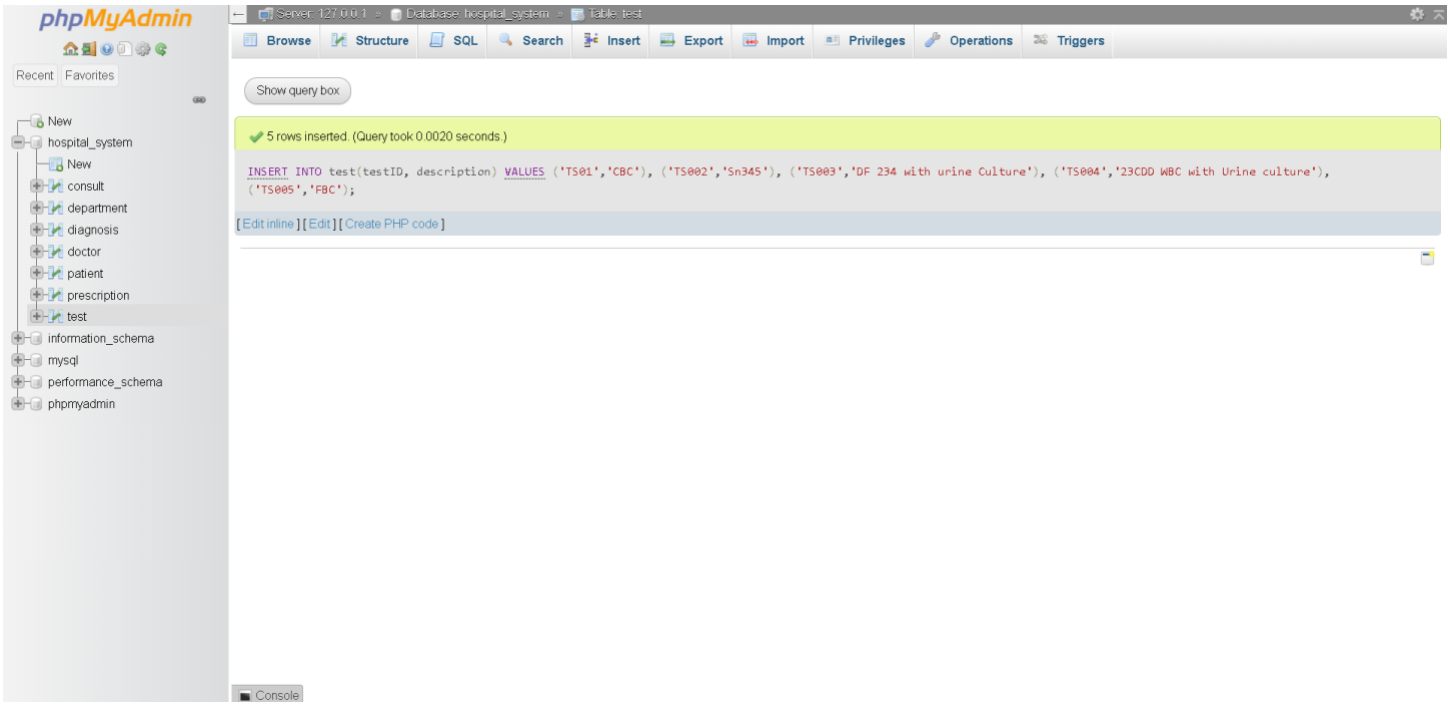
Output table



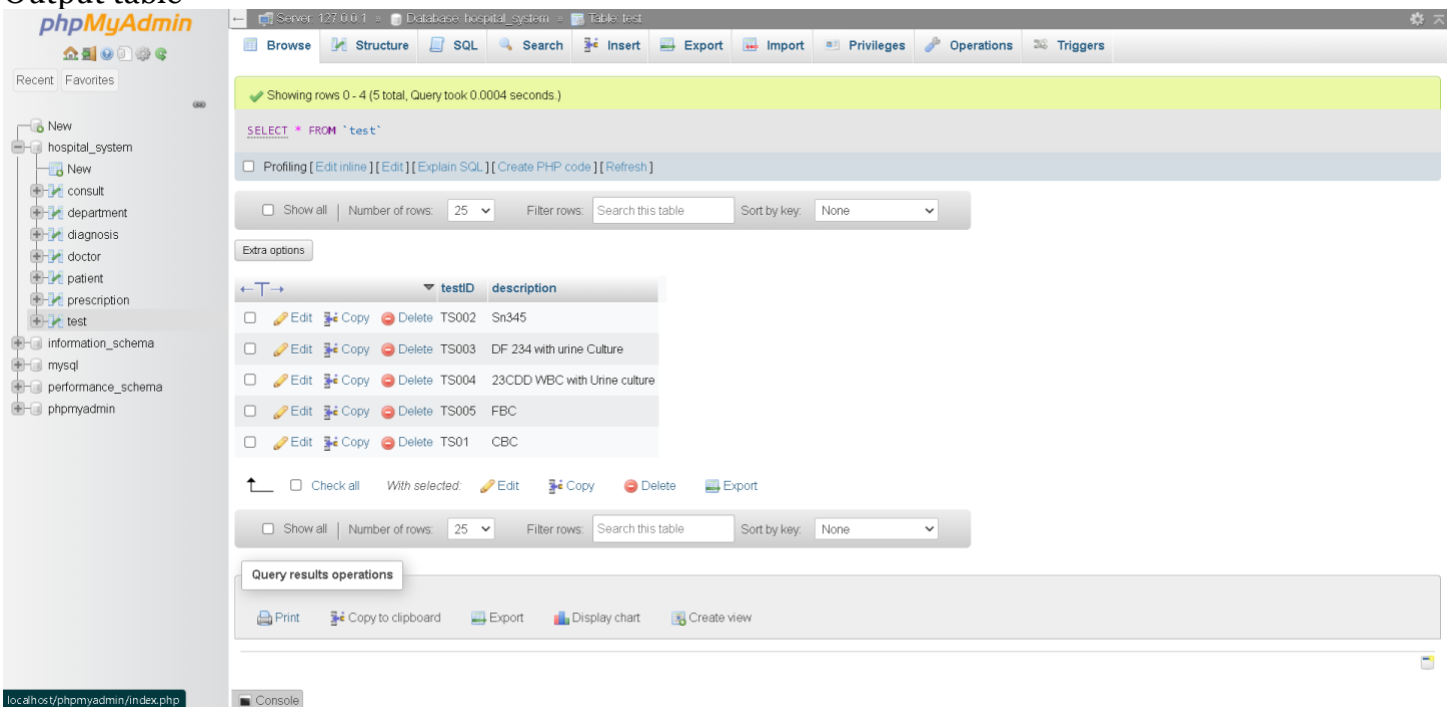
3.Test table
Query



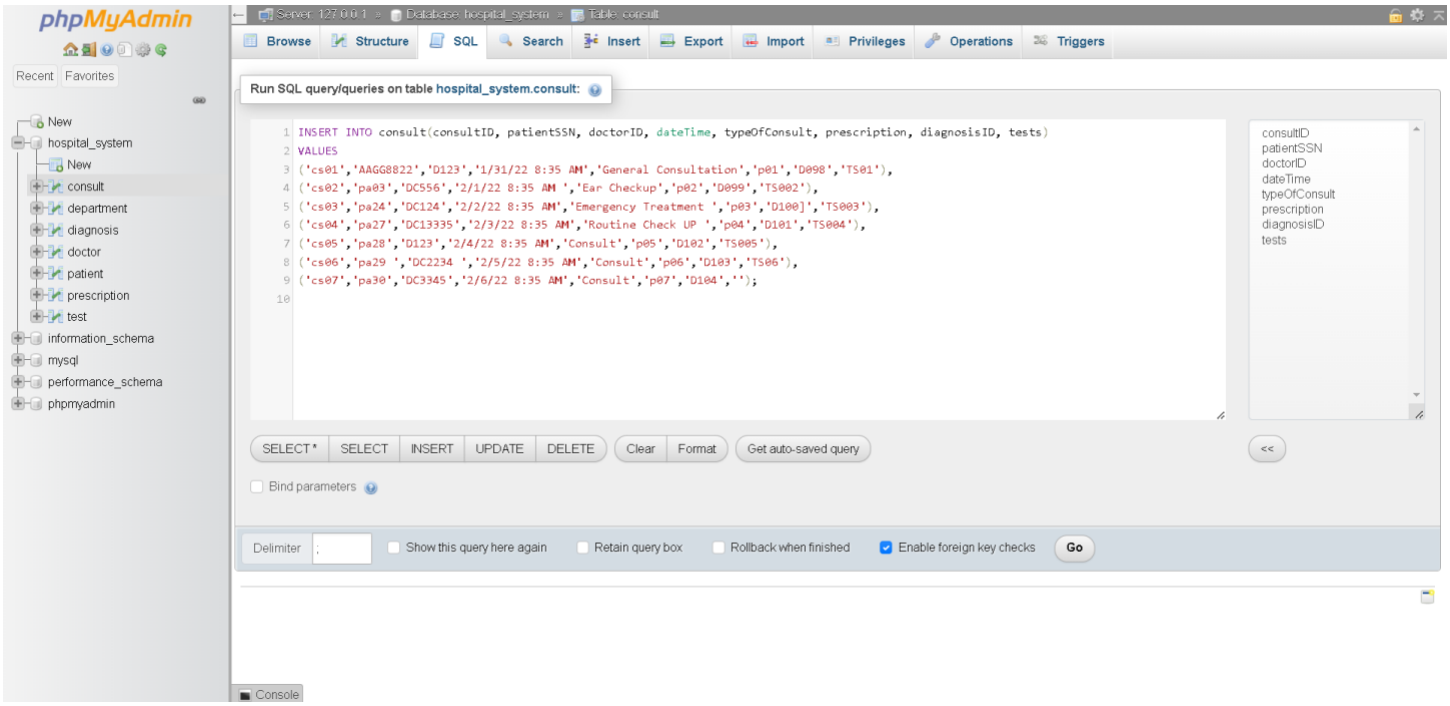
Result



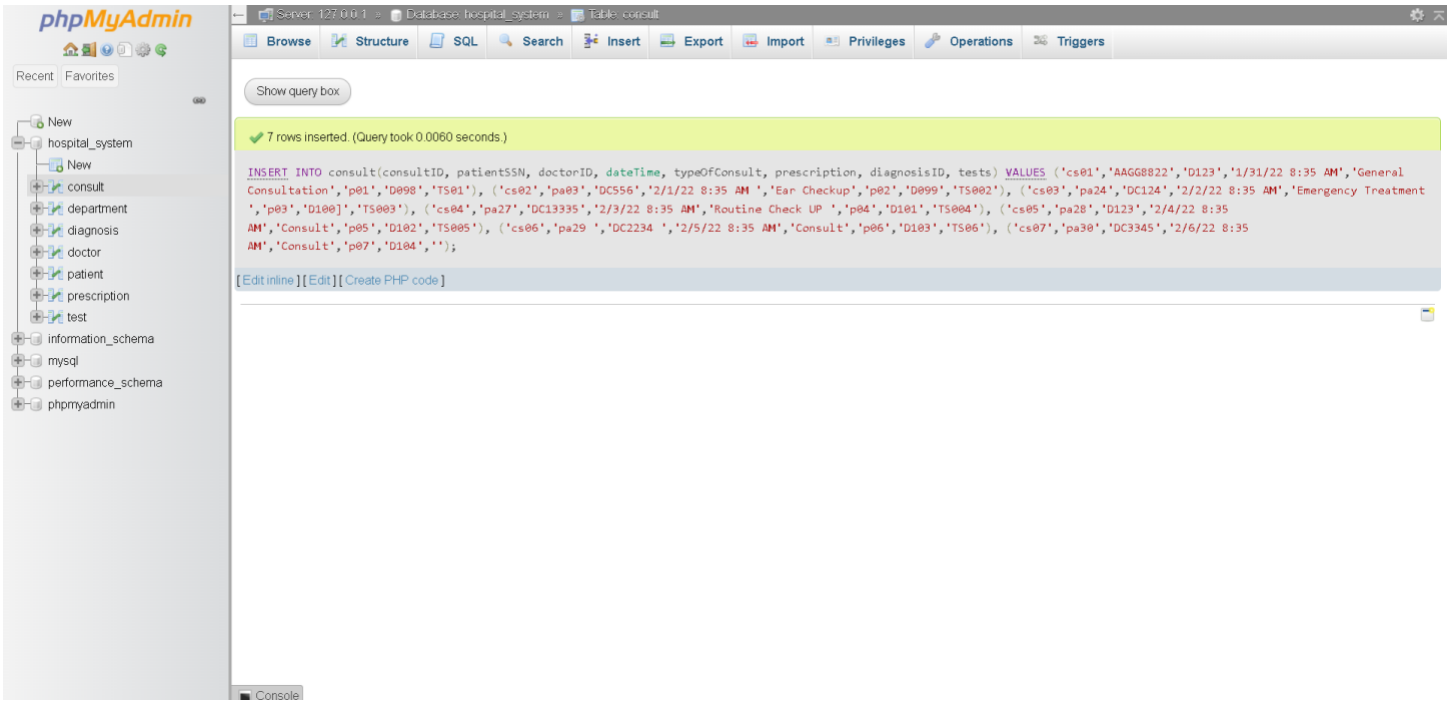
Output table



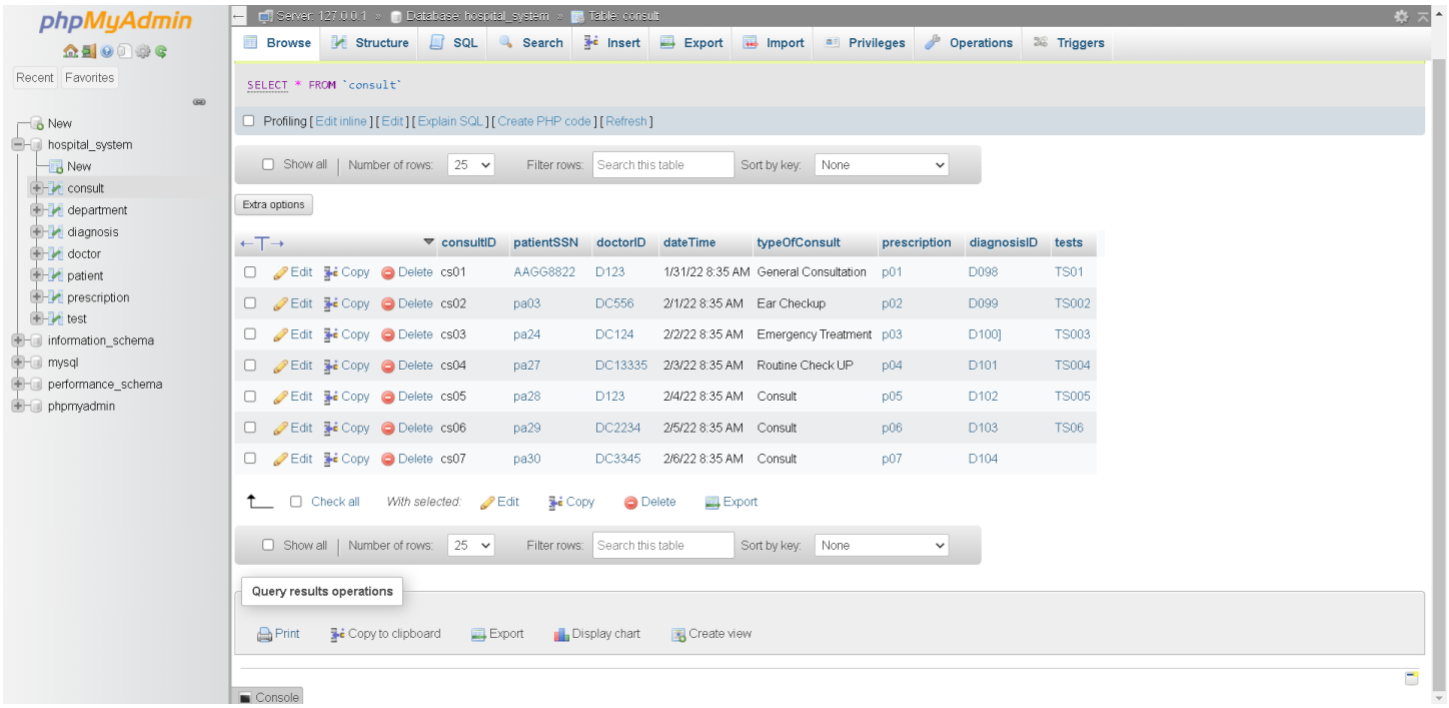
4.Consult table
Query



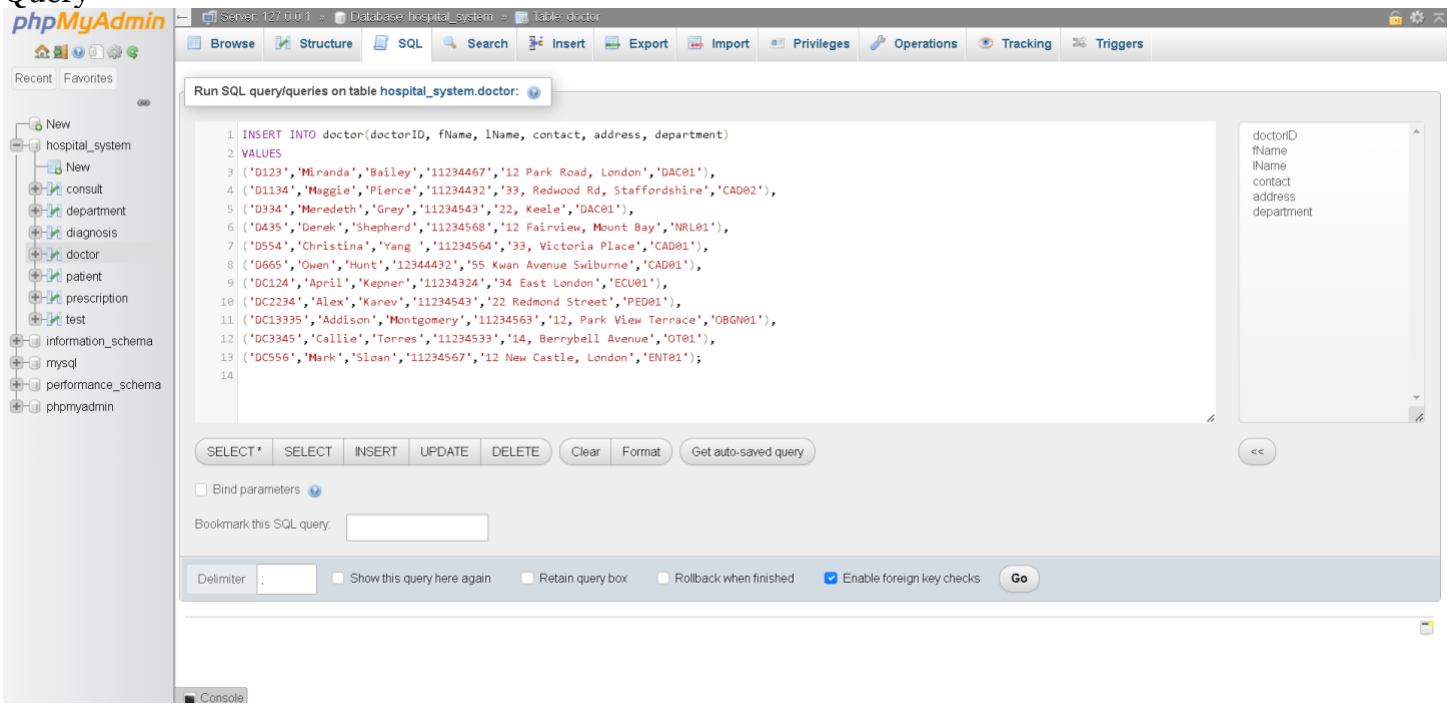
Result



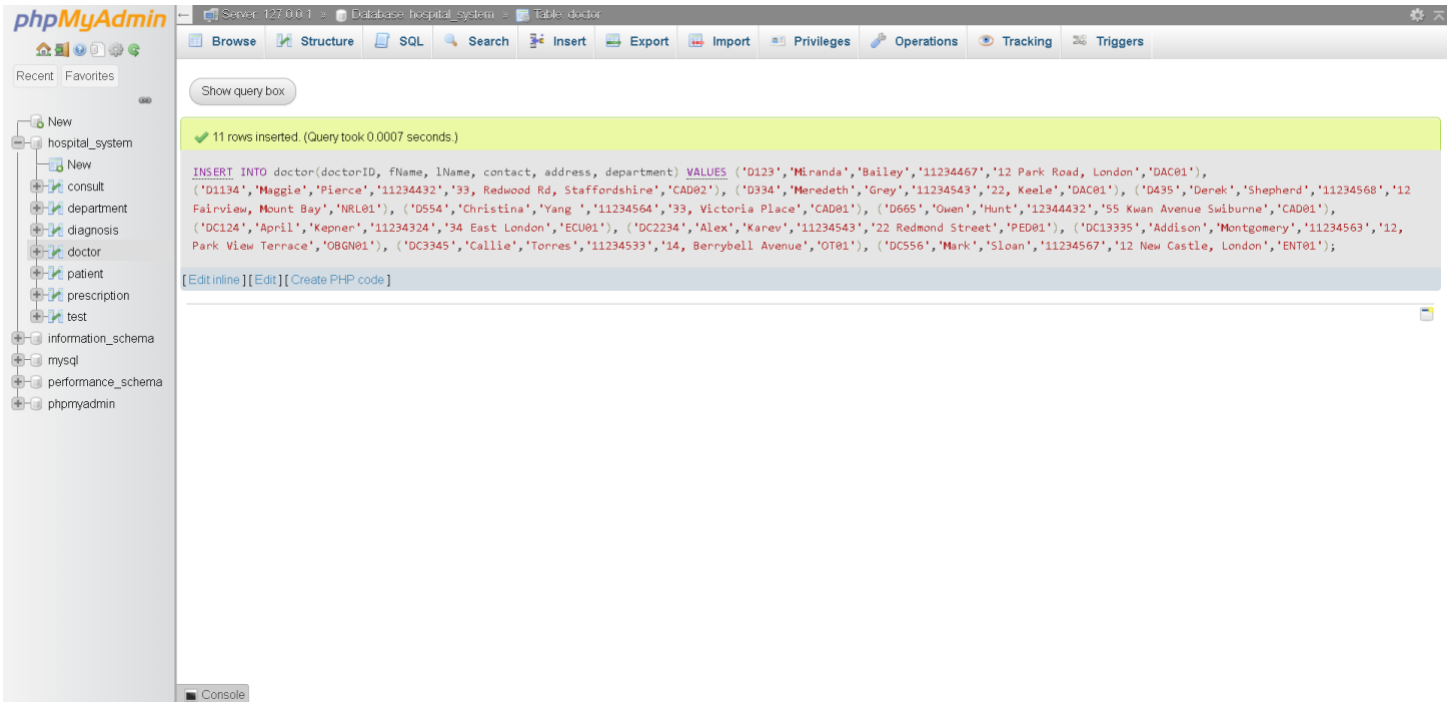
Output table



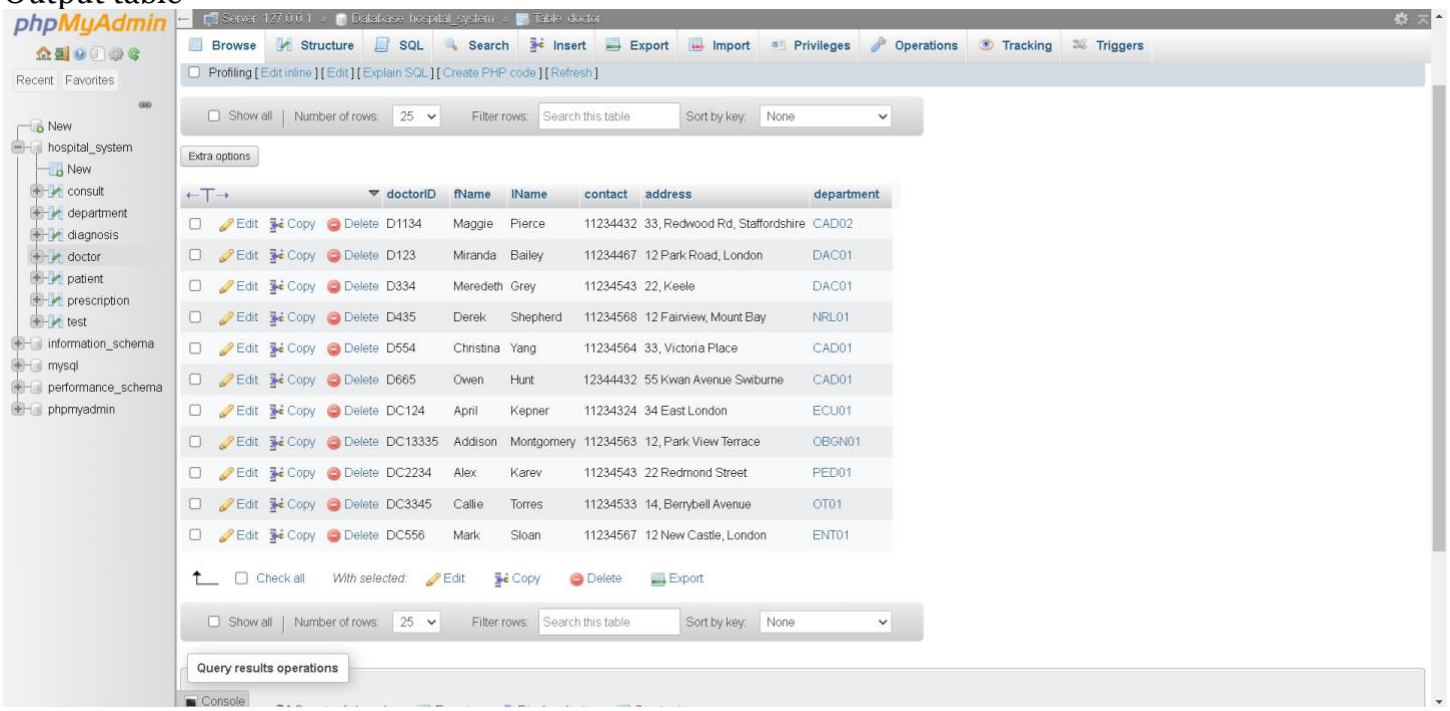
5.Doctor table
Query



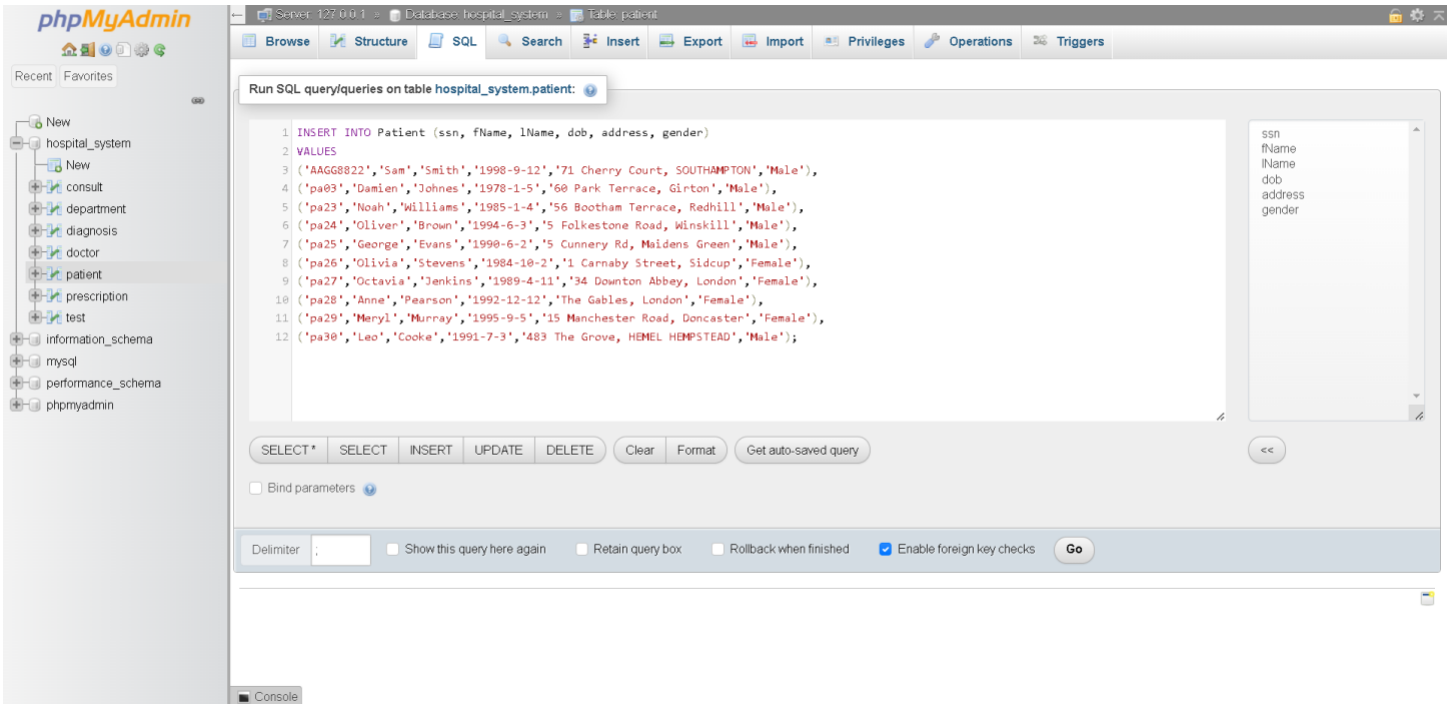
Result



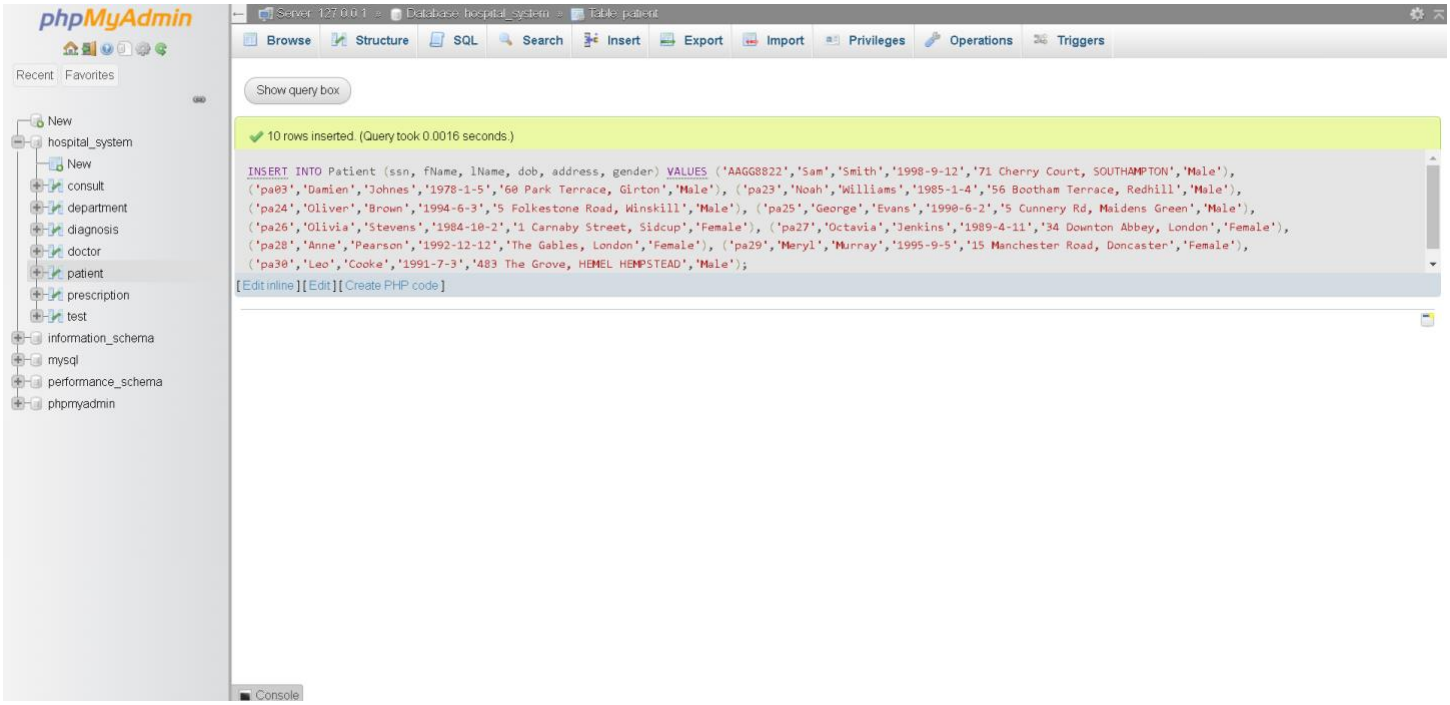
Output table



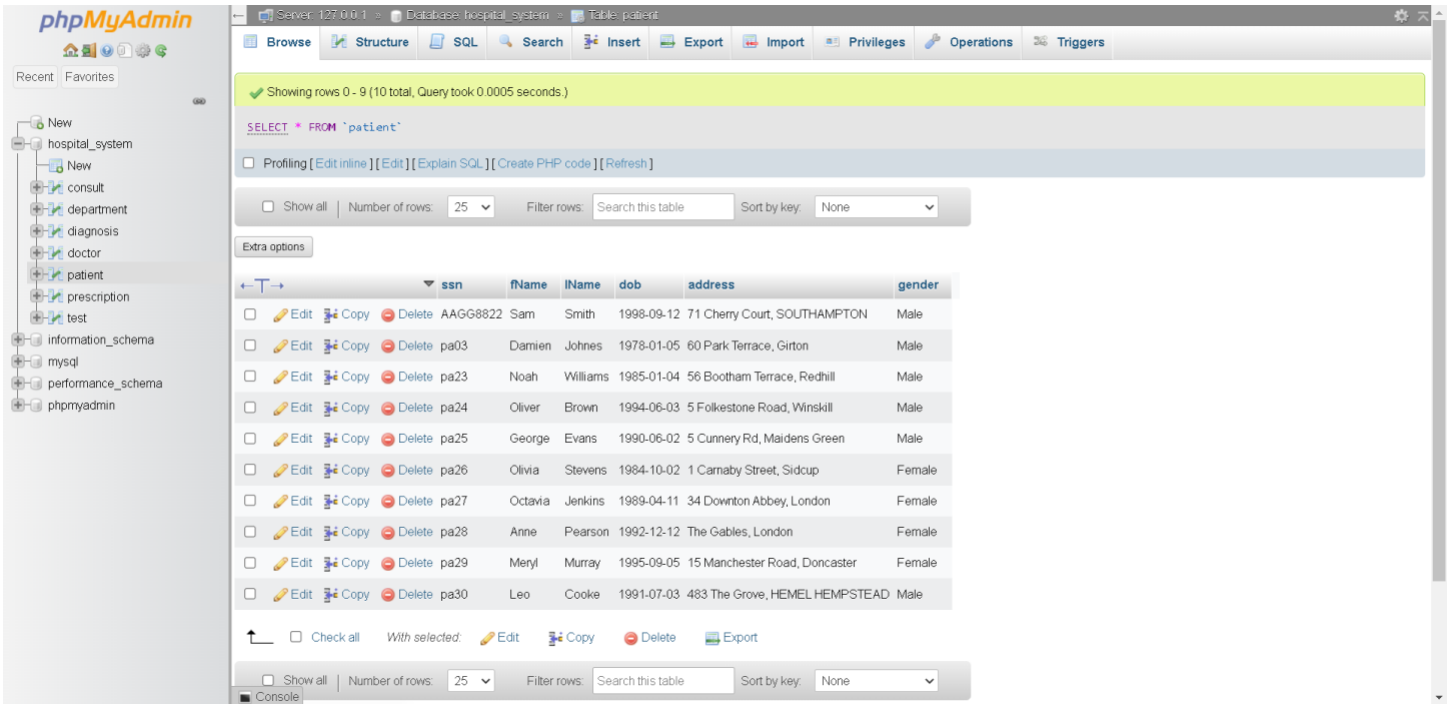
6.Patient table
Query



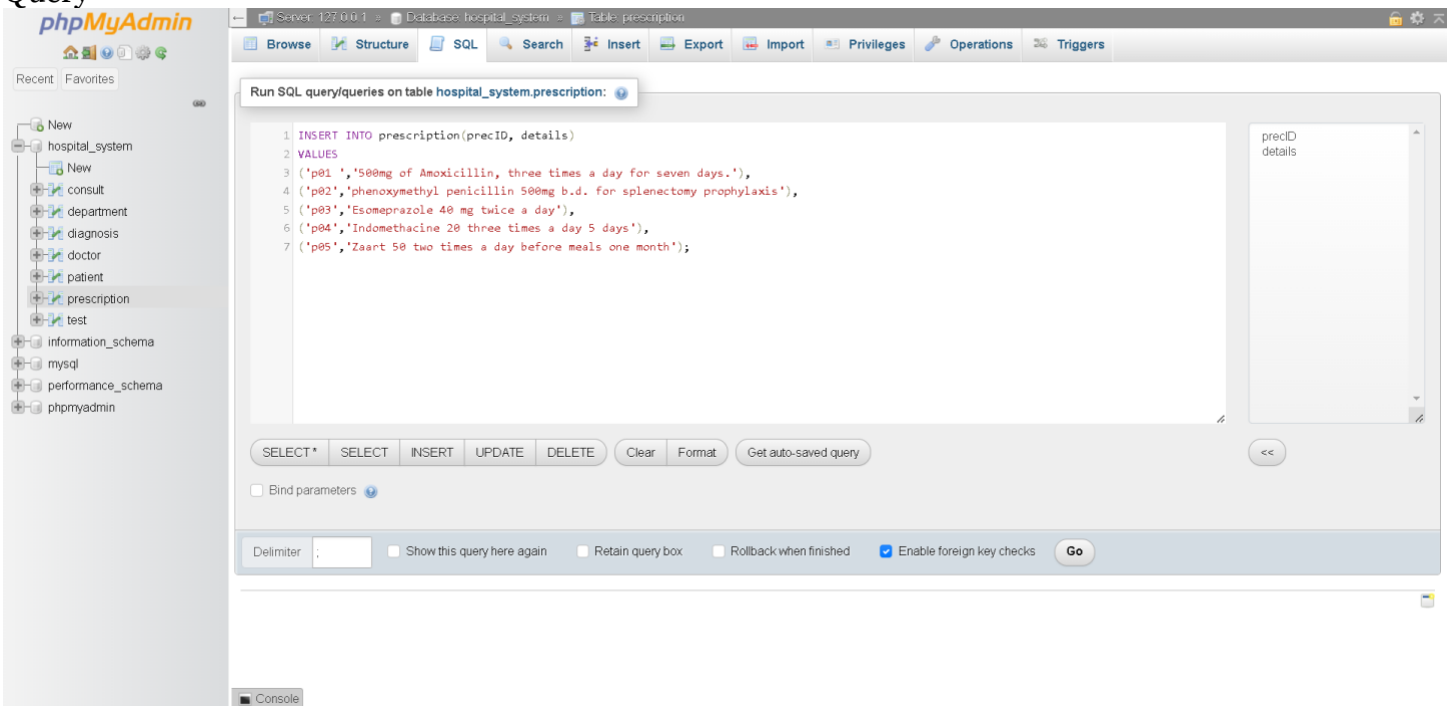
Result



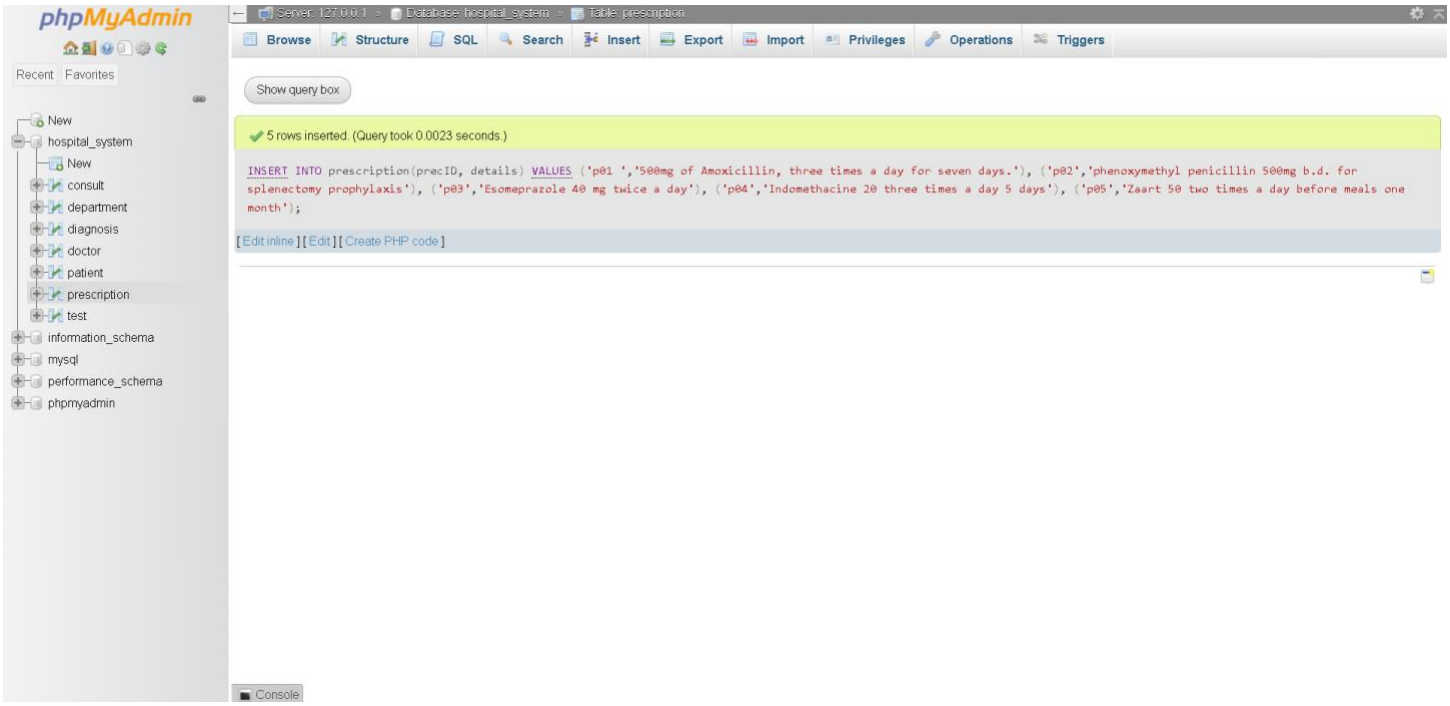
Output table



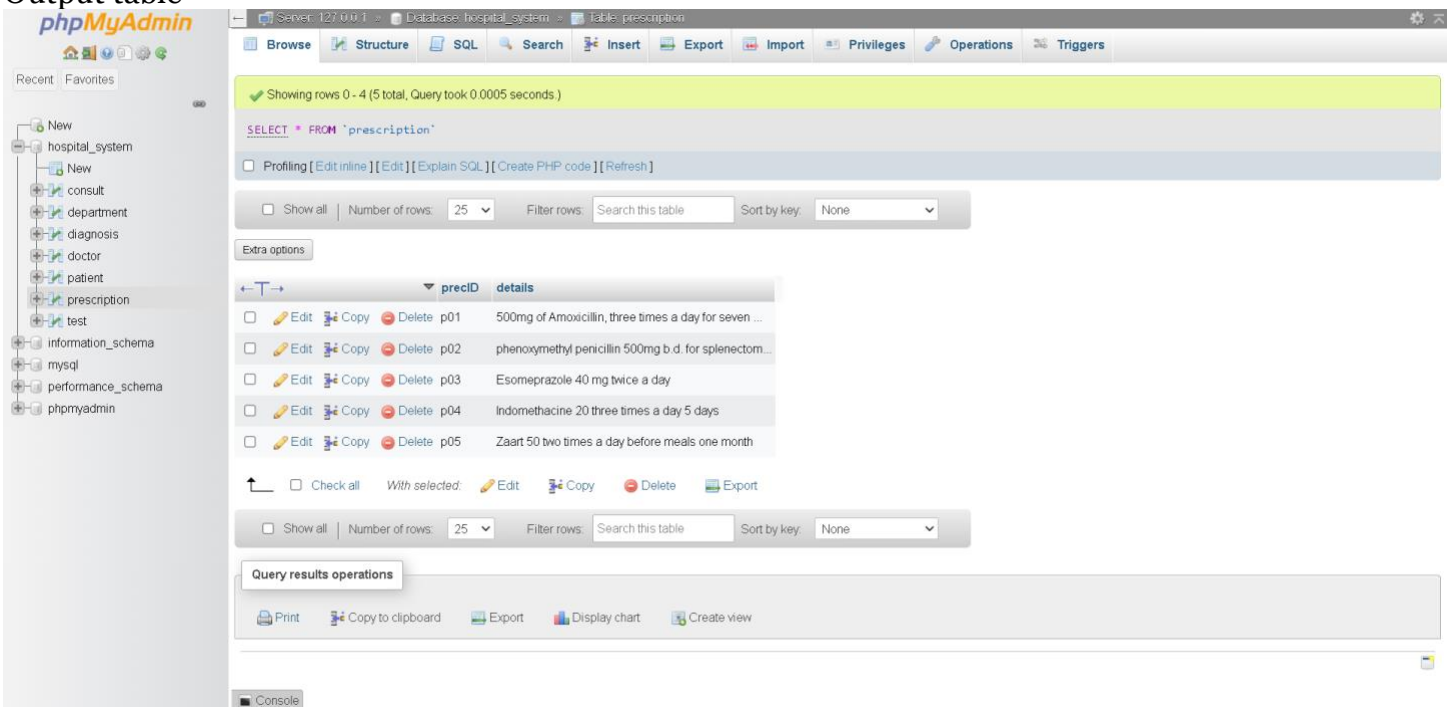
7.Prescription table
Query



Result



Output table



In conclusion

the DOC327 - Working with Data coursework was very informative, and we picked up a lot of knowledge that covered the entire subject. During this training, we learned about ER diagrams, DDL SQL statements, relationship types, structural constraints, and many other things that will undoubtedly be useful in our future careers in information technology. This lesson served as a link to our degree program, which is a crucial component.

From this coursework, we learned how databases function in the ICT sector and how crucial it is to maintain human everyday labor. Our team would like to express its sincere gratitude to the module leader and lecturer for their assistance in helping us reach this objective.

Workload Matrix

Task Breakdown	20221031	20221044	20221025	20221041	20221024
Acknowledgement		✓			
Drawing Er diagram				✓	
Part A 1st Question			✓	✓	
Part A 2nd Question					✓
Part B Question 1	✓	✓			
Part B Question 2	✓				
Editing and formatting the document		✓			